

# **A ROLE-AWARE EXPLAINABLE AI DASHBOARD FOR STUDENT AT-RISK CLASSIFICATION IN LEARNING MANAGEMENT SYSTEMS**

A Thesis Submitted in Partial Fulfilment of the Requirements for the Degree of  
Master of Science in Computer Science

by

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## ABSTRACT

This thesis presents the design, development, and technical evaluation of a role-aware explainable artificial intelligence (XAI) dashboard for predicting at-risk students within Learning Management System (LMS) environments. While predictive models using LMS behavioural and assessment data have been investigated for early identification of students at risk of failing or withdrawing, existing systems typically present these predictions as isolated risk scores, offering little insight into the underlying reasoning and no differentiation between the needs of different institutional stakeholders. Guided by a Knowledge Engineering approach grounded in the CommonKADS methodology, this study designed a dashboard that delivers explanations tailored separately to instructors, academic advisors, and administrators from a single underlying prediction system.

An ensemble machine-learning pipeline was trained and evaluated on the Open University Learning Analytics Dataset (OULAD), comprising 32,593 student-module-presentation observations across 22 module presentations, with SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) integrated as the explanation layer. The XGBoost classifier used in the prototype, selected with recall as the primary deployment criterion, achieved an out-of-fold accuracy of 87.7%, a precision of 0.866, a recall of 0.909, and an F1-score of 0.887, correctly identifying the large majority of at-risk students in the held-out data, while Random Forest obtained a marginally higher F1-score of 0.898. Because the features were computed over the full course presentation, these results describe retrospective classification using the available OULAD feature set and do not establish performance at a predefined early-warning point. Explainability analysis showed partial agreement between SHAP and LIME rankings, indicating that presenting both methods side by side provides a technically cross-checked representation of model behaviour and provides a basis for future practitioner evaluation.

The study's contributions include an implemented role-aware XAI dashboard architecture, a documented and benchmarked predictive model, empirical evidence on the behaviour and agreement of SHAP and LIME on a real-world dataset, and a set of candidate design guidelines for explainable learning analytics dashboards. Because the evaluation was conducted technically against a secondary dataset rather than through direct engagement with practitioners, the findings establish the technical feasibility and soundness of the approach and lay the groundwork for future empirical validation of its impact on practitioner trust, understanding, and intervention decision-making.

**Keywords:** explainable AI, learning analytics, at-risk prediction, role-aware dashboard, SHAP, LIME, CommonKADS, OULAD.

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# CHAPTER 1

## INTRODUCTION

### 1.1 Background of the Study

The rise of digital technology has transformed the operational landscape of higher education institutions, producing a new generation of learning environments powered by Learning Management Systems (LMS) such as Moodle, Canvas, and Blackboard (Kuzilek et al., 2017). These platforms have restructured the mechanics of course delivery, assessment submission, and student-instructor communication at scale. Their institutional adoption expanded sharply in the wake of the COVID-19 pandemic, as universities were compelled to move instruction online and thereby generated substantially larger volumes of student behavioural data than had previously been available for analysis (Rebelo Marcolino et al., 2025).

With this expansion of digital learning, interest in learning analytics (LA) has grown steadily. LA is defined as the measurement, collection, analysis, and reporting of data about learners and their contexts with the purpose of understanding and optimising learning (Siemens & Long, 2011). One of its most important applications is the early identification of students at risk of failing or withdrawing. Prior research has investigated predictive models trained on LMS behavioural logs, demographic records, and assessment histories for the early identification of these students, with the aim of affording institutions the opportunity to intervene before outcomes become irreversible (Jin et al., 2024; Kuzilek et al., 2017). The Open University Learning Analytics Dataset (OULAD), capturing interaction data from 32,593 student-module-presentation observations across 22 module presentations (Kuzilek et al., 2017), has become one of the most widely used benchmarks for this line of research (Gunasekara & Saarela, 2025; Jin et al., 2024; Wang, 2025). Wang (2025) reports that a hybrid ensemble integrating decision trees, random forests, support vector machines, and artificial neural networks attained 96.9% classification accuracy on OULAD and 98.8% on the UCI student performance dataset. Figures of this kind depend on the evaluation protocol under which they were obtained, so they are reported here as context rather than as a benchmark directly comparable with the results of this thesis (see Chapter 4, Section 4.6). Rebelo Marcolino et al. (2025) showed that CatBoost with multi-objective optimisation produces strong dropout prediction performance on Moodle log data.

Despite these technical advances, the question of what happens after a student is flagged as at-risk has remained largely unaddressed. Ali et al. (2023) document in their review of over 410 XAI studies, the outputs of many AI models are difficult to comprehend owing to their black-box nature. The field of Explainable AI (XAI) addresses this limitation through techniques such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), which enable practitioners to identify the features that drove a particular prediction. In educational settings, such transparency matters considerably because a prediction about a student's future can directly affect both that student's academic trajectory and the professional judgement of the instructor or advisor expected to act upon it.

Different stakeholders within an LMS, instructors, academic advisors, and administrators, each require different kinds of explanation from the same underlying model. An instructor monitoring weekly engagement does not require the same level of detail as an administrator reviewing withdrawal rates across an entire cohort (Kostopoulos et al., 2024). The literature reviewed for this

study did not identify a dashboard that delivers these role-specific explanations from a shared prediction system (Afzaal et al., 2021; Cabral et al., 2025; Kaliisa et al., 2023; Susnjak et al., 2022). This study addressed that gap through the design and technical evaluation of a role-aware XAI dashboard that delivers tailored model explanations to each stakeholder group.

## **1.2 Problem Statement**

With the increasing adoption of AI-driven student prediction systems in higher education, concerns have shifted from model accuracy to issues of transparency and practical usability. Many prediction systems generate risk scores without explaining the reasoning behind them, leaving instructors, advisors, and administrators without sufficient context to make informed interventions (Susnjak et al., 2022). Although techniques such as SHAP and LIME can provide interpretable explanations, their integration into real-world learning analytics dashboards remains limited (Ahmed et al., 2025; Ali et al., 2023). As a result, most existing systems continue to present risk indicators in isolation. This limitation is further amplified by the heterogeneous information needs of institutional stakeholders, which the literature suggests call for explanations suited to the recipient's role and expertise (Haque et al., 2023; Kostopoulos et al., 2024), yet current dashboards rarely support role-specific perspectives. In addition, evaluation approaches for explainable AI in education remain underdeveloped, with limited domain-specific frameworks extending beyond general measures such as trust (Al-Ansari, 2024). The problem addressed in this thesis is therefore framed as a design and technical research problem: how a role-aware explainable AI dashboard for at-risk student prediction can be designed, implemented, and technically evaluated, guided by a Knowledge Engineering approach grounded in the CommonKADS methodology. Because the study does not collect data from practitioners, it does not establish a demonstrated usability problem or measure practitioner decision-making; these questions are left to future practitioner-facing research.

## **1.3 Research Aim and Objectives**

### **Aim**

The aim of this study is to investigate the extent to which a role-aware explainable artificial intelligence dashboard can provide prediction for student at-risk within Learning Management System environments. The Open University Learning Analytics Dataset (OULAD) is used as the empirical basis of the study.

### **Specific Objectives**

1. To examine, through a structured review of relevant literature, the predictive features, machine learning models, and XAI techniques most effective for student at-risk prediction in LMS environments, and to identify the knowledge engineering and design principles that govern role-aware explainable dashboard development.
2. To develop and evaluate an ensemble machine learning predictive model for at-risk student identification using the OULAD benchmark dataset, and to integrate SHAP and LIME explainability techniques to generate both global and instance-level explanations of model behaviour.

3. To design and implement a role-aware XAI dashboard delivering differentiated explanatory views for instructors, academic advisors, and institutional administrators, and to evaluate the technical adequacy and explanatory behaviour of the underlying predictive and explainability components against the OULAD dataset.

## **1.4 Research Questions**

1. How can explainable AI techniques (specifically SHAP and LIME) be systematically integrated into a knowledge engineering pipeline to improve transparency in student at-risk prediction models?
2. What dashboard design features and explanation structures, grounded in the CommonKADS Agent Model and the empirical learning analytics dashboard literature, can be specified to support the distinct information needs of instructor, advisor, and administrator roles within an LMS?
3. What do the predictive performance and explainability findings indicate about the technical potential of the proposed role-aware explainable dashboard to support practitioner interpretation and decision-making?

## **1.5 Contributions of the Study**

This thesis advances academic discourse on explainable AI, learning analytics, and knowledge engineering by producing an implemented dashboard architecture and verified technical evidence on the predictive and explanatory behaviour of role-aware XAI dashboards for AI-generated student risk predictions in higher education. The study makes the following contributions:

1. An implemented role-aware XAI dashboard architecture demonstrating how SHAP and LIME explanations can be differentiated across instructor, advisor, and administrator interfaces within a single LMS-integrated prediction system, underpinned by the CommonKADS communication and design models.
2. A documented ensemble predictive model trained on OULAD with performance benchmarks against prior studies, providing evidence on predictive accuracy and on the agreement between the two explanation methods.
3. Verified technical evidence on model performance and on SHAP/LIME explanation behaviour against a real-world benchmark dataset (OULAD), including the degree of agreement and divergence between the two explanation methods, providing an empirical basis for future practitioner-facing evaluation of trust, interpretation, and intervention decision quality.
4. A set of candidate design guidelines for explainable learning analytics dashboards, derived from the CommonKADS methodology and the implemented system, proposed as a basis for future empirical validation with practitioners.

## **1.6 Scope and Limitations**

The scope of this study was limited to the design and evaluation of a role-aware XAI dashboard for student at-risk prediction, focusing on higher education institutions operating LMS platforms that generate interaction log data. The primary dataset employed for model development was the publicly available OULAD dataset, capturing 32,593 student-module-presentation observations across 22 module presentations at the Open University (Kuzilek et al., 2017). The modelling

component was limited to binary at-risk classification using behavioural and demographic features extractable from the LMS. Because these features were computed over the full course presentation, the study evaluates retrospective classification using the available OULAD feature set and does not establish performance at a predefined early-warning point in a presentation.

The dashboard was designed to differentiate explanatory content for three stakeholder role groups: instructors, academic advisors, and administrators. The study did not extend to student-facing interfaces, as the primary concern was with institutional decision-makers rather than self-regulated learners. Because the study relied exclusively on the secondary OULAD dataset, the predictive and explainability components underlying the dashboard were evaluated technically rather than through direct engagement with practitioners at an institution, and no primary data were collected from instructors, advisors, or administrators. Conclusions about the appropriateness of the role-differentiated design for real practitioners are accordingly drawn from existing literature rather than from an original user study, and are not generalised to any specific institutional context without further empirical validation.

The study did not assess the long-term real-world impact of the interventions triggered by the dashboard, since this would have required both primary practitioner data and longitudinal data collection beyond the scope of a master's thesis relying on a single secondary dataset. Limitations included the absence of primary practitioner data with which to empirically validate the trust, usability, and decision-quality implications discussed in Chapter 5, reliance on a single secondary dataset (OULAD) whose demographic and institutional context may not generalise to other settings, and the consequent inability to assess how the role-differentiated design would perform in live practitioner use. In addition, the explanation stability and fidelity checks, the probability calibration analysis, and the subgroup fairness assessment specified in the methodology were not completed within the final evaluation (Chapter 3, Section 3.4.1). Accordingly, the findings of this thesis are reported against a three-level evidence model that distinguishes what was demonstrated by the technical evaluation, what is supported by design, and what is not empirically established.

## **1.7 Structure of the Thesis**

The remainder of this thesis is organised into six chapters. Chapter 2 presents a structured review of the literature on learning analytics, at-risk student prediction, explainable AI, and role-aware dashboard design, and establishes the knowledge engineering foundations of the study. Chapter 3 sets out the research methodology, detailing the CommonKADS-based knowledge engineering approach, the modelling pipeline, and the technical evaluation design. Chapter 4 describes the design and implementation of the ensemble predictive model, the SHAP and LIME explanation layer, and the role-aware dashboard as a working prototype. Chapter 5 reports the results of the technical evaluation of the predictive model and the SHAP/LIME explainability components against the OULAD dataset. Chapter 6 discusses the implications of these findings with reference to existing literature, revisits the research questions, and sets out the limitations of the study. Chapter 7 concludes the thesis by summarising the study, answering the research questions, stating its contributions, and outlining recommendations and directions for future research.

## CHAPTER 2

### LITERATURE REVIEW

#### 2.1 Learning Analytics in Higher Education

Since the early 2010s, learning analytics has emerged as an established field of research. Despite this maturity, advances in prediction-oriented learning analytics research have not been matched by equal attention to the human, ethical, and pedagogical considerations necessary for effective institutional implementation. Siemens & Long (2011) offer the field's most widely cited definition of LA as "the measurement, collection, analysis, and reporting of data about learners and their contexts", yet this definition does not address the question of who benefits from that analysis or how insights should be communicated to different institutional actors. Verbert et al. (2013) argue that dashboard design must center the practitioner as an active user rather than a passive recipient of algorithmic outputs, a position that resonates strongly with the role-aware approach of this research. The growth of LMS platforms in higher education has produced an abundance of clickstream logs, assessment histories, forum activity, and demographic records, creating the data needed for predictive analytics but not necessarily for their effective institutional use (Kuzilek et al., 2017).

Jin et al. (2024) conducted a PRISMA-compliant systematic review of 17 OULAD-based studies, classifying machine learning approaches by predictive target: academic performance, at-risk status, and engagement level. Their findings confirm that ensemble methods, particularly gradient boosting variants and random forests, consistently outperform simpler classifiers. However, Jin et al. (2024) note that the overwhelming focus on model accuracy leaves the practitioner-facing dimension of these systems largely unaddressed. Leitner et al. (2019) review the challenges of LA implementation in higher education institutions and identify a persistent disconnect between the technical sophistication of predictive models and their uptake in institutional decision-making, attributing this gap in part to the absence of role-appropriate explanation interfaces. Together, these studies demonstrate that while predictive modelling techniques have advanced considerably, comparatively less attention has been given to designing explainable learning analytics systems that communicate predictions according to the differing information needs of institutional stakeholders. This study addresses that gap by proposing a role-aware explainable learning analytics framework for higher education.

Beyond these systematic reviews, a broader body of research has attempted to clarify what learning analytics is for and whose interests it should serve. Ferguson (2012) traces the emergence of the field from earlier traditions in educational data mining and academic analytics, arguing that its distinctiveness lies not in the data or algorithms it employs but in its explicit orientation toward improving learning. This orientation has since been reinforced by Gašević, Dawson, and Siemens (2015), who caution that the technical maturity of predictive tooling has outpaced its pedagogical grounding, and that models which optimize for statistical accuracy without a defensible theory of learning risk producing insights that institutions cannot meaningfully act upon. This study takes this critique seriously by anchoring its dashboard design not in raw predictive performance but in the decision needs of specific institutional actors.

The reference model proposed by Chatti, Dyckhoff, Schroeder, and Thüs (2012) provides a useful conceptual framework for situating this study within the wider field. Their four-dimensional

taxonomy, covering the data and environments analysed, the stakeholders addressed, the objectives pursued, and the methods employed, places importance on the stakeholder differentiation as a first-class design concern rather than an afterthought. Viberg, Hatakka, Bälter, and Mavroudi (2018), in a systematic review of 252 higher education learning analytics studies, report that only a small fraction of published work provides evidence of measurable improvement in learning outcomes, and that even fewer studies engage with the differentiated information needs of non-student stakeholders. This finding provides direct empirical support for the problem framing adopted in this study.

The gap between analytical capability and institutional adoption is now well documented. Tsai and Gašević (2017) synthesise policy-level evidence from European and Australasian universities and identify a persistent shortage of practitioner-facing decision support as one of the primary barriers to sustained adoption. Complementary evidence is provided by Larrabee Sønderlund, Hughes, and Smith (2019), whose systematic review of learning analytics interventions concludes that the empirical case for improved learner outcomes remains modest, largely because the interventions triggered by analytical insights are inconsistently designed and inconsistently delivered. Read together with Leitner et al. (2019), this literature suggests that the effectiveness of learning analytics depends less on the sophistication of the underlying models and more on whether the resulting insights are communicated in a form that institutional actors can interpret and act upon, which is the central premise of the study.

## **2.2 Student At-Risk Prediction Models**

The prediction of student at-risk status has attracted sustained research attention, with investigations spanning logistic regression, decision trees, ensemble methods, and deep learning architectures. Wang (2025) reports that a hybrid ensemble combining decision trees, random forests, support vector machines, and artificial neural networks achieves 96.9% accuracy on OULAD (and 98.8% on the UCI student performance dataset), while Rebelo Marcolino et al. (2025) demonstrate that CatBoost with NSGA-II multi-objective optimisation attains comparably high dropout prediction performance on Moodle log data. These results are consistent with the earlier findings of Jin et al. (2024), who identify ensemble gradient boosting models as the dominant approach in recent OULAD literature. Guevara-Reyes et al. (2025) extend this comparison by examining interpretability trade-offs across model families, finding that more complex ensembles gain accuracy at the cost of intrinsic interpretability, a trade-off that motivates the post-hoc XAI integration in this study.

A systematic review of 33 machine learning studies by Albreiki et al. (2021) finds that predictive accuracy has improved substantially across the literature, but that this technical progress has not been accompanied by improvements in how predictions are communicated: the overwhelming majority of reviewed systems generate risk outcomes without offering educators any interpretive account of why a particular student was flagged. Ahmed et al. (2025) further demonstrate that ensemble models with integrated SHAP and LIME explanations are both technically feasible and accurate when applied to multi-institutional academic performance datasets, yet their deployment in live institutional dashboards remains rare. This convergence of findings across independent review studies establishes a clear and well-evidenced gap between the technical capability to predict student at-risk status and the practitioner-facing communication of those predictions.

The increased focus on prediction within learning analytics has deep theoretical roots that are not always made explicit in the technical literature. Tinto (1975) proposed an influential model of student departure in which academic and social integration jointly determine persistence, and subsequent empirical work has operationalised his constructs through the behavioural traces captured by modern LMS platforms. Aljohani, Fayoumi, and Hassan (2019) show the relationship by demonstrating that clickstream indicators of engagement, submission timeliness, and forum participation align closely with Tinto's academic integration dimension and yield reliable early warnings on the OULAD dataset. This theoretical grounding is important for the present study because it clarifies which behavioural features carry pedagogical meaning and are therefore worth surfacing in explanation views.

A broader empirical literature has explored a wide range of algorithmic families for at-risk prediction, and their findings converge on several stable patterns. Hussain, Zhu, Zhang, and Abidi (2018) applied logistic regression, support vector machines, and neural networks to LMS engagement records and reported that engagement-based features consistently outperformed static demographic features across all model families. Adnan et al. (2021) extended this line of enquiry by evaluating multiple classifiers at different points within a course and demonstrated that useful predictions can be generated after as little as 20% of the course has elapsed, an important finding for institutional intervention windows. Chen & Cui (2020) argue that time-series representations of student behaviour further improve early prediction, particularly when the model can capture patterns of disengagement that unfold gradually across weeks rather than surfacing in a single indicator.

Deep learning approaches have also been investigated, though with mixed results relative to their computational cost. Waheed et al. (2020) trained deep artificial neural networks on OULAD and reported strong classification performance, but they also noted that the marginal accuracy gains over well-tuned ensembles were modest and did not justify the additional opacity that deeper architectures introduce. Hassan et al. (2019) reached a similar conclusion for withdrawal prediction, showing that recurrent architectures capture temporal engagement patterns effectively but produce outputs that are difficult to interpret without dedicated post-hoc explanation techniques. These results reinforce the argument that predictive accuracy alone is an insufficient basis for model selection in institutional contexts, and that interpretability and explanation-readiness should be treated as the primary design criteria.

A further strand of the literature examines the fairness and generalizability of at-risk models, a concern that is directly relevant to any system that will influence intervention decisions. Baker & Hawn (2022) provide a comprehensive review of algorithmic bias in education and document systematic disparities in model performance across demographic subgroups, particularly where training data underrepresents specific student populations. Yu, Li, Fischer, Doroudi, and Xu (2020) show, using large-scale institutional data, that predictive models can produce systematically different error rates across gender, ethnic, and first-generation student groups even when overall accuracy is high. Taken together, these studies suggest that the aggregate accuracy figures reported in benchmark comparisons on OULAD should be interpreted with caution, and that any operational dashboard should surface model behaviour at a level of detail that allows practitioners to identify and question such disparities. The role-aware explanation design pursued in this study is one mechanism through which this becomes possible.

Several methodological refinements recur in this literature and are relevant to the design choices made later in this thesis. Class imbalance is commonly addressed using the Synthetic Minority Oversampling Technique (SMOTE), originally introduced by Chawla, Bowyer, Hall, and Kegelmeyer (2002), and the fifteen-year retrospective analysis of Fernández, García, Herrera, and Chawla (2018) confirms that SMOTE and its principled variants remain competitive with more recent resampling methods on tabular data. Because naive oversampling can amplify noise near class boundaries, borderline and adaptive variants of SMOTE have also been proposed, and Prati, Batista, and Silva (2015) describe a protocol for analysing the comparative behaviour of such treatment methods. Santos et al. (2018) further recommend that resampling occur only on the training partition of each cross-validation fold, so that oversampling before splitting does not produce optimistic bias. Class-weighted loss functions are a common alternative, since they preserve the marginal distribution of the training data.

Hyperparameter optimisation and validation design are a second recurring concern. Bergstra, Bardenet, Bengio, and Kegl (2011) provide empirical evidence that sequential model-based optimisation, such as Bayesian optimisation with the Tree-structured Parzen Estimator, outperforms grid and random search in high-dimensional configuration spaces at equivalent computational budgets. Many OULAD studies report results from stratified random splits, and Jin et al. (2024) note that evaluation protocols vary considerably across this literature. Krstajic, Buturovic, Leahy, and Thomas (2014) criticise protocols in which observations from the same higher-level unit appear in both training and validation folds, since this form of leakage produces optimistic estimates. Chicco and Jurman (2020) recommend the Matthews correlation coefficient as a more informative summary than accuracy or F1 alone under class imbalance, and area under the precision-recall curve and balanced accuracy are commonly reported as secondary metrics for the same reason.

Probability calibration is a third consideration for systems that display risk probabilities rather than binary decisions, since miscalibrated probabilities may mislead practitioners into overreacting to weak signals. Niculescu-Mizil and Caruana (2005) compare isotonic regression and Platt scaling for this purpose, and Guo, Pleiss, Sun, and Weinberger (2017) quantify calibration error using reliability diagrams and expected calibration error. Kim et al. (2024) note that trust in AI-generated outputs is sensitive not only to accuracy but also to the perceived reliability of the confidence values displayed to users.

Finally, feature design matters for early prediction. Adnan et al. (2021) provide evidence that trend-based features, which capture period-over-period change in engagement and performance, improve early warning even when absolute values are stable. These refinements (class imbalance handling, hyperparameter optimisation, calibration, and trajectory features) were considered when the methodology of this study was planned; Chapter 3, Section 3.4 explains which were used in the final evaluation and records the remainder as planned but not implemented extensions.

## **2.3 Explainable Artificial Intelligence: A Comparative Analysis**

Explainable AI has emerged as an important addition to high-accuracy models, and the educational domain illustrates both the urgency and the complexity of this challenge. Arrieta et al. (2020), in a comprehensive study of XAI methods spanning over 400 papers, distinguish between interpretable models and explanation techniques, arguing that the choice between these approaches depends on the deployment context and the expertise of the intended recipient, a distinction that earlier XAI

surveys by Gunning and Aha (2019) did not systematically address. Ali et al. (2023) extend this analysis by documenting that user-type-specific explanation tailoring remains an open problem across the XAI literature, with most systems defaulting to generic, technically framed outputs that presuppose a level of ML expertise that practitioners in education typically do not possess.

Within the educational domain, SHAP and LIME dominate current applications, but their comparative strengths and limitations are contested. Nnadi et al. (2024) evaluated SHAP, LIME, Anchors, Accumulated Local Effects (ALE), and counterfactual explanations on a student adaptability dataset, finding SHAP most consistent for feature attribution stability across model types. By contrast, Salih et al. (2024) provides a rigorous critique of both techniques, noting that SHAP's computational overhead limits real-time deployment and that LIME's local approximation boundary is sensitive to sampling parameters in ways that affect reproducibility. Gunasekara & Saarela (2025), who apply SHAP and LIME to models trained on the OULAD dataset, help explain these differences: prior studies have reported complementary strengths, with SHAP commonly used for global attribution and LIME for local approximations. Islam et al. (2025) further demonstrate that combining SHAP, Shapash, ELI5, and LIME with XGBoost enables fine-grained performance explanation suitable for institutional reporting, though their work stops short of evaluating how non-technical stakeholders receive and interpret such explanations.

What is noticeably absent from all of these studies is a consideration of how explanations should be formatted differently for practitioners with different roles, levels of expertise, and decision-making responsibilities. Afzaal et al. (2021) present one of the earliest examples of an XAI dashboard delivering SHAP-based explanations within an LMS, directed at students rather than institutional staff. While their work demonstrates the technical feasibility of embedding XAI in an LMS context, it does not address the instructor, advisor, or administrator perspectives that are central to institutional at-risk intervention. The gap between what XAI techniques can deliver and what specific stakeholder roles actually require is the central research gap addressed in this study.

The theoretical foundations of explainable AI have been the subject of previous research that complements the empirical evaluations discussed above. Miller (2019), drawing on decades of research in cognitive and social psychology, argues that explanations are inherently social, contrastive, and selective, and that XAI systems which ignore these properties tend to produce technically accurate but psychologically unsatisfying accounts of model behaviour. Lipton (2018) offers a parallel critique of the interpretability literature, showing that the term is used inconsistently and that different notions of interpretability, including simulatability, decomposability, and post-hoc transparency, imply different design choices. These conceptual contributions have direct implications for the study because they establish that a role-aware dashboard must decide not only which explanations to display but also which underlying notion of interpretability each stakeholder actually requires.

A related literature has examined the limits of post-hoc explanation techniques themselves. Adadi & Berrada (2018) survey a broad taxonomy of XAI methods and observe that the field lacks agreed evaluation protocols, leaving many claims about explanation quality difficult to verify. Guidotti et al. (2018) provide a complementary survey of black-box explanation methods and note that fidelity, the degree to which an explanation faithfully reflects the underlying model, is often traded against comprehensibility in ways that are rarely reported. Rudin (2019) advances the strongest version of this critique by arguing that high-stakes decisions should be supported by intrinsically

interpretable models rather than by post-hoc explanations of opaque ones. Although the study adopts a post-hoc approach for consistency with prior OULAD literature, these critiques motivated the fidelity and stability checks specified for the explanation pipeline in Chapter 3, Section 3.5, which were proposed as safeguards but not completed in the final evaluation, and they suggest caution when relying on SHAP or LIME outputs in isolation.

Within the educational domain specifically, Khosravi et al. (2022) map the state of explainable AI in education and identify six recurring stakeholder groups, including learners, teachers, and administrators, whose explanation needs are structurally different. Their analysis reinforces the central design proposition of this study by showing that a single, undifferentiated explanation view is unlikely to serve any of these groups well. Conati, Barral, Putnam, and Rieger (2021) provide empirical support for a personalised approach through a case study within an intelligent tutoring system, demonstrating that learners with different levels of prior knowledge derive substantially different value from the same underlying explanations. These findings extend the case for role-aware design from a normative argument to an evidence-based design requirement.

Counterfactual explanations have also emerged as an increasingly important complement to feature-attribution methods such as SHAP and LIME. Wachter, Mittelstadt, and Russell (2018) argue that counterfactuals answer the question of what would have to change for a prediction to be different, which is precisely the question that a practitioner facing an intervention decision needs to answer. Susnjak, Ramaswami, and Mathrani (2022) integrate counterfactual reasoning into a learner-facing dashboard that is deployed for trials at a higher education institution, but their design addresses a single stakeholder group. The comparative literature therefore establishes both the technical maturity and the educational value of counterfactual explanations while confirming that their role-differentiated presentation remains an open problem.

## **2.4 Dashboards in Learning Analytics**

Learning analytics dashboards (LADs) serve as the primary interface through which analytical insights reach institutional stakeholders, yet the evaluation literature reveals an ongoing design limitation. Susnjak et al. (2022) note that most existing LADs confine themselves to descriptive reporting of past behaviour, and they respond to this limitation by proposing a learner-facing dashboard, described by its authors as the first of its kind in the breadth of analytics it integrates, that combines descriptive, predictive, and prescriptive analytics with counterfactual explanations within a single interface. However, this dashboard is designed for a single user group (learners) and does not differentiate between the varied information needs of instructors, advisors, and administrators. This single-role limitation is shared by Chavan & Mitra (2022), who apply iterative prototyping to an instructor-facing video analytics dashboard and demonstrate improved usability, but whose design does not generalise across multiple institutional stakeholder groups.

Kaliisa & Dolonen (2022) evaluate a teacher-facing LA dashboard through ten semi-structured interviews, finding that instructors prioritise summary-level alerts over granular statistical breakdowns, a finding that has direct implications for how the instructor interface of the present study should be designed. Kaliisa et al. (2023) contribute a four-dimensional design checklist covering purpose, stakeholder involvement, design, and evaluation that guided the dashboard development methodology of this study. Cabral et al. (2025), in a recent systematic review of AI-powered learning analytics dashboards, identify limited XAI integration as one of the most persistent research gaps in the field, noting that XAI components are absent from most of the

reviewed dashboards. Kaliisa et al. (2023) further report that, although teachers are involved in the design of teacher-facing dashboards, this involvement is mainly at the exploratory or problem-definition stage, with little input beyond it, which risks producing designs that do not fully reflect the needs of the practitioners they are intended to serve.

The dashboard literature reviewed above sits within a longer research tradition that has attempted to systematise the design space of learning analytics dashboards. Schwendimann et al. (2017), in a systematic review of 55 dashboards, propose a taxonomy that distinguishes dashboards by their data sources, intended users, indicators, and visualisation styles, and note that only a small proportion of reviewed systems were grounded in explicit theories of learning or decision making. Bodily and Verbert (2017) offer a complementary review focused specifically on student-facing dashboards and educational recommender systems, and observe that the field has developed rich technical methods for surfacing behavioural indicators while making comparatively little progress on how those indicators should be interpreted by their intended users. Read alongside the more recent review by Cabral, Pinto, and Goncalves (2025), these earlier reviews show that the limited explanation of analytical outputs has been a recognised gap in the literature for some time.

Empirical evaluations of dashboard effectiveness have arrived at similarly cautious conclusions. Matcha, Gašević, and Pardo (2020) synthesise the findings of 21 empirical studies of learning analytics dashboards through a self-regulated learning lens and report that most systems are neither theoretically grounded nor evaluated against learning outcomes, with the majority of studies limiting themselves to usability measurement or short-term satisfaction surveys. Sedrakyan, Malmberg, Verbert, Järvelä, and Kirschner (2020) argue that this evaluation deficit is partly a consequence of design decisions that treat the dashboard as a terminal reporting artefact rather than as one component of a broader feedback loop, and they call for designs that explicitly model how analytical outputs enter the practitioner's decision-making process. The study responds to this call by embedding the dashboard within a CommonKADS communication model that treats each stakeholder role as a distinct decision-making agent with its own knowledge requirements.

## **2.5 Role-Aware System Design**

Beyond the general dashboard design literature, a smaller but growing body of work explicitly addresses the role-aware presentation of analytical insights. The CommonKADS methodology, first consolidated by Schreiber et al. (2000) and developed further by Studer, Benjamins, and Fensel (1998), provides one of the clearest theoretical foundations for this approach by treating agents, their tasks, and the knowledge they require as distinct modelling artefacts. Applied to learning analytics, this perspective implies that the design of an instructor interface, an advisor interface, and an administrator interface must begin from an explicit analysis of the tasks each role performs and the knowledge each role brings to those tasks, rather than from a uniform visual template applied across roles. This is the modelling stance adopted throughout this study.

Human-centred XAI research reinforces the same principle from a different direction. Liao & Varshney (2022) and Ehsan et al. (2021) have argued that the effectiveness of an explanation depends on the fit between the explanation content, the recipient's expertise, and the decision they are trying to make, and that generic explanations tend to satisfy no one particularly well. Kouki, Schaffer, Pujara, O'Donovan, and Getoor (2020) provide empirical evidence for this claim in the context of hybrid recommender systems, showing that personalised explanations that adapt to user characteristics outperform undifferentiated explanations on measures of perceived usefulness and

satisfaction. Although these studies are not situated within education, their findings can be applied to the differentiated instructor, advisor, and administrator interfaces proposed in this study.

Empirical work within learning analytics has begun to converge on similar conclusions. Kaliisa, Jivet, and Prinsloo (2023) argue that dashboard design must move beyond a single archetypal user and explicitly engage with the plurality of stakeholders whose decisions the system will inform. Gutierrez et al. (2020), in an evaluation of the LADA advising dashboard, demonstrate that even a single stakeholder role encompasses substantial variation in information needs, with senior advisors preferring cohort-level views and less experienced advisors relying more heavily on individual case detail. Kaliisa et al. (2023) bring these observations together in a checklist that treats stakeholder involvement, design, and evaluation as dimensions to be planned together, a position that shapes both the design and the evaluation components of the present study.

Comparative analysis across these bodies of work reveals a consistent pattern. The technical components required to generate predictions and SHAP or LIME explanations are now relatively mature and widely available. However, methodological approaches for translating shared computational outputs into role-appropriate interface content remain less developed. This study contributes to closing this methodological gap by combining the CommonKADS agent and communication models with the Kaliisa et al. (2023) design checklist, thereby providing a structured route from stakeholder analysis to role-differentiated interface content that is currently absent from the reviewed literature.

## **2.6 Research Gap**

The literature reviewed in this chapter collectively establishes a well-supported research gap in the intersection of student at-risk prediction, explainable AI, and role-aware learning analytics dashboard design. Several specific and inter-related dimensions of this gap warrant more attention and study.

Firstly, while ensemble predictive models have achieved high accuracy on OULAD and comparable datasets (Guevara-Reyes et al., 2025; Jin et al., 2024; Rebelo Marcolino et al., 2025; Wang, 2025), none of these studies addresses the practitioner-facing communication of predictions. The evaluation criteria across this body of work remain overwhelmingly technical, with F1-score, accuracy, and AUC dominating reporting, and with the human interpretability of model outputs receiving little or no consideration.

Second, the XAI literature (including Arrieta et al., 2020; Ali et al., 2023; Salih et al., 2024; Nnadi et al., 2024; Islam et al., 2025; and Gunasekara and Saarela, 2025) demonstrates the technical feasibility of post-hoc explanation methods in educational settings but stops short of examining how explanations should be structured differently for different institutional stakeholder roles. The only role-specific XAI dashboard identified in the literature reviewed for this study, that of Afzaal et al. (2021), is directed at students, not at the instructors, advisors, and administrators who are responsible for institutional intervention decisions.

Thirdly, the dashboard design literature (Cabral et al., 2025; Chavan & Mitra, 2022; Kaliisa & Dolonen, 2022; Susnjak et al., 2022) consistently demonstrates that existing LADs serve a single stakeholder type or provide undifferentiated, role-agnostic outputs. The literature reviewed in this study did not identify a study that has designed, implemented, and empirically evaluated a

dashboard that serves three distinct LMS stakeholder roles (instructors, advisors, and administrators) from a single shared prediction system with role-specific XAI explanations.

Furthermore, from a methodological standpoint, the knowledge engineering aspect of XAI dashboard development has received little attention. The representation of explanation knowledge, specifically how SHAP and LIME outputs are modelled as knowledge structures and delivered to role-differentiated recipients, is a knowledge engineering problem that existing studies have not conceptualised as such. The application of a structured KE methodology such as CommonKADS (Schreiber et al., 2000) to this design context represents an original methodological contribution.

Finally, the evaluation frameworks used in existing XAI studies (Al-Ansari, 2024; Kim et al., 2024) are predominantly generic and do not account for the role-specific differences in how trust, understanding, and decision quality manifest across different LMS stakeholder groups. This study addresses these dimensions through a technically evaluated artefact and a proposed framework for subsequent expert validation, providing a theoretically grounded, role-differentiated XAI dashboard within a CommonKADS knowledge engineering framework.

Collectively, these five dimensions of the research gap reflect a common underlying issue: the field has developed its technical, methodological, and evaluation components in relative isolation, and the small number of studies that combine them tend to do so for a single stakeholder role. What is required, and what this current study provides, is an integrated approach in which the predictive model, the explanation layer, the role-aware interface, and the evaluation framework are all designed against a shared knowledge engineering specification. Based on these findings, the study extends previous work by bringing these areas together to address a gap that has received limited attention in earlier research.

## **2.7 Chapter Summary**

This chapter has reviewed related literature on learning analytics, student at-risk prediction, explainable artificial intelligence, and learning analytics dashboard design, and has traced how each of these literatures contributes to, but does not by itself address, the design problem addressed by this study. The conceptual and empirical reviews confirm that ensemble predictive models can identify at-risk students with high accuracy on benchmark datasets such as OULAD, that post-hoc explanation techniques including SHAP and LIME are technically mature, and that learning analytics dashboards are widely deployed. At the same time, the critical review of these literatures has shown that predictive accuracy has not translated into institutional impact, that explanation

techniques are rarely tailored to their recipients, and that dashboards continue to be designed for a single archetypal user despite well-documented differences in stakeholder information needs.

The research gap identified in the previous section follows directly from this synthesis. The approach used in the remainder of the thesis, namely the use of the CommonKADS knowledge engineering framework alongside the Kaliisa et al. (2023) design checklist and the human-centred evaluation framework of Kim et al. (2024), is a direct response to the shortcomings surfaced by this review. The chapter therefore provides both the theoretical grounding and the empirical justification for the role-aware explainable AI dashboard developed and technically evaluated in the chapters that follow.

## **CHAPTER 3**

### **METHODOLOGY**

#### **3.1 Research Design**

This chapter sets out what was planned for the study and why. Chapter 4 reports what was actually built and how, Chapter 5 reports what happened when the system was tested, and Chapter 6 discusses what those results mean. Where the final evaluation departed from the original plan, this chapter describes the final method and records the planned but unimplemented elements in Section 3.4.1.

This study adopts the CommonKADS Knowledge Engineering (KE) methodology as its primary knowledge-engineering and design framework (Schreiber et al., 2000; Studer, Benjamins, & Fensel, 1998). CommonKADS was selected because the central design problem in this study is fundamentally a knowledge engineering problem: how should the knowledge produced by an AI prediction system, specifically the SHAP and LIME explanations of at-risk predictions, be acquired, represented, and delivered to three distinct stakeholder roles in a form that supports decision making for each? CommonKADS was specifically developed for the analysis and design of knowledge-intensive systems, making it directly appropriate for a system whose core function is the acquisition of student behavioural knowledge from LMS data, the representation of that knowledge as predictive model outputs and XAI explanations, and the communication of that knowledge to role-differentiated users (Schreiber et al., 2000).

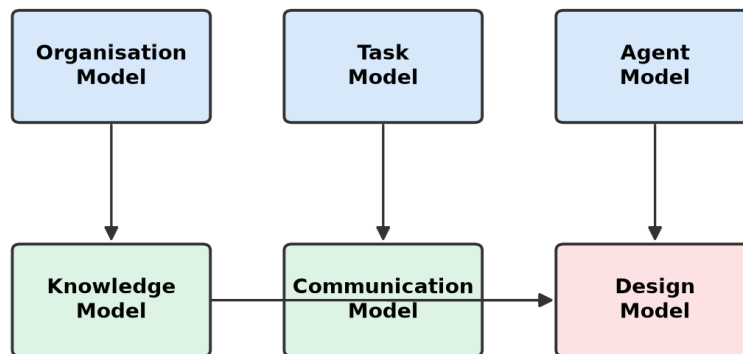
The suitability of CommonKADS for this study is further supported by applications of the methodology in educational analytics. Rodríguez-Ortiz et al. (2022) report that CommonKADS-derived models substantially improved the traceability of design decisions in a decision support system for academic performance monitoring, providing empirical support for its use in educational analytics contexts. This indicates that the methodological framework adopted here is not merely inherited from earlier expert-systems research but continues to be applied within the wider field of intelligent decision support.

Schreiber et al. (2000) structure CommonKADS around six interacting analytical components (covering institutional context, task decomposition, agent characterisation, domain knowledge formalisation, stakeholder communication patterns, and technical system design respectively), each of which maps directly onto a distinct design requirement of the present study. For the present study, the Organisation Model captures the institutional context of LMS use and identifies how at-risk prediction information currently flows between instructors, advisors, and administrators, as well as identifying where that flow breaks down. The Task Model decomposes the overall at-risk prediction and explanation task into sub-tasks: risk assessment, explanation generation, role assignment, and explanation delivery. The Agent Model characterises the knowledge, capabilities, and explanation needs of each stakeholder role, informing the differentiated interface designs. The Knowledge Model formalises the domain knowledge underlying at-risk prediction, including the behavioural and demographic features identified by the predictive model and their relative contributions as quantified by SHAP. The Communication Model specifies the interaction patterns between each agent role and the system. The Design Model maps these knowledge engineering components onto the technical architecture. This structured approach ensures that the dashboard design is based on a systematic analysis of what knowledge each stakeholder role needs, rather than a technically driven design that presumes uniform information needs (Studer et al., 1998). Table 3.1 summarises how each of the six models was instantiated in this study, the evidence on which it draws, and the design decision that resulted.

*Table 3.1. Instantiation of the CommonKADS models in this study.*

| <b>CommonKADS Model</b> | <b>Study-specific instantiation</b>                                                                                                                                                                     | <b>Evidence/source</b>                                                                                                                                                                   | <b>Resulting design decision</b>                                                                                |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Organisation Model      | Institutional context of LMS use: how at-risk information reaches instructors, advisors, and administrators, and where that flow breaks down (risk scores without explanation; one view for all roles). | Published accounts of learning analytics governance and stakeholder workflows (Tsai & Gašević, 2017; Leitner et al., 2019; Kaliisa et al., 2023); no organisational study was conducted. | Three staff roles are served from one shared prediction system; students are outside the scope of the study.    |
| Task Model              | Decomposition into risk assessment, explanation generation, role assignment, and explanation delivery; the intervention decision is left to the human agent.                                            | Task decomposition patterns for learning analytics dashboards (Verbert et al., 2013).                                                                                                    | Layered architecture separating data, prediction, explanation, role assignment, and presentation (Section 3.2). |

| <b>CommonKADS Model</b> | <b>Study-specific instantiation</b>                                                                                     | <b>Evidence/source</b>                                                                                                                                                              | <b>Resulting design decision</b>                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Agent Model             | Knowledge, capabilities, and explanation needs of instructors, advisors, and administrators.                            | Published evidence on role-specific explanation needs (Haque et al., 2023; Kostopoulos et al., 2024); populated from literature, not from knowledge elicitation with practitioners. | Three role-specific information briefs with individual, caseload, and cohort granularity (Section 3.6, Table 3.2). |
| Knowledge Model         | Domain knowledge for at-risk prediction: behavioural, assessment, and demographic features and their SHAP attributions. | OULAD feature taxonomies (Jin et al., 2024; Kuzilek et al., 2017); SHAP (Lundberg & Lee, 2017).                                                                                     | Engagement, assessment, and demographic feature families with a human-readable feature dictionary (Section 3.4).   |
| Communication Model     | Interaction pattern between each role and the system; explanations are selective and phrased for the recipient.         | Miller (2019).                                                                                                                                                                      | Progressive disclosure, side-by-side SHAP and LIME presentation, and role-gated views (Sections 3.5 and 3.6).      |
| Design Model            | Mapping of the upstream models onto the technical architecture.                                                         | Schreiber et al. (2000); the five models above.                                                                                                                                     | Five-layer architecture and role-based access control, implemented as described in Chapter 4.                      |



*Context models (top) inform knowledge & communication models, which are operationalised in the Design Model (dashboard architecture).*

*Figure 3.1. CommonKADS six-model suite adopted in this study (adapted from Schreiber et al., 2000).*

Each of the six models is instantiated concretely in the present study rather than treated as a purely conceptual scaffold. The Organisation Model is elaborated through a review of published accounts of institutional learning analytics governance (Tsai & Gašević, 2017; Leitner et al., 2019) together with the documented workflows of stakeholder groups reported by Kaliisa et al. (2023), and is used

to produce an as-is description of how at-risk information currently traverses the institution, set out in the following paragraph. The Task Model draws on the task decomposition patterns proposed by Verbert et al. (2013) for learning analytics dashboards, refined to accommodate the additional sub-tasks introduced by post-hoc explanation. The Agent Model is informed by the role-specific information needs identified by Kostopoulos et al. (2024) and by Haque, Islam, and Mikalef (2023), who synthesise practitioner-facing evidence on how explanation preferences differ by expertise. Because no practitioners participated in the study, the Agent Model was populated from published literature rather than from knowledge elicitation with practitioners, which is the acquisition activity CommonKADS would ordinarily expect (Schreiber et al., 2000). The resulting agent characterisations are therefore a well-sourced hypothesis about the three roles rather than a documented account of them, and this limitation applies to every role-specific design decision that follows from them. The Knowledge Model draws on the feature taxonomies documented in the OULAD literature (Jin et al., 2024; Kuzilek et al., 2017) and integrates them with feature-attribution outputs generated by SHAP (Lundberg & Lee, 2017). The Communication Model draws on Miller's (2019) work on the pragmatics of explanation, which argues that effective explanations are contrastive, selective, and audience-sensitive. The Design Model then translates these upstream models into the layered technical architecture developed in Section 3.2. This approach ensures that CommonKADS guides system design rather than serving only as a descriptive framework.

**Organisation Model.** Because no institutional partner was involved, the Organisation Model was constructed from published accounts rather than from an organisational study. In the as-is situation described in this literature, at-risk predictions reach institutional staff as risk scores presented in isolation, without an account of the reasoning behind them (Susnjak et al., 2022), dashboards are typically designed for a single archetypal user (Kaliisa et al., 2023), and a shortage of practitioner-facing decision support is one of the barriers to institutional adoption (Tsai & Gašević, 2017). The flow of at-risk information therefore breaks down at two points: when a prediction reaches a practitioner without an explanation that can be checked against what the practitioner knows, and when a single view requires each role to aggregate or disaggregate the information before it becomes usable. In the to-be situation specified for this study, instructors, advisors, and administrators receive the same underlying prediction and explanation at the granularity relevant to the decision each role makes, while the intervention decision remains with the practitioner.

The secondary theoretical lens is the Technology Acceptance Model (TAM) developed by Davis (1989), which frames user adoption of information systems around perceived usefulness and perceived ease of use. TAM (Davis, 1989) informs the design goals for each role-specific interface, its two core constructs, perceived usefulness and perceived ease of use, providing a simple but well validated basis for reasoning about what a usable, adoptable dashboard interface should offer each stakeholder role, complementing the user-need framing established in the CommonKADS Agent and Communication Models. Because this study is scoped to secondary OULAD data rather than primary practitioner data collection (Section 3.8), TAM is used solely as a design lens and not as an empirical framework in this study; a TAM-based practitioner evaluation is identified in Chapter 7 as future work. The research proceeds through four iterative phases: knowledge acquisition and organisation analysis; knowledge and system modelling; artifact design and development; and technical evaluation.

The choice of TAM as a secondary lens is deliberate rather than incidental. Venkatesh and Bala (2008) extended the original TAM into TAM3 by consolidating a broad body of empirical

evidence on the antecedents of perceived usefulness and perceived ease of use, and their meta-analytic findings continue to support the predictive value of these constructs for professional software adoption. Marangunic and Granic (2015) reviewed 85 TAM studies and concluded that the model remains the most consistently supported framework for evaluating information system acceptance across educational and workplace contexts. Using TAM in parallel with the CommonKADS Agent and Communication Models therefore gives the design process a well validated pair of design lenses, perceived usefulness and perceived ease of use from TAM, and role-specific informational need from the Agent and Communication Models, complementing the human-centred XAI evaluation framework of Kim et al. (2024), which identifies these same constructs as central to how AI explanations are ultimately received once a direct practitioner evaluation is conducted.

The research is additionally informed by the design science paradigm articulated by Hevner et al. (2004) and later refined into the Design Science Research Methodology by Peffers et al. (2007). Both formulations argue that a research contribution centred on a purposeful artefact is legitimated by the rigour of the construction process and by the demonstrable utility of the artefact when evaluated against a well-defined problem. This position aligns closely with the intent of the present study, which contributes both a dashboard system and design guidelines derived from the evaluation. The iterative process proposed by Peffers et al. (2007) is adopted alongside CommonKADS. It comprises of problem identification, objective definition, design and development, demonstration, evaluation, and communication. CommonKADS guides the knowledge modelling activities within each phase, while DSRM provides the overall structure for conducting the research.

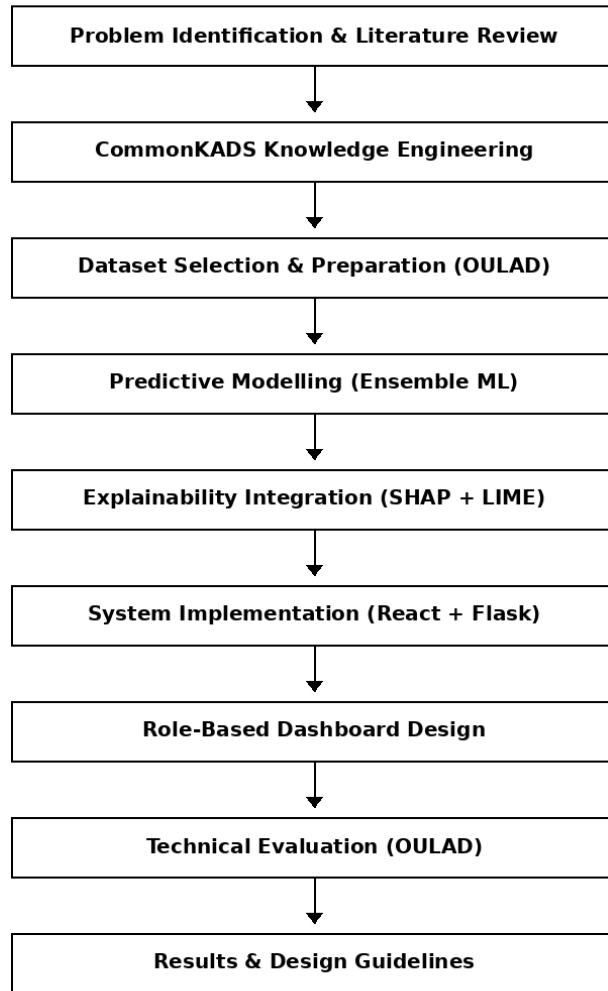
The relationship between these frameworks is therefore hierarchical. The Design Science Research Methodology (DSRM) is the overarching research paradigm and process; CommonKADS is the knowledge-engineering and design framework applied within that process; TAM is a theoretical lens for anticipated technology acceptance; and the Kaliisa et al. (2023) checklist provides dashboard design guidance. None of these frameworks other than DSRM is used as the overarching methodology, and TAM is not evaluated empirically.

The four iterative phases identified above are conducted with explicit feedback loops between them rather than as a strict linear progression. In this study, knowledge acquisition was conducted from published literature rather than through user-based elicitation, so findings from the literature review fed into the modelling phase where an agent characterisation was refined, and outputs of the technical evaluation could inform late adjustments to the design phase. This iterative structure is consistent with the recommendations of Verbert et al. (2013), who caution that dashboard design in learning analytics must accommodate revisions arising from user studies rather than freezing design decisions at an early stage; because no user studies were conducted here, such revisions are left to the future practitioner evaluation outlined in Chapter 7.

### **3.2 System Architecture**

The overall system architecture adopted in this study is illustrated in Figure 3.2. The process begins with problem identification and literature review, followed by CommonKADS knowledge engineering, dataset selection and preparation using OULAD, predictive modelling with ensemble machine learning, explainability integration using SHAP and LIME, system implementation using React and Flask, role-based dashboard design, technical evaluation of

predictive performance and explanation behaviour against OULAD, and concludes with the development of design guidelines.



*Figure 3.2. Research process flowchart, from problem identification through evaluation and derivation of design guidelines.*

The architecture of this system is driven by the need to address two established limitations in the literature. First, existing student at-risk prediction systems emphasise predictive accuracy while offering limited interpretability, thereby constraining their usefulness in practice (Ali et al., 2023). Second, most learning analytics dashboards adopt uniform interface designs despite clear evidence that different stakeholder roles require different forms of analytical insight (Kaliisa et al., 2023;

Kostopoulos et al., 2024). The architecture is therefore structured to separate data processing, prediction, explanation, and user interaction, ensuring that each component directly addresses these limitations.

The overall system is organized into five separate architectural layers, namely the Data Layer, the Prediction Engine, the Explainability Module, the Role Assignment and Personalisation Layer, and the Presentation Layer. Communication between layers is mediated by well-defined interfaces implemented as RESTful endpoints, which allows each layer to be revised, replaced, or scaled without cascading changes throughout the system. This architectural design is consistent with the guidance of Bass, Clements, and Kazman (2021) and aligns with the recommendations of Adadi & Berrada (2018), who argue that explainable AI systems benefit from architectures in which the explanation component is treated as a core part of the system rather than being added after model development.

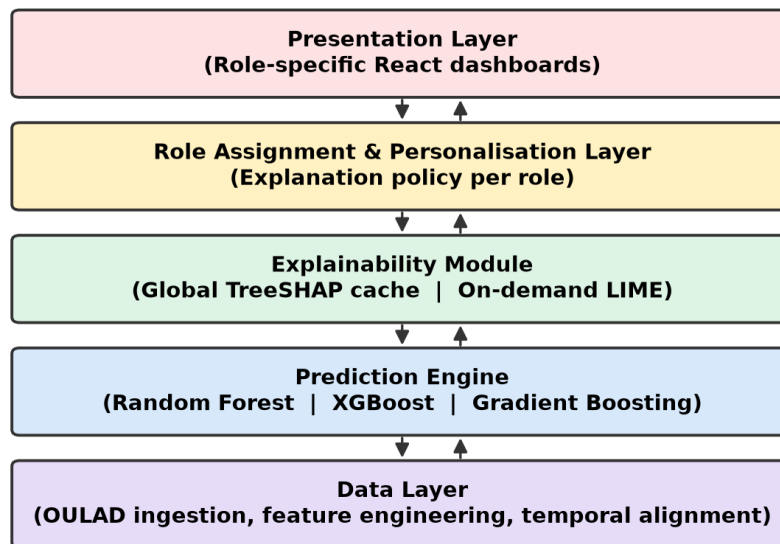


Figure 3.3. Layered system architecture separating data, prediction, explanation, personalization and presentation concerns.

A layered architectural approach is adopted because modular separation improves system clarity, scalability, and integration of explainability mechanisms in AI systems (Mohseni et al., 2021; Arrieta et al., 2020). This is particularly necessary in this study, as post-hoc explanation techniques such as SHAP and LIME operate independently of the predictive model and require flexible integration within the system pipeline (Ali et al., 2023; Salih et al., 2024).

The choice of a layered rather than pipeline-only architecture is also motivated by the need to accommodate concurrent use. Instructors, advisors, and administrators may access the dashboard simultaneously and request explanations at different granularities, and computationally expensive explanation queries could otherwise delay simpler cohort-level requests. Separating the explanation service from the presentation logic, and pre-computing the most expensive SHAP outputs, is intended to avoid this bottleneck; the specific implementation is described in Chapter 4.

The Data Layer is defined as a separate component to ensure consistency in feature extraction and preprocessing. This is justified by evidence that LMS behavioural logs, demographic variables, and assessment data are the most reliable predictors of student risk (Guevara-Reyes et al., 2025; Jin et al., 2024; Kuzilek et al., 2017; Wang, 2025). Isolating this layer supports reproducibility and reduces variability in feature engineering, which has been identified as a source of inconsistency in prior studies (Jin et al., 2024).

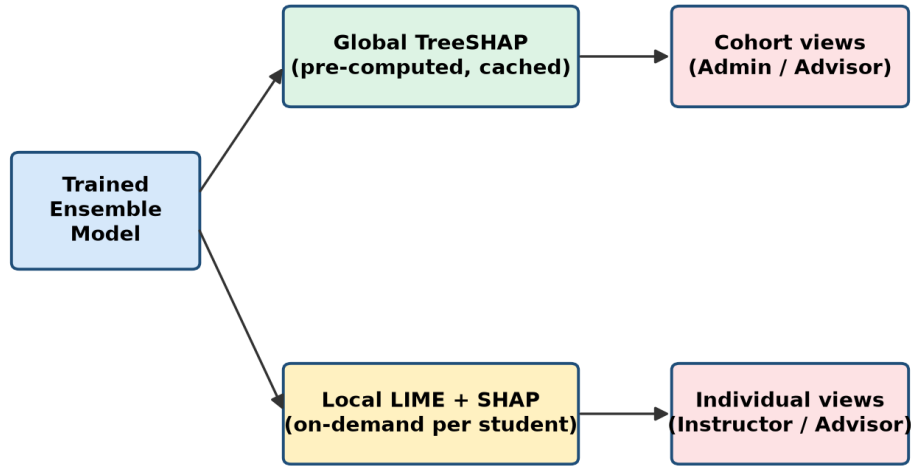
The Data Layer is additionally responsible for temporal alignment of interaction records with assessment deadlines, a step that is essential for the construction of time-window features. Following the temporal feature construction strategies documented by Hlostá, Zdrahal, and Zendulka (2017), interaction counts are aggregated over rolling windows anchored to submission dates rather than over calendar weeks, because deadline-anchored windows have been shown to yield stronger early-warning signals than calendar-anchored windows on OULAD. This treatment also allows the same feature pipeline to be reused for both training and inference without leakage of future information into historical windows, a concern emphasised by Krstajic, Buturovic, Leahy, and Thomas (2014) in their analysis of cross-validation pitfalls in predictive modelling.

The Prediction Engine is implemented as an independent layer to support the use of ensemble learning methods. Ensemble models have consistently demonstrated superior predictive performance in at-risk classification tasks (Rebelo Marcolino et al., 2025; Wang, 2025). However, their increased complexity reduces interpretability, thereby necessitating the inclusion of a dedicated explanation component (Guevara-Reyes et al., 2025).

The Explainability Module addresses this requirement by converting model outputs into interpretable representations. The combined use of SHAP and LIME is based on prior studies reporting complementary strengths, with SHAP commonly used for global attribution and LIME for local explanations of individual predictions (Gunasekara & Saarela, 2025; Nnadi et al., 2024; Salih et al., 2024). This ensures that both cohort-level and instance-level interpretation needs are adequately addressed.

The Explainability Module is implemented with two internal sub-services. The first, a global explanation service, is responsible for computing and caching TreeSHAP values (Lundberg et al., 2020) across the training population; these values underpin the cohort-level visualisations used by administrators and, at a coarser grain, by advisors. The second, a local explanation service, is responsible for generating on-demand LIME approximations for individual predictions (Ribeiro, Singh, & Guestrin, 2016) and for producing a corresponding local SHAP explanation using the tree-path-dependent estimator, so that the two techniques can be compared side by side within the interface. Storing global SHAP outputs while computing LIME on demand reflects the difference in computing time between the two methods reported by Salih et al. (2024), in which SHAP has a higher computing time than LIME. Global SHAP outputs can be cached and reused, whereas LIME, which relies on perturbation and repeated model evaluations, must be regenerated for each new instance.

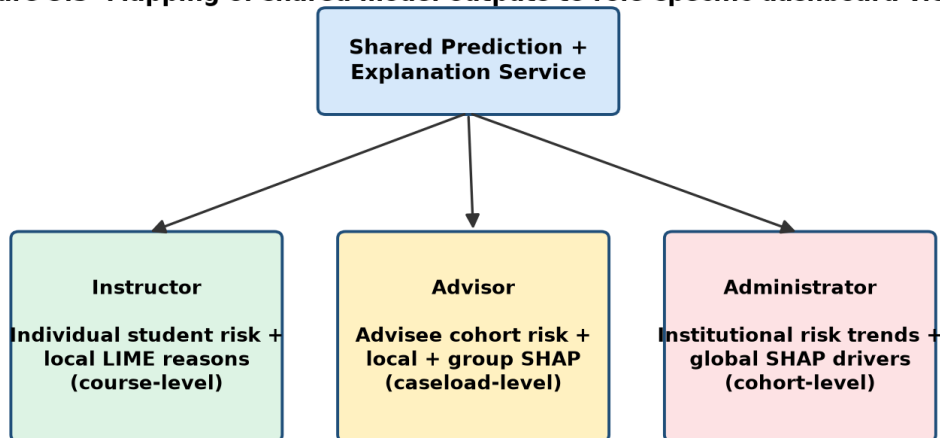
**Figure 3.4 Dual-service explainability pipeline (SHAP for global, LIME for local)**



*Figure 3.4. Dual-service explainability pipeline: global TreeSHAP is pre-computed and cached, while LIME (and local SHAP) are generated on demand.*

The Role-Based Interface Layer is introduced to resolve the limitation of role-agnostic dashboard design. Prior studies show that explanation effectiveness depends on alignment with user roles and decision contexts (Gutiérrez et al., 2020; Haque et al., 2023; Kostopoulos et al., 2024). Accordingly, the system maps shared explanation outputs to role-specific visualisations for instructors, advisors, and administrators. This allows a single predictive model to support multiple stakeholders without redundancy, and is intended to support each role's decision-making; its effect on usability was not measured in this study.

**Figure 3.5 Mapping of shared model outputs to role-specific dashboard views**



*Figure 3.5. Mapping of shared prediction and explanation outputs to instructor, advisor and administrator dashboard views.*

The mapping between shared explanation outputs and role-specific visualisations is implemented as a configurable presentation policy rather than as hard-coded logic. This design decision follows the recommendation of Sokol and Flach (2020) that explanation systems adopt a modular fact-sheet based approach in which the choice of explanation type, granularity, and format is decoupled from the underlying model. Each stakeholder role in the present system is associated with a policy specifying the explanation granularity (cohort, group, or individual), the visualisation modality (aggregated bar chart, per-student breakdown, or narrative summary), and the level of technical detail exposed. This makes it possible to introduce additional roles in future work without altering the prediction or explanation services.

The Presentation Layer is treated as a distinct concern rather than as a thin rendering shell over the Role Assignment policy. Its responsibilities include the composition of visual components, the management of interaction state, and the translation of explanation artefacts into forms that respect the cognitive constraints of the intended audience. Hernandez-Leo et al. (2019) emphasise that the visualisation layer of a learning analytics system is where design decisions become directly observable to end users, and that weaknesses at this layer often mask the value of otherwise sound analytical pipelines. Accordingly, the Presentation Layer in this study is developed as a component library organised around three primitives, namely the risk indicator, the feature attribution panel, and the cohort comparison view, each of which is parameterised by the presentation policy handed down from the Role Assignment layer.

A second reason for isolating the Presentation Layer is the need to accommodate accessibility and progressive disclosure. Miteva and Stefanova (2022), in their model of a learning analytics visualisation system, note that dashboards intended for heterogeneous audiences benefit from layered disclosure in which summary indicators are shown first and technical detail is revealed on demand. The Presentation Layer implements this pattern by rendering compact risk summaries as the default view for every role and by exposing SHAP and LIME visualisations only when the user requests a deeper explanation. This behaviour is delegated entirely to the interface tier so that the Explainability Module remains agnostic to the visualisation choices made downstream.

The interaction among these layers follows a structured data flow in which LMS data is processed, transformed into predictions, interpreted through post-hoc explanations, and delivered through role-specific visualisation logic. This design ensures consistency, flexibility, and alignment with stakeholder needs. The architecture is therefore justified as a knowledge engineering solution that integrates modular design, explainability, and role-awareness to address the limitations of existing systems. Implementation details of each layer, including feature dictionaries, model manifests, and caching, are reported in Chapter 4 rather than here.

A supporting architectural diagram is produced as part of the CommonKADS Design Model to make the described data flow explicit. In the diagram, the Data Layer feeds pre-processed feature vectors into the Prediction Engine, which returns risk probabilities to both the Explainability Module and the Role Assignment Layer. The Explainability Module returns cached global SHAP artefacts and freshly generated local explanations to the Presentation Layer, which composes them into role-appropriate views. Authentication and authorisation are enforced at the Presentation

Layer, so that a practitioner is only ever exposed to the interface and explanation granularity associated with their role. This end-to-end trace is intended to satisfy the reproducibility criterion proposed by Gundersen and Kjensmo (2018), who argue that machine learning research artefacts should permit an independent reviewer to reconstruct the pipeline from raw data to end-user output.

### 3.3 Dataset

The Open University Learning Analytics Dataset (OULAD) is adopted as the primary dataset for model development (Kuzilek et al., 2017). This selection is justified by its extensive use in the literature, which enables direct comparison with prior studies and strengthens the validity of the results (Gunasekara & Saarela, 2025; Jin et al., 2024; Wang, 2025).

OULAD has, in the years since its release, become one of the most widely used benchmarks for at-risk prediction research and has therefore accumulated an unusually rich body of comparative evidence. Hlostá, Zdrahal, and Zendulka (2017) provide one of the earliest baseline analyses of the dataset and document the class distributions, the temporal structure of the interaction logs, and the operational meaning of the pass, fail, and withdrawn labels. Subsequent contributions from Waheed et al. (2020) and from Adnan et al. (2021) have extended this baseline with deep learning treatments and time-series formulations, respectively, and their published performance figures provide credible reference points against which the models developed in this study can be benchmarked. The decision to adopt OULAD is therefore also a decision to situate the present study within a live comparative research programme.

OULAD contains anonymised demographic data and aggregated Virtual Learning Environment interaction records for 32,593 student-module-presentation observations across 22 module presentations (Kuzilek et al., 2017). These data types correspond directly to the feature categories most frequently identified as predictive of student at-risk status (Jin et al., 2024; Kuzilek et al., 2017). The dataset is therefore appropriate for the binary classification task adopted in this study.

The dataset is composed of seven relational tables covering student registrations, demographic attributes, module presentations, assessments, assessment results, VLE resources, and VLE interaction logs, and these tables are joined on student and module identifiers to produce the analytical dataset used for modelling. Waheed et al. (2020) report that the interaction table alone contains over ten million records, providing the volume necessary for stable estimation of high-capacity ensemble models. The demographic attributes include age band, gender, region, highest prior education, index of multiple deprivation band, disability status, and number of previous attempts, all of which have been shown in prior work to carry non-trivial predictive signal (Adnan et al., 2021; Kuzilek et al., 2017). Because several of these attributes are also protected characteristics under educational equity frameworks, their inclusion is treated as a design decision with fairness implications, discussed further in Section 3.9.

The use of OULAD is further justified by its scale, completeness, and ethical suitability. Its anonymised structure eliminates concerns relating to personally identifiable information, while its size supports the development of robust and generalisable models. The dataset is therefore selected on the basis of relevance, comparability, and methodological consistency with existing research.

Data preparation proceeded through five stages. The first stage performs schema validation and referential integrity checks across the seven source tables. The second stage constructs the

analytical target variable by collapsing the four-class final result field into a binary at-risk indicator, where Withdrawn and Fail are mapped to the positive class and Pass and Distinction to the negative class, following the convention adopted by Jin et al. (2024) in their systematic review of OULAD-based studies. The third stage generates cumulative and rolling-window interaction features anchored on assessment deadlines, following the temporal aggregation strategy described by Hlosta et al. (2017). The fourth stage encodes the categorical attributes, including module, gender, region, highest education, index of multiple deprivation band, age band, and disability, using one-hot representations (Chapter 5, Section 5.2). The fifth stage records the course presentation (code\_presentation) of every observation so that cross-validation folds can be grouped by presentation, mitigating the risk of optimistic estimates arising from ignoring the panel structure of the data, a concern raised by Rebelo Marcolino et al. (2025) in their optimisation study of Moodle log data.

Missing values are handled explicitly rather than imputed globally. Interaction counts for weeks in which a student is not yet registered are treated as structural zeros, whereas missing demographic fields are retained through a dedicated indicator variable so that missingness itself is available as a signal to the model. This treatment is consistent with the guidance of Van Buuren (2018), whose analysis of missing data mechanisms demonstrates that indicator encoding is preferable to blind imputation when missingness may be informative, a condition that is plausible for administrative educational data.

### **3.4 Predictive Modelling**

The predictive modelling task is formulated as a binary classification problem, where students are classified as at-risk if they withdraw or fail their module. This formulation is consistent with established approaches in the literature and supports practical intervention scenarios (Jin et al., 2024).

The binary formulation is preferred over multi-class or regression alternatives for two principled reasons. First, the immediate screening decision supported by the classifier is binary: whether a student should be flagged for further consideration. Albreiki et al. (2021) report that the majority of OULAD-based studies reviewed in their systematic analysis adopt a comparable binary framing. The intervention that may follow a flag is not binary, since an instructor or advisor might contact the student, monitor, refer, provide resources, adjust teaching, escalate, or do nothing, and that decision remains with the practitioner. Second, a binary target avoids the label ambiguity that arises when Pass and Distinction are treated as distinct classes, since prior work has shown that the boundary between these two categories is more sensitive to marking conventions across module presentations than to underlying behavioural differences (Kuzilek et al., 2017).

Feature engineering is guided by prior studies identifying submission timeliness, VLE interaction frequency, assessment performance trends, and demographic indicators as key predictors of student outcomes (Guevara-Reyes et al., 2025; Wang, 2025). The selection of these features ensures alignment with existing empirical findings and supports interpretability.

The engineered feature set is organised into three thematic families to preserve interpretability at the presentation layer. The engagement family captures aggregate and time-windowed counts of clicks by VLE activity type, following the taxonomy of resource categories documented in the OULAD schema (Kuzilek et al., 2017). The assessment family captures scores achieved on tutor-marked and computer-marked assessments, together with submission latency computed as the

interval between assessment release and submission, following the timeliness features shown by Hussain, Zhu, Zhang, and Abidi (2018) to be strongly associated with subsequent module outcomes. The demographic family captures the encoded demographic attributes described in Section 3.3. Each engineered feature is documented with a human-readable label and a short explanatory phrase in a feature dictionary, so that the presentation layer can render meaningful labels rather than variable names when surfacing SHAP and LIME attributions.

Three ensemble models are compared: Random Forest, XGBoost, and Gradient Boosting Classifier. The use of ensemble methods is justified by their demonstrated superiority in predictive performance across multiple studies (Rebelo Marcolino et al., 2025; Wang, 2025). F1-score is used for comparative reporting, because it provides a balanced measure of performance in the presence of class imbalance (Jin et al., 2024). The model deployed in the prototype, however, is selected with recall as the primary criterion and F1-score as a secondary metric, because recall is prioritised to reduce false negatives in the intended screening context: a student who is at risk but not flagged receives no support from the dashboard, whereas a student who is flagged unnecessarily is reviewed by a practitioner and can be set aside.

The three candidate models are chosen because they represent distinct points along the bias-variance and interpretability spectra recognised in the machine learning literature. Random Forest, as formulated by Breiman (2001), reduces variance by averaging deep decorrelated trees and remains widely regarded as a strong general-purpose baseline in tabular classification. XGBoost, introduced by Chen and Guestrin (2016), applies regularised gradient boosting on decision trees and has become the reference implementation of boosting for structured data, particularly in the presence of heterogeneous feature types. Gradient Boosting Classifier, as implemented in scikit-learn following the original formulation of Friedman (2001), provides a more conservative boosting baseline against which the XGBoost configuration can be compared. This three-way comparison follows the pattern established by Rebelo Marcolino et al. (2025) and by Guevara-Reyes et al. (2025), both of whom argue that credible ensemble evaluations require at least one bagging model and at least two boosting models to isolate the effects of the ensembling strategy from those of the boosting refinements.

The final evaluation used four-fold GroupKFold cross-validation grouped by course presentation (`code_presentation`), so that observations from the same presentation never appear in both the training and the validation partition of a fold, avoiding the leakage pattern criticised by Krstajic et al. (2014). Because OULAD contains four course presentations, this produces four folds, and the folds are not stratified, since stratification is not compatible with grouping on a variable that takes only four values. The XGBoost model deployed in the prototype was evaluated using a fixed configuration (300 estimators, maximum depth 5, learning rate 0.05, subsample 0.8, and column subsample 0.8) without additional hyperparameter optimisation or resampling, and Random Forest and Gradient Boosting were evaluated under the identical preprocessing pipeline and protocol. Accuracy, precision, recall, and F1-score, computed over out-of-fold predictions at a decision threshold of 0.5, are the primary metrics, and ROC-AUC is also recorded by the training code.

### **3.4.1 Planned but Not Implemented Methodological Extensions**

The original design of the study specified several refinements to the modelling and evaluation pipeline that were not implemented in the final evaluation. They are recorded here so that this

chapter describes the final method accurately while remaining transparent about the original plan. The literature motivating each refinement is reviewed in Chapter 2, Section 2.2.

**Class imbalance handling.** SMOTE, including borderline and adaptive variants applied within the training folds, and class-weighted loss functions were specified as ways of addressing class imbalance. Neither was applied, because the final evaluation used the observed class distribution, which is only mildly imbalanced (Chapter 5, Section 5.2).

**Hyperparameter optimisation.** Bayesian optimisation with the Tree-structured Parzen Estimator was specified for hyperparameter search. The prototype model instead used the fixed configuration described above.

**Probability calibration.** A comparison of isotonic regression and Platt scaling, assessed through reliability diagrams and expected calibration error, was specified because the dashboard displays risk probabilities. Calibration was not assessed in the final evaluation.

**Trajectory features.** A trajectory feature family capturing period-over-period change in engagement and performance was specified. The implemented feature set does not include it.

**Secondary metrics.** Area under the precision-recall curve, balanced accuracy, and the Matthews correlation coefficient were specified as secondary metrics and were not computed in the final evaluation.

**Cross-validation protocol.** The original design described a stratified five-fold protocol. It was superseded by the four-fold GroupKFold protocol described above, which is the method used in the final evaluation.

**Explanation stability and fidelity, and subgroup fairness.** The stability and fidelity checks for explanations (Section 3.5) and the subgroup performance and fairness assessment (Section 3.9) were proposed but were not completed within the final evaluation.

### **3.5 Explainability Integration**

The integration of explainable AI techniques is necessary to address the limited interpretability of predictive models in educational contexts. Prior research indicates that risk predictions without explanation provide limited support for decision-making (Ahmed et al., 2025; Ali et al., 2023).

The design of the explanation subsystem is grounded in the distinction drawn by Doshi-Velez and Kim (2017) between transparency-oriented and post-hoc interpretability. Transparency-oriented approaches require the model itself to be intrinsically understandable, which typically constrains model choice to linear models or shallow trees at a substantial cost in accuracy. Post-hoc approaches, in contrast, decouple the interpretability question from the modelling question and allow high-capacity models to be deployed alongside a dedicated explanation layer. Adadi & Berrada (2018) argue that in domains such as education, where the accuracy of the underlying model has direct consequences for the students subject to intervention, the post-hoc route is generally preferable provided that the resulting explanations are subjected to explicit fidelity and stability testing. This position underpins the decision in the present study to retain ensemble models and to invest architectural effort in the explanation layer.

SHAP is used for global model interpretation, while LIME is used for local explanations. This combination is justified by studies reporting that SHAP provides consistent feature attribution,

while LIME offers flexible and interpretable explanations for individual instances (Gunasekara & Saarela, 2025; Nnadi et al., 2024; Salih et al., 2024). The use of both techniques ensures that explanations are available at different levels of analysis.

SHAP is grounded in the game-theoretic framework of Shapley values and is implemented in this study using the TreeSHAP algorithm proposed by Lundberg et al. (2020), which computes exact Shapley values for tree ensembles in polynomial time and thereby renders the method tractable for the OULAD-scale feature space. LIME, introduced by Ribeiro et al. (2016), fits a sparse linear surrogate to the behaviour of the model in the local neighbourhood of a given prediction and returns the surrogate coefficients as feature-level explanations. The two methods are complementary rather than redundant: SHAP is formulated to satisfy properties such as local accuracy and consistency (Lundberg & Lee, 2017), although these properties depend on the exact SHAP formulation and its assumptions, including the assumption of feature independence discussed by Salih et al. (2024), whereas LIME provides a more intuitive linear approximation that many practitioners find easier to reason about in isolation. Salih et al. (2024) argue explicitly for the joint use of both techniques as a form of methodological triangulation, and this argument is adopted here.

The explanation subsystem was also designed to incorporate safeguards against the known failure modes of post-hoc methods. Slack et al. (2020) demonstrate that SHAP and LIME can be manipulated by adversarial classifiers to conceal discriminatory behaviour, and although the present study does not confront an adversarial setting, the same underlying sensitivity to perturbation choices motivates the reporting of stability metrics. The methodology proposed quantifying stability using the Jaccard similarity of the top-k feature sets returned by repeated LIME runs on the same instance with different random seeds, following the protocol of Visani, Bagli, Chesani, Poluzzi, and Capuzzo (2022), and quantifying fidelity using the local squared error between the surrogate and the black-box model within a locality kernel of controlled bandwidth, with instances falling below preregistered thresholds flagged in the interface. These stability and fidelity checks were proposed safeguards rather than completed evaluation procedures: they were not completed in the final evaluation, and the explanation evaluation reported in Chapter 5 is limited to cross-method agreement between SHAP and LIME.

SHAP values are pre-computed during model training and stored to support efficient retrieval. LIME explanations are generated on demand for individual predictions, enabling real-time interaction within the dashboard. Explanation quality was evaluated through cross-method agreement between SHAP and LIME; the fidelity and stability metrics described above were specified but not completed.

The presentation of explanations is shaped by the pragmatics of explanation set out by Miller (2019), whose synthesis of social science research on how humans give and receive explanations identifies three properties that any effective explanation must satisfy. Explanations should be contrastive, in that they should clarify why a particular outcome was predicted rather than a plausible alternative. Explanations should be selective, in that they should surface the small number of factors that most strongly account for the outcome rather than the full attribution vector. Explanations should be social, in that they should be phrased in terms accessible to the intended recipient. The presentation layer draws on the selective and social properties by rendering per-instance explanations as signed feature attributions limited to a small number of leading features, with human-readable feature labels, at a level of aggregation suited to the associated stakeholder

role. The explanations implemented in the prototype are feature attributions rather than contrastive or counterfactual explanations, and natural-language comparative statements were not implemented, so the contrastive property remains a design aspiration rather than a feature of the current system. This aligns the explanation experience with the empirical evidence of Poursabzi-Sangdeh et al. (2021), who show that human comprehension of model outputs degrades sharply when unfiltered attribution vectors are displayed in full.

### 3.6 Dashboard Design

The dashboard design process was structured by the four-stage checklist of Kaliisa et al. (2023), which guides developers from initial purpose specification and stakeholder engagement through iterative prototyping and rigorous evaluation planning, a sequence directly applicable to the three-role design problem of this study. This checklist was selected because it was developed explicitly for learning analytics contexts and has been applied in subsequent LA dashboard research (Cabral et al., 2025). Three role-specific interface variants were designed and implemented as a web application using React.js with a Python Flask backend. Interface decisions were guided by the heuristic criteria set out below and checked against the CommonKADS Agent Model, consistent with the secondary-data scope of this study (Section 3.8); no documented iterative heuristic review, with recorded iterations, reviewers, or severity ratings, was conducted. The design process was structured by the CommonKADS Design Model, so that each interface variant maps directly to the knowledge delivery requirements identified for its corresponding agent role.

Each of the three interface variants is specified against a distinct information brief derived from the CommonKADS Agent Model. The instructor interface prioritises class-level engagement over the current teaching period and per-student explanations of risk drivers at the granularity of assessments and weekly activity, following the information needs identified by Molenaar and Knoop-van Campen (2019) in their study of teacher-facing dashboards. The advisor interface prioritises longitudinal views across multiple modules for a caseload of students, foregrounding demographic context in a form suitable for one-to-one consultations, in line with the design pattern used by Gutiérrez et al. (2020) in the LADA advising system. The administrator interface prioritises cohort-level aggregations, module-over-module comparisons, and equity-relevant breakdowns, following the institutional decision-support patterns documented by Tsai and Gašević (2017). Grouping the interfaces around these role-specific briefs prevents the dashboard from defaulting to a single view that is optimal for none of the intended users. Table 3.2 summarises the resulting role-based design.

*Table 3.2. Role-based dashboard design: decisions, information needs, components, and explanation granularity.*

| Role       | Decision                                                  | Information needed                                                                                                                          | Dashboard component                                                | Explanation granularity |
|------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------|
| Instructor | Whether to act within a course that is currently running. | Class-level engagement over the current teaching period and per-student risk drivers at the granularity of assessments and weekly activity. | Per-student reasoning trail with SHAP and LIME shown side by side. | Individual              |
| Advisor    | How to conduct a                                          | Views across multiple                                                                                                                       | Cohort-averaged                                                    | Caseload                |

| Role          | Decision                                                          | Information needed                                                                                   | Dashboard component                                                             | Explanation granularity |
|---------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------|
|               | conversation with a particular student and prioritise a caseload. | modules for a caseload of students, with demographic context, suitable for one-to-one consultations. | reasoning trail for students at or above the watch threshold.                   |                         |
| Administrator | Where to direct institutional attention and resource.             | Cohort-level aggregations and module-over-module comparisons.                                        | Institution-wide model performance summary and full-population reasoning trail. | Cohort                  |

Interface conventions follow the established human-computer interaction heuristics of Nielsen (1994), particularly the principles of visibility of system status, match between system and the real world, recognition rather than recall, and consistency and standards. Colour usage is designed to remain informative for practitioners with common forms of colour vision deficiency by relying on colour-blind safe palettes and by encoding critical categorical distinctions redundantly through shape and label, following the accessible visualisation guidance synthesised by Munzner (2014). Typography, spacing, and interactive controls were designed with reference to the WCAG 2.1 conformance criteria of the World Wide Web Consortium, which is consistent with the accessibility-first orientation increasingly expected of higher education software.

No documented iterative heuristic review, with recorded iterations, reviewers, or severity ratings, was conducted in this study. The Nielsen (1994) criteria were instead applied as design guidance during implementation. A formal heuristic evaluation, together with a formative think-aloud study with representative practitioners of the kind recommended by Verbert et al. (2020), is identified in Chapter 7 as future work once the dashboard is evaluated with an institutional partner.

A system architecture diagram was produced during the design phase to illustrate the data flow from raw LMS input through the prediction engine, explainability module, and role-based interface layer. This diagram serves as the principal artefact of the CommonKADS Design Model (Figure 3.3 and Figure 4.1).

The design plan also specified three supporting artefacts: a use-case diagram enumerating the primary interactions available to each role, produced in the Unified Modelling Language following the notation of Booch, Rumbaugh, and Jacobson (2005); an entity-relationship diagram capturing the persistent data model that underlies the Data Layer; and a sequence diagram illustrating the runtime interaction between the Presentation Layer, the Prediction Engine, and the Explainability Module for a representative on-demand explanation request. These supporting diagrams are not reproduced in this thesis, so the architecture diagrams in Figures 3.3 and 4.1 are the principal design documentation presented, in line with the design-artefact expectations of Hevner et al. (2004) for a design science contribution.

### 3.7 Implementation Plan

The system was implemented using Python for all data processing, model training, and explainability computation, with scikit-learn, XGBoost, SHAP, and LIME as the primary libraries.

The selection of these libraries reflects considered technical judgement rather than convenience. Scikit-learn is adopted for its consistent estimator interface and its well-tested implementations of

cross-validation, preprocessing, and model evaluation utilities, as documented by Pedregosa et al. (2011). XGBoost is adopted for the reasons articulated by Chen and Guestrin (2016), including its regularised objective, sparsity-aware split finding, and out-of-core training capability. SHAP is adopted through its official reference implementation to guarantee alignment with the theoretical guarantees of TreeSHAP established by Lundberg et al. (2020). LIME is adopted through the reference implementation of Ribeiro et al. (2016). The library versions used in the evaluation are reported in Chapter 5, Section 5.7 and in Appendix A, in line with the reproducibility recommendations of Gundersen and Kjensmo (2018).

The web application was built using React.js for the frontend and Python Flask as the API layer, a combination selected because it allows for rapid prototyping of interactive visualisation components alongside a well-tested machine learning pipeline. The code was organised into separate modules for the prediction, explainability, and visualisation components (Chapter 4, Section 4.3). The three role-specific interfaces were implemented as web pages within a single application, with role-based authentication determining which view a practitioner accesses upon login.

The frontend is organised as a component-based application using the composition principles articulated in the React documentation of Meta Open Source (2024) and the maintainability guidance of Martin (2017). Visualisation components are implemented using the D3-based ecosystem, drawing on the grammar of interactive graphics described by Satyanarayan, Moritz, Wongsuphasawat, and Heer (2017), which supports the declarative construction of coordinated views necessary for exploratory dashboards. The Flask backend exposes REST endpoints for prediction, global explanation retrieval, and local explanation generation, and authentication is implemented using JSON Web Tokens following the specification of Jones, Bradley, and Sakimura (2015). Role-based access control is enforced at the API layer as well as at the interface layer, consistent with the defence-in-depth principle articulated by Saltzer and Schroeder (1975), so that the interface layer is not the sole enforcement point for role separation.

Software quality was intended to be safeguarded through a layered testing strategy, with unit tests for the preprocessing pipeline, model wrappers, and explanation utilities, integration tests for the API contracts between the frontend and the backend, and regression tests for the explanation subsystem to detect unexpected shifts in SHAP and LIME outputs following library upgrades, which matters because Slack et al. (2020) demonstrate that explanation methods are sensitive to implementation details that may change silently across versions. Continuous integration, following the practice recommended by Humble and Farley (2010) for reliable software delivery, was also planned. This testing and continuous integration pipeline is not demonstrated in this thesis, and it is therefore described as planned practice rather than as completed infrastructure. Similarly, the codebase, the trained model artefacts, and the derived experimental data are intended for deposit in a public repository in accordance with the FAIR data principles articulated by Wilkinson et al. (2016), but this deposit is not reported here as completed.

### **3.8 Evaluation Plan**

Consistent with the scope of this study (Section 1.6), evaluation is conducted as a technical evaluation of the predictive and explanatory components against the OULAD dataset, rather than as a primary study involving instructor, advisor, or administrator participants. The evaluation plan below specifies what is measured, how, and against what standard of comparison.

Predictive performance is evaluated using GroupKFold cross-validation grouped by course presentation (code\_presentation), so that every prediction is produced by a model that has not seen the presentation it is evaluated on. Because OULAD contains four course presentations, this produces four folds. Metrics are computed over out-of-fold predictions across the full population: accuracy, precision, recall, and F1-score are reported as the four primary metrics, giving a small number of clearly interpretable, comparable results rather than a large battery of secondary statistics. The three ensemble models introduced in Section 3.4, Random Forest, XGBoost, and Gradient Boosting, are compared under an identical pipeline and protocol using F1-score for comparative reporting, and the model deployed in the prototype is selected on recall, with F1-score as a secondary metric, as set out in Section 3.4. Performance is additionally reported separately for each held-out course presentation, giving a direct empirical indication of how much variability to expect across cohorts. Because the features are computed over the full course presentation, this evaluation measures retrospective classification using the available OULAD feature set; it does not establish performance at a predefined early-warning point, which would require features to be restricted to the information available at each point in a presentation.

Explainability is evaluated in two complementary ways. Global feature importance is computed using TreeSHAP (Lundberg et al., 2020) once, for the final fitted model, across all 32,593 observations, ranking features by mean absolute SHAP value; this explains the final fitted model rather than providing a fold-specific explanation evaluation. Local, instance-level explanations are computed using LIME (Ribeiro et al., 2016) on a representative sample of instances. The two methods are compared directly: aggregate rankings are compared for overlap in their leading features, and instance-level overlap is quantified by comparing each sampled instance's top-ranked SHAP features against its top-ranked LIME features, giving a concrete measure of how consistently the two explanation methods agree at the level of an individual prediction. This comparison follows the recommendation of Lundberg and Lee (2017) that global and local attribution methods be validated against one another rather than relied upon in isolation.

The technical evaluation is completed by a verification step, in which the metrics and figures reported during system development (Chapter 4) are independently reproduced from the trained model artefact, and by a structured comparison with the methodology specified earlier in this chapter, so that any point where the implemented system extends or refines the originally specified approach is identified and reported rather than left undocumented.

This technical evaluation establishes whether the predictive and explanatory components function correctly, perform well, and behave consistently, which is the necessary foundation for the dashboard's practitioner-facing value. Direct evaluation of that practitioner-facing value, instructor, advisor, and administrator trust, understanding, and decision quality when using the dashboard, requires primary data collection from practitioners and is outside the scope of this thesis; it is identified in Chapter 7 as the recommended next phase of this research programme, building directly on the CommonKADS Agent Model, the TAM-based design lens (Section 3.1), and the technical foundation established here.

The findings of this evaluation are reported against a three-level evidence model that is used throughout the remainder of the thesis. Findings that were demonstrated are those measured directly in the technical evaluation, such as predictive performance and cross-method agreement. Findings supported by design are properties of the dashboard that are implemented and traceable to stated requirements and to the literature but were not tested with users. Findings not empirically

established are those that this study did not measure, including practitioner trust, usability, and decision quality, probability calibration, subgroup fairness, and early-warning performance.

### **3.9 Ethical Considerations**

Because this study relies exclusively on the secondary OULAD dataset (Section 3.3) and does not involve primary data collection from students, instructors, advisors, or administrators, the ethical considerations addressed here concern data handling and algorithmic fairness rather than human-subjects research procedures such as informed consent or participant recruitment. No personally identifiable student data is used at any stage, as OULAD is fully anonymised by design (Kuzilek et al., 2017), and all data processing is limited to the anonymised identifiers already present in the published dataset.

Algorithmic fairness is treated as an integral ethical consideration rather than as a separate concern. Because OULAD contains demographic attributes that overlap with protected characteristics, the evaluation plan proposed subgroup performance and fairness assessment, with model performance disaggregated by gender, age band, disability status, and index of multiple deprivation band, and fairness assessed using demographic parity difference and equal opportunity difference following the framework of Hardt, Price, and Srebro (2016). These analyses were not completed within the final evaluation. The plan specified that any subgroup performance disparities should be reported openly rather than concealed, in line with the transparency norm advocated by Mitchell et al. (2019) in their model card proposal, and it responds to the empirical evidence of Yu, Li, Fischer, Doroudi, and Xu (2020) that predictive models in higher education can exhibit systematically different error rates across student sub-groups even when aggregate accuracy is high. Completing this assessment is identified as a first-tier item of future work in Chapter 7.

The study also considers the ethical implications of deploying a role-aware XAI dashboard in an authentic institutional setting, even though such deployment, and the practitioner evaluation that would accompany it, is beyond the scope of the present thesis. Prinsloo and Slade (2017) argue that learning analytics systems should be evaluated against the principles of student agency, transparency, and the avoidance of harm. In line with this argument, the design of the dashboard deliberately avoids features that would automate high-stakes decisions about students, keeping the practitioner as the final decision-maker in any intervention. Slade and Prinsloo (2013) further note that learning analytics interventions carry a risk of paternalism if student voice is not represented in the design process; while student participants are outside the scope of the present study, this limitation is acknowledged here, and student consultation is recommended as a prerequisite for institutional deployment. A future practitioner evaluation of the kind outlined in Chapter 7 would additionally require institutional ethical approval and informed consent from participating instructors, advisors, and administrators before data collection could begin.

### **3.10 Chapter Summary**

This chapter has presented the methodology through which the study designs, implements, and technically evaluates a role-aware explainable AI dashboard for student at-risk prediction in Learning Management Systems, using the Open University Learning Analytics Dataset as its sole empirical basis. The research follows the Design Science Research Methodology of Peffers et al. (2007) as its overarching process, with the CommonKADS Knowledge Engineering methodology as its knowledge-engineering and design framework, the Technology Acceptance Model as a

design lens only, and the Kaliisa et al. (2023) checklist as dashboard design guidance. The system architecture is organised into loosely coupled data, prediction, explanation, personalisation, and presentation layers, each justified against the technical and practitioner-facing limitations identified in the literature. The Open University Learning Analytics Dataset provides the empirical foundation for model development, prepared through a documented five-stage pipeline that preserves temporal integrity and groups observations by course presentation. Random Forest, XGBoost, and Gradient Boosting classifiers are compared using four-fold GroupKFold cross-validation grouped by course presentation, with a fixed model configuration and no resampling; F1-score is used for comparative reporting, and recall is the primary criterion for selecting the model deployed in the prototype. Explainability is delivered through the combined use of TreeSHAP and LIME, with agreement between the two methods treated as the core explanation evaluation criterion; the stability and fidelity checks, calibration analysis, and subgroup fairness assessment originally specified were not completed and are recorded as planned extensions in Section 3.4.1. The dashboard is developed against the Kaliisa et al. (2023) checklist and rendered in three role-specific variants derived from the CommonKADS Agent Model, which was populated from published literature rather than from practitioners. Evaluation is technical and retrospective: it verifies predictive performance on features computed over the full course presentation, confirms explanation behaviour and cross-method agreement, and compares the implementation against the originally planned methodology, and its findings are reported using the three-level evidence model set out in Section 3.8. Ethical considerations, including data anonymisation, algorithmic fairness, and the implications of eventual institutional deployment, are addressed across the methodology, and the primary practitioner evaluation of trust, understanding, and decision quality that would require human-subjects procedures is identified as the recommended next phase of this research programme in Chapter 7. Overall, these design decisions provide a coherent and reproducible plan for producing an evidence-based answer to the research questions established in Chapter One.

## CHAPTER 4

### SYSTEM ARCHITECTURE AND IMPLEMENTATION

#### 4.1 Introduction

This chapter presents the technical architecture and implementation of the role-aware explainable at-risk prediction dashboard described conceptually in Chapter 3. Where Chapter 3 established the CommonKADS-derived design rationale (the Organisation, Task, Agent, Knowledge, Communication, and Design Models), this chapter documents how that design was realised as a working prototype: the specific frameworks and libraries used at each architectural layer, the model development environment, the Learning Management System (LMS) integrations built and tested for the prototype, and the containerised prototype deployment topology.

The system was implemented and validated end to end against the actual Open University Learning Analytics Dataset (OULAD) described in Chapter 3, rather than synthetic or placeholder data, so that the architectural decisions reported here reflect a system that has been tested against real data volumes (32,593 student-module-presentation observations) rather than a purely theoretical design.

This chapter corresponds directly to the second and third specific objectives from Chapter 1, Section 1.3, namely the development of an ensemble predictive model with integrated SHAP and LIME explanations and the implementation of a role-aware dashboard delivering differentiated explanatory views to instructors, academic advisors, and institutional administrators. Chapter 3 specified how these objectives were to be met at the level of knowledge engineering models and analytical procedure; the present chapter reports the artefact that resulted from following that specification. This distinction is important because the contribution claimed earlier (Chapter 1, Section 1.5) is an artefact contribution, and Schreiber et al. (2000) argue that a knowledge engineering contribution of this kind is supported by the traceability of the modelling process from conceptual model to operational system, not by the novelty of any individual component. Studer et al. (1998) similarly treat the working system as the operational endpoint of a staged knowledge engineering process that runs from analysis through conceptual modelling to implementation, and this chapter occupies that position within the study.

The chapter is also intended to establish the technical preconditions for the evaluation reported in the following chapter. Because the technical evaluation reported in Chapter 5 depends on the predictive and explanatory outputs the system actually produces when run against the OULAD dataset, the implementation choices documented here determine what can meaningfully be evaluated from the secondary data available to this study. Kim et al. (2024) note that human-centred XAI evaluations are frequently difficult to interpret because the systems under study are described only in general terms, leaving it unclear which interface property is responsible for a given behaviour. Each implementation decision reported in this chapter is stated in sufficient detail for the evaluation findings to be attributed to identifiable design choices, not to the system as an undifferentiated whole.

## 4.2 Architectural Overview

The system follows a five-layer architecture in line with the Design Model from Chapter 3, Section 3.1: a Data Layer responsible for feature engineering, a Prediction Engine responsible for risk classification, an Explainability Module responsible for generating SHAP and LIME explanations, an Interface/API layer enforcing authentication and role-based access control, and a Presentation Layer providing three role-differentiated dashboard views. Figure 4.1 shows the architecture, including the model development environment (Google Colab) and the two Learning Management System integration paths (Moodle and Canvas) evaluated during prototype testing.

The retention of a layered structure in the implemented system reflects the architectural reasoning behind Chapter 3, Section 3.2, not simply development convenience. Bass, Clements, and Kazman (2021) argue that layering is justified when the layers correspond to distinct rates of change, and this condition holds in the present system because the feature engineering logic, the trained model artefact, the explanation services, and the interface components are each expected to be revised on different schedules. Mohseni, Zarei, and Ragan (2021) make the related point for explainable systems specifically, observing that architectures which treat explanation as a separate architectural component rather than as an appendage to the model are substantially easier to evaluate against user-facing criteria.

**Figure 4.1: System Architecture**

*All services run locally under Docker Compose*

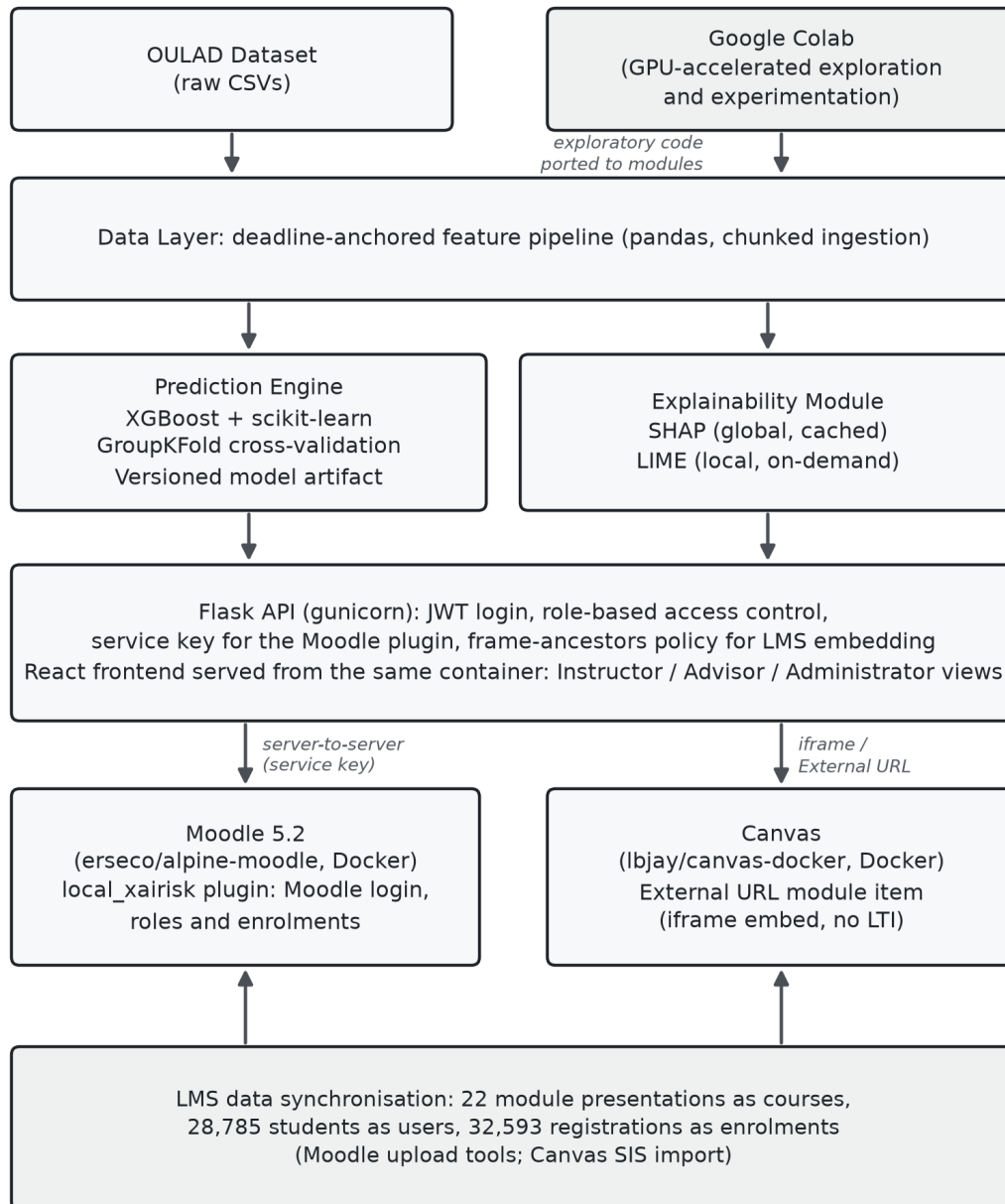


Figure 4.1. System architecture, showing the data, prediction, explainability, interface, and presentation layers, the model development environment, and the two LMS integration paths tested.

### 4.3 Model Development Environment

Before the prototype pipeline was built, model development and hyperparameter experimentation were conducted in Google Colab, which provided free access to GPU-accelerated compute for

iterating on feature engineering strategies and comparing candidate ensemble configurations without needing local hardware. Colab notebooks were used to explore the OULAD schema, prototype the deadline-anchored rolling-window feature strategy from Chapter 3, Section 3.4, and validate that the chosen feature set produced acceptable separation between at-risk and not-at-risk classes before the pipeline was ported into the prototype codebase.

This separation between an exploratory notebook environment and a prototype codebase follows standard applied machine learning practice: Colab was used for rapid, temporary experimentation, while the code that actually runs in the prototype was rewritten as versioned, testable Python modules (`data_layer`, `prediction_engine`, `explainability`) independent of the notebook environment, so that the prototype pipeline does not depend on Colab's session-based execution model or its GPU availability at inference time.

This arrangement also serves the reproducibility requirement from Chapter 3, Section 3.7. Gundersen and Kjensmo (2018) demonstrate that a substantial proportion of published machine learning work cannot be reconstructed by an independent reviewer because the experimental environment, the data preparation steps, and the final model are not documented as a single traceable workflow. Porting the exploratory work into versioned modules addresses this concern directly, since the prototype pipeline can be executed end to end from raw OULAD tables to a scored prediction without reference to any notebook state. Sculley et al. (2015) describe the accumulation of exactly this kind of untracked dependency as a principal source of technical debt in machine learning systems, and the separation maintained here is intended to reduce this type of dependency at the point where it would otherwise be easiest to introduce.



```

    "module_presentation_length", "final_result", "at_risk",
    return df[keep]

def build_assessment_features(assessments, student_assessment):
    merged = student_assessment.merge(assessments, on="id_assessment", how="left", suffixes=("", "_assessment"))
    merged["date_submitted"] = pd.to_numeric(merged["date_submitted"], errors="coerce")
    merged["date"] = pd.to_numeric(merged["date"], errors="coerce")
    merged["score"] = pd.to_numeric(merged["score"], errors="coerce")
    merged["days_late"] = merged["date_submitted"] - merged["date"]
    grp = merged.groupby(["code_module", "code_presentation", "id_student"])
    feats = grp.agg(
        n_assessments_submitted=("id_assessment", "count"),
        mean_score=("score", "mean"),
        min_score=("score", "min"),
        mean_days_late=("days_late", "mean"),
        n_late_submissions=("days_late", lambda s: (s > 0).sum()),
        n_banked=("is_banked", "sum"),
    ).reset_index()
    return feats, merged

def aggregate_daily_clicks(chunksize=2_000_000):
    partial_sums = []
    reader = pd.read_csv(
        RAW / "studentVle.csv", chunksize=chunksize,
        dtype={"code_module": "category", "code_presentation": "category"},
    )
    for chunk in reader:
        chunk["date"] = pd.to_numeric(chunk["date"], errors="coerce")
        daily = chunk.groupby(
            ["code_module", "code_presentation", "id_student", "date"], observed=True
        )["sum_click"].sum().reset_index()

```

Figure 4.2. Screenshot from the prototype development: Google Colab pre-training of the OULAD dataset.

## 4.4 Technology Stack

Table 4.1 summarises the frameworks and libraries used at each architectural layer, together with their specific role in the implemented system.

The libraries listed were selected against the criteria in Chapter 3, Section 3.7, not simply on availability. Scikit-learn is used for cross-validation, preprocessing, and evaluation because of the consistency of its estimator interface and the maturity of its model selection utilities (Pedregosa et al., 2011). XGBoost is used for the prototype classifier for the reasons given by Chen and Guestrin (2016), particularly its regularised objective and its handling of heterogeneous tabular features, which are characteristic of the joined OULAD tables. The SHAP reference implementation is used so that the attributions displayed in the interface retain the properties established for TreeSHAP by Lundberg et al. (2020), and the LIME reference implementation of Ribeiro, Singh, and Guestrin (2016) is used for the same reason on the local explanation path. Retaining the reference implementations rather than reimplementing either method is important because Slack et al. (2020) show that post-hoc explanation outputs are sensitive to implementation details that may otherwise change without notice.

Table 4.1. Frameworks and libraries used by architectural layers.

|            |        |                                                |
|------------|--------|------------------------------------------------|
| Data Layer | pandas | Chunked ingestion of the ~450MB studentVle.csv |
|------------|--------|------------------------------------------------|

|                          |                                                |                                                                                                                                                         |
|--------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          |                                                | clickstream file; deadline-anchored feature aggregation; versioned CSV output                                                                           |
| Prediction Engine        | scikit-learn, XGBoost                          | GroupKFold cross-validation grouped by course presentation; gradient-boosted ensemble classifier; preprocessing pipeline (imputation, one-hot encoding) |
| Model persistence        | joblib                                         | Serialises the fitted pipeline (preprocessing + model) as a single versioned artifact alongside a JSON manifest                                         |
| Explainability           | SHAP, LIME                                     | TreeSHAP for cached global/cohort-level explanations; LIME for on-demand local explanations; both computed against the same fitted model                |
| API layer                | Flask, PyJWT, Werkzeug                         | REST API; JSON Web Token issuance and verification; password hashing for the demo account login flow                                                    |
| Web server               | gunicorn                                       | Serves the Flask API and the built React application from a single container                                                                            |
| LMS plugin (Moodle)      | PHP, Moodle plugin API                         | local_xairisk plugin adding the three role views to Moodle; uses Moodle login, capabilities, and enrolments, and calls the API server to server         |
| Frontend                 | React, Vite                                    | Component-based single-page application implementing the three role-differentiated views                                                                |
| LMS embedding (Canvas)   | Canvas REST API                                | External URL module item in every course, embedding the dashboard in an iframe                                                                          |
| LMS data synchronisation | pandas; Moodle upload tools; Canvas SIS import | Loads every OULAD module presentation as a course, and its students as enrolled users, in each LMS                                                      |
| LMS (tested)             | Moodle 5.2 (ersec/alpine-moodle)               | Native plugin integration, run locally in Docker with PostgreSQL                                                                                        |
| LMS (tested)             | Canvas (lbjay/canvas-docker)                   | Alternative LMS integration path, tested via non-LTI External URL module embedding                                                                      |

|                  |                           |                                                                                                           |
|------------------|---------------------------|-----------------------------------------------------------------------------------------------------------|
| Containerisation | Docker, Docker<br>Compose | Single Compose file running the API, Moodle with its<br>database, and Canvas for the prototype deployment |
|------------------|---------------------------|-----------------------------------------------------------------------------------------------------------|

## 4.5 Data Layer Implementation

The Data Layer ingests the seven OULAD source tables (studentInfo, studentRegistration, studentVle, vle, studentAssessment, assessments, and courses) and produces a single feature table of one row per (student, module, presentation) triple. The studentVle clickstream file, at approximately 450MB, is read using pandas' chunked-iteration interface rather than loaded into memory in full, with per-chunk aggregation to daily click totals performed before concatenation. The resulting feature table comprised 32,593 rows and 28 columns, one row for each of the 32,593 student-module-presentation observations in OULAD (Kuzilek et al., 2017), as reported in Chapter 1.

As set out in Chapter 3, Section 3.4, VLE engagement features are computed as deadline-anchored rolling windows (7, 14, and 28 days prior to each assessment submission deadline) rather than fixed calendar-week windows, and a leakage guard is enforced programmatically: any window construction that would draw on activity dated on or after its associated deadline raises an assertion error rather than silently including it. A versioned feature dictionary (JSON) is emitted alongside every processed feature batch, recording the data type, description, and provenance of every column, so that the Prediction Engine and Explainability Module can validate their inputs against a defined schema.

The leakage guard implements, as an executable check, the methodological requirement from Chapter 3, Section 3.3. Krstajic et al. (2014) show that leakage of future information into historical windows is among the most common sources of optimistic performance estimates in predictive modelling, and that such leakage is frequently introduced silently during feature construction rather than during model fitting. Enforcing the constraint programmatically rather than by convention means that any subsequent revision of the window logic raises an error instead of producing an inflated result. The choice of deadline-anchored rather than calendar-anchored windows follows Hlosta, Zdrahal, and Zendulka (2017), who report that windows aligned to submission dates yield stronger early-warning signals on OULAD, and it also preserves the property that the same feature code can be used for both training and inference.

The versioned feature dictionary serves the schema contract function described by Hernandez-Leo et al. (2019), who argue that stable contracts between the acquisition and analysis layers are a precondition for iterative dashboard refinement, since interface work cannot proceed if the semantics of the underlying features shift beneath it. In the implemented system the dictionary also carries the human-readable label and short explanatory phrase associated with each engineered feature, so that the reasoning trail (described later, Section 4.9) renders meaningful labels rather than internal variable names. This is a presentation requirement with an explanatory basis: Miller (2019) observes that explanations are understood by users in a social context and must be phrased in terms accessible to the recipient, and an attribution expressed against an unlabelled column name does not satisfy that condition for any of the three stakeholder roles. Missing values are handled as described earlier in Chapter 3, Section 3.3, with structural zeros distinguished from informative missingness through indicator encoding, following Van Buuren (2018).

## 4.6 Prediction Engine Implementation

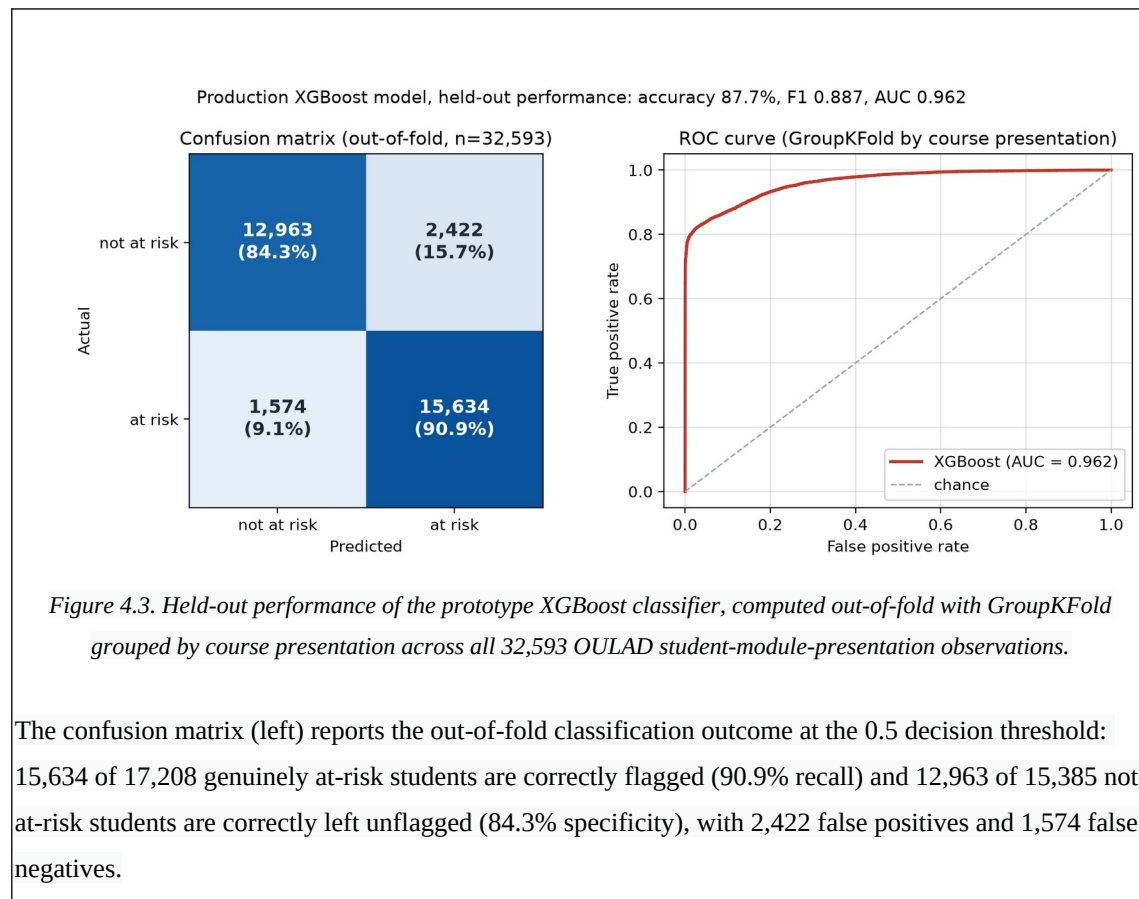
The prototype classifier is a gradient-boosted ensemble (XGBoost), selected with recall as the primary deployment criterion (Chapter 3, Section 3.4) and evaluated using GroupKFold cross-validation grouped by course presentation (of which OULAD contains four: 2013B, 2013J, 2014B, and 2014J), ensuring that no fold is evaluated on a presentation the model was also trained on. Held-out, out-of-fold performance across the full dataset was 87.7% accuracy, 0.887 F1-score, and 0.962 AUC. The fitted preprocessing pipeline and model are persisted as a single versioned artifact together with a manifest recording the training date, the feature dictionary version used, and the evaluation metrics, so that a prediction returned by the API can always be traced back to the exact model version and feature contract that produced it.

The grouping of folds by course presentation is a stricter protocol than the stratified splits reported in much of the OULAD literature, and this affects how the reported figures should be interpreted. Because no presentation appears in both the training and the evaluation partition of a fold, the out-of-fold estimates approximate transfer to an unseen course presentation under the retrospective feature construction used in this study. The consequence is that these figures are not directly comparable with the highest accuracies reported on OULAD in prior work, including the 96.9% reported on OULAD by Wang (2025) for a hybrid ensemble, since those studies do not in general isolate presentations across folds. Jin et al. (2024) note in their systematic review that evaluation protocols vary considerably across OULAD studies, and that reported accuracies are not straightforwardly comparable across the literature as a result. Krstajic et al. (2014) make the general form of the same argument, showing that ignoring the higher-level unit structure of a dataset produces optimistic estimates.

The asymmetry of the error profile also connects to the modelling rationale from Chapter 3, Section 3.4. F1-score was adopted there as the primary metric for comparative reporting precisely because accuracy is uninformative under class imbalance, and recall as the primary criterion for selecting the prototype model, and Chicco and Jurman (2020) argue that summary metrics should be interpreted alongside the full confusion matrix rather than in place of it. Reporting recall and specificity separately, as is done above, shows the trade-off that a single threshold imposes, and it is this trade-off that the dashboard exposes to practitioners rather than resolving on their behalf. The probabilities surfaced in the interface are treated with corresponding caution, since Niculescu-Mizil and Caruana (2005) and Guo et al. (2017) both demonstrate that discrimination and calibration are distinct properties, and Kim et al. (2024) observe that practitioner trust responds to the perceived reliability of displayed confidence values and not only to aggregate accuracy.

A further consideration follows from the fairness position taken earlier, in Chapter 3, Section 3.9. Because the feature set includes demographic attributes that overlap with protected characteristics, aggregate out-of-fold performance is not on its own a sufficient account of model behaviour. Baker and Hawn (2022) document systematic disparities in educational model performance across demographic subgroups, and Yu et al. (2020) show that error rates can differ substantially across gender, ethnic, and first-generation groups even where overall accuracy is high. The persistence of the model manifest alongside the fitted artefact is what makes disaggregated reporting of this kind tractable, since any subsequent audit can reconstruct the exact feature contract and model version against which subgroup performance was computed, in line with the model card proposal of

Mitchell et al. (2019). This disaggregated reporting was not completed within the final evaluation (Chapter 5, Section 5.10).



## 4.7 Explainability Module Implementation

Following the computational asymmetry argument from Chapter 3, Section 3.5, the explainability module is split into two independently-operating services. A global service computes TreeSHAP values once per model version, using the final fitted model, across its full training population of all 32,593 observations, and caches the result. These global values explain the final fitted model and are not fold-specific explanations produced within the cross-validation (Chapter 5, Section 5.6); this cache serves the coarser, cohort-level views required by the advisor and administrator roles without recomputation on every request. A local service computes both a local SHAP explanation and an independent LIME explanation on demand for a single prediction, reconstructing the exact feature vector the model was originally scored against, so that the two explanation methods can be displayed side by side in the instructor view and any agreement or disagreement between them is directly visible to the end user.

Displaying the two methods side by side rather than selecting between them puts into practice the methodological triangulation argued for earlier, in Chapter 3, Section 3.5. Prior studies have reported complementary uses of the two methods, with SHAP commonly used for global attribution and LIME for local explanations, and Gunasekara and Saarela (2025) apply both to models trained on OULAD, while Salih et al. (2024) report a higher computing time for SHAP than for LIME, which motivates the caching strategy adopted here. LIME is relatively lightweight in

some contexts, but it involves perturbation and repeated model evaluations and must be regenerated for each instance. Presenting both attributions simultaneously also makes visible any divergence between them, which provides additional information: where the two methods identify the same drivers, the practitioner has a stronger basis for acting on the explanation than either method provides alone.

The same design decision responds to the known limitations of post-hoc explanation discussed back in Chapter 2 (Section 2.3). Rudin (2019) argues that high-stakes decisions should rest on intrinsically interpretable models rather than on post-hoc accounts of opaque ones, and Slack et al. (2020) demonstrate that both SHAP and LIME are sensitive to perturbation choices in ways that can be exploited. The present study retains the post-hoc route for consistency with the prior OULAD literature, but the side-by-side presentation is the mechanism through which that choice is made transparent to the practitioner. The stability and fidelity checks specified earlier (Chapter 3, Section 3.5), including the Jaccard similarity of top-k feature sets across repeated LIME runs proposed by Visani et al. (2022), were not completed in the final evaluation. In keeping with the selectivity property identified by Miller (2019), and with the evidence of Poursabzi-Sangdeh et al. (2021) that comprehension degrades when unfiltered attribution vectors are displayed in full, the reasoning trail shows a small number of leading contributions rather than the complete attribution vector.

## **4.8 API and Security Architecture**

The Flask API enforces role-based access control on every request, independently of the Learning Management System in front of it, following a defence-in-depth principle. A request is accepted either with a JSON Web Token issued by the API's own login, or, for server-to-server calls from the Moodle plugin, with a shared service key known only to the plugin and the API; the Moodle plugin first checks the user's Moodle capability for the requested view, and the API then checks the forwarded role against the endpoint, so that role separation does not collapse to a single point of failure in the Learning Management System. Three roles are recognised throughout the system including the instructor, advisor, and administrator, each receiving a different level of explanatory detail from the same underlying prediction, in keeping with the Agent Model from Chapter 3, Section 3.6.

For access outside Moodle, including the Canvas embedding (covered in Section 4.10), the API additionally exposes a conventional username/password login endpoint, with three demonstration accounts corresponding one-to-one with the three application roles. Passwords are stored as salted hashes rather than in plaintext.

The decision to enforce role separation at both the Learning Management System and the API reflects the defence-in-depth principle of Saltzer and Schroeder (1975), cited earlier in Chapter 3, Section 3.7, under which no single component should constitute the sole enforcement point for an access control decision. The consequence in this system is that a forwarded role header is treated as an unverified claim requiring corroboration, not as an established fact, which is the appropriate posture given that the dashboard is designed to be embedded within a Learning Management System and, as a result, to receive requests that originate outside its own trust boundary. Token issuance and verification follow the JSON Web Token specification of Jones, Bradley, and Sakimura (2015), and the separation between the HS256 tokens used for the dashboard's own sessions and the service key used by the Moodle plugin keeps the two trust relationships distinct.

Role separation is also a substantive requirement of the study rather than a purely technical concept. The Agent Model, summarised earlier in Chapter 3, Section 3.6, treats each stakeholder role as a distinct decision-making agent with its own knowledge requirements, and Kostopoulos, Davrazos, and Kotsiantis (2024), in their review of explainable decision support systems, note that such systems can be tailored to the individual needs of their users. If a practitioner could obtain a level of explanatory detail associated with a different role, the differentiation that the study evaluates would not hold in practice, and the evaluation reported in the following chapter would be measuring responses to an interface other than the one specified. Gating the visible view by the authenticated session is, as a result, a requirement of the evaluation design as well as a security measure.

## 4.9 Frontend Implementation

The presentation layer is a single-page React application with three role-gated views. Rather than allowing free navigation between roles, the deployed prototype gates the visible view by the authenticated session: a user must sign in, and the role returned by that sign-in determines which of the three views is rendered, with no client-side mechanism to switch roles without re-authenticating. Each view reuses a single 'reasoning trail' component which is a visualisation of signed SHAP/LIME contributions at three different levels of aggregation: individual-student detail for the instructor view, cohort-averaged detail for the advisor view, and full-population detail for the administrator view, so that the same visual format is used across all three levels of the Agent Model rather than requiring three unrelated visualisations.

The decision to reuse a single reusable visual component across the three views follows from the CommonKADS position adopted throughout the study. The Agent Model differentiates the roles by the granularity and framing of the knowledge each requires, not by the visual vocabulary in which that knowledge is expressed, and Sokol and Flach (2020) argue on similar grounds that explanation type, granularity, and format should be treated as separable configuration parameters. Reusing the reasoning trail at three levels of aggregation also reduces the effort required to interpret the information for practitioners who occupy more than one role in practice, which is common in smaller institutions, and it aligns with the recognition-rather-than-recall and consistency heuristics of Nielsen (1994), referenced earlier in Chapter 3, Section 3.6.

The content of each view is derived from the role-specific information briefs set out earlier, in Chapter 3, Section 3.6. The instructor view foregrounds per-student detail because Molenaar and Knoop-van Campen (2019) show that teachers act on dashboard information at the level of the individual learner, and because Kaliisa and Dolonen (2022) find that instructors prioritise summary-level alerts over granular statistical breakdowns, which is why the full attribution is reached through the reasoning trail rather than presented as the default. The advisor view is cohort-averaged over students at or above the watch threshold, following the caseload orientation of the LADA advising dashboard reported by Gutiérrez et al. (2020). The administrator view is full-population because institutional decision support is concerned with distributional rather than individual behaviour, in line with the patterns documented by Tsai and Gašević (2017). Progressive disclosure is applied uniformly across the three views, with compact risk summaries rendered by default and the attribution detail exposed on request, following the layered disclosure pattern described by Springer and Whittaker (2020) and adopted earlier, in Chapter 3, Section 3.2.

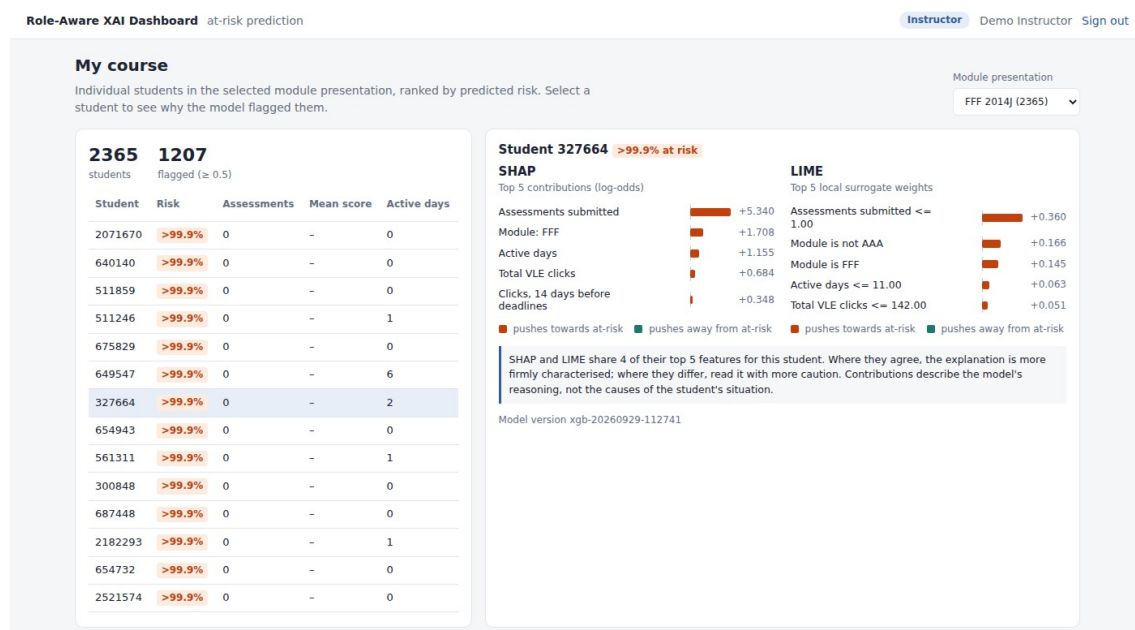


Figure 4.4. Screenshot from the prototype development: instructor view, per-student SHAP and LIME reasoning trail, shown side by side for a single OULAD student.

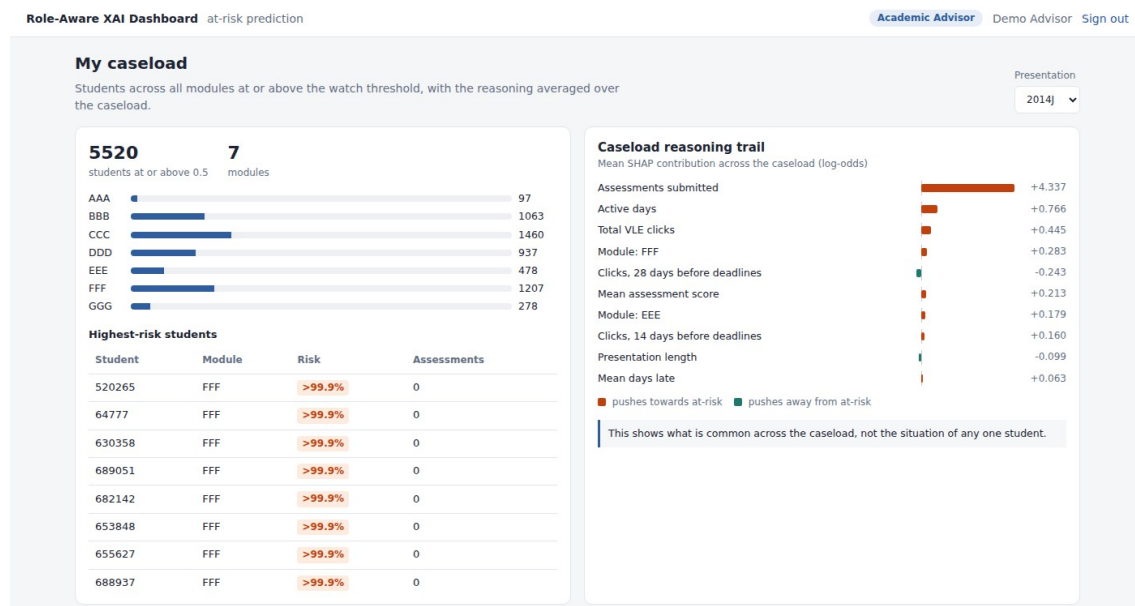


Figure 4.5. Screenshot from the prototype development: advisor view, cohort-averaged reasoning trail for students flagged at or above the watch threshold.

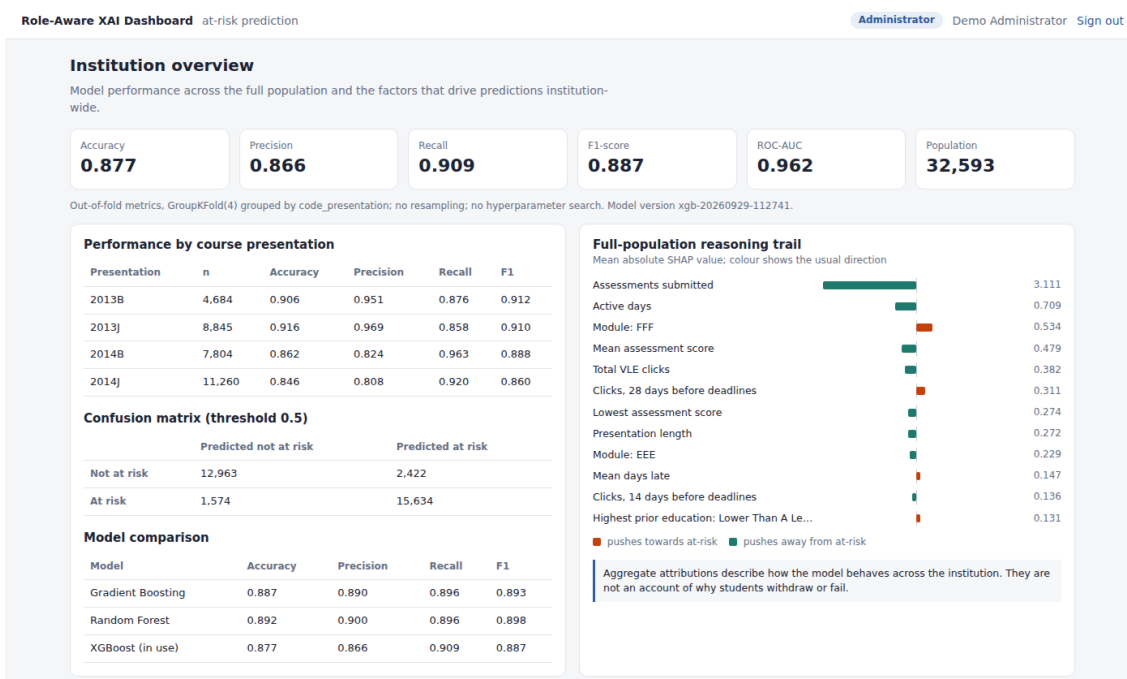
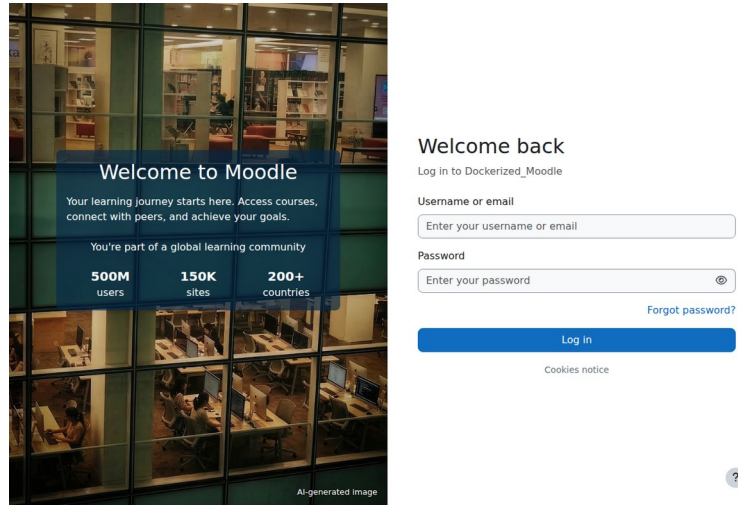


Figure 4.6. Screenshot from the prototype development: administrator view, institution-wide model performance summary and full-population reasoning trail.

(a) Moodle login page



(b) Dashboard sign-in inside a Canvas course

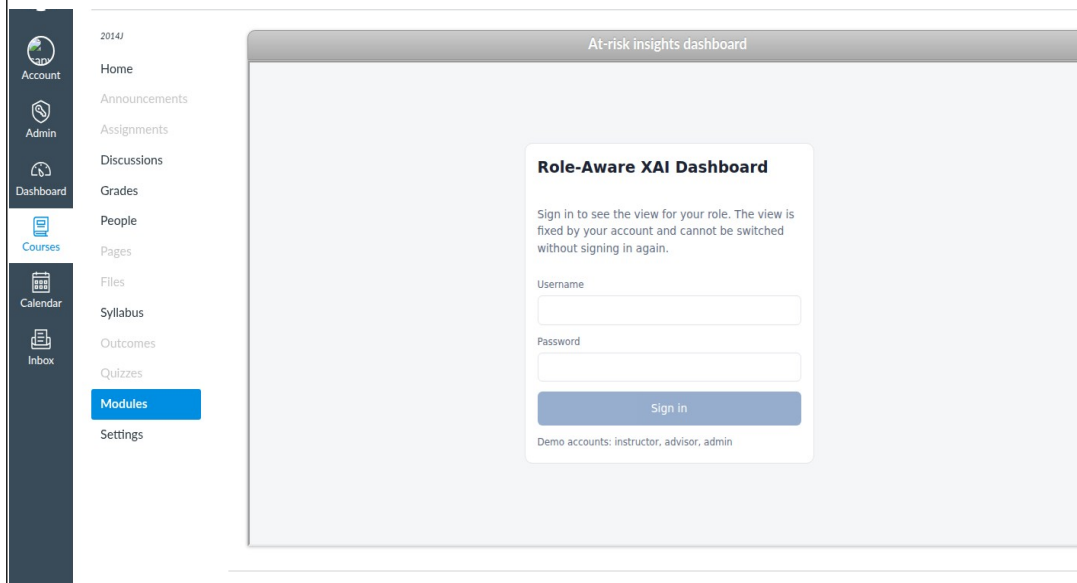


Figure 4.7. Screenshot from the prototype development: authentication and role resolution, (a) Moodle login page (role and course context come from Moodle), (b) dashboard sign-in inside a Canvas course

## 4.10 Containerisation and LMS Integration Testing

Two Learning Management System integration paths were evaluated locally prior to any institutional deployment, both run in Docker to allow rapid, temporary testing without setting up institutional infrastructure.

Testing these integration paths was treated as a substantive part of the implementation rather than as a deployment detail. The literature reviewed earlier, in Chapter 2 (Section 2.1), identifies

institutional uptake, rather than predictive capability, as the main constraint on learning analytics: Tsai and Gašević (2017) attribute limited adoption in part to the shortage of practitioner-facing decision support, and Leitner, Ebner, and Ebner (2019) document a persistent disconnect between the technical sophistication of predictive models and their use in institutional decision making. A dashboard that cannot be reached from within the environment in which instructors and advisors already work is unlikely to overcome that constraint, and demonstrating that the system launches from a Learning Management System with the launching user's role and course context intact is, as a result, evidence of technical integration feasibility, supporting the contribution stated earlier in Chapter 1, Section 1.5.

#### **4.10.1 Moodle**

A lightweight Moodle instance (the community-maintained `ersec/alpine-moodle` image, running Moodle 5.2, substantially smaller than the official Bitnami or MoodleHQ development images) was run locally via Docker, together with a PostgreSQL container. Rather than an LTI 1.3 launch, which would require a publicly resolvable HTTPS endpoint, the dashboard was integrated as a native Moodle local plugin (`local_xairisk`). Every OULAD module presentation was loaded as a Moodle course (22 courses, with short names such as FFF-2014J), every student as a Moodle user (28,785 users), and every registration as a course enrolment (32,593 enrolments), using Moodle's own course and user upload tools. The plugin adds an At-risk insights page to each course for users with the teacher role, an Advisor caseload page for an Academic advisor system role created by the plugin, and an Institution overview page for managers. Login, role, and course context are therefore taken from Moodle itself: an instructor sees only the students of a course they teach, and the plugin calls the API server to server with the shared service key described in Section 4.8, so that no dashboard credentials reach the browser. Access control was checked by attempting cross-role and cross-course requests, such as an instructor of module AAA opening the insights page of an FFF course, and each of these requests was refused.

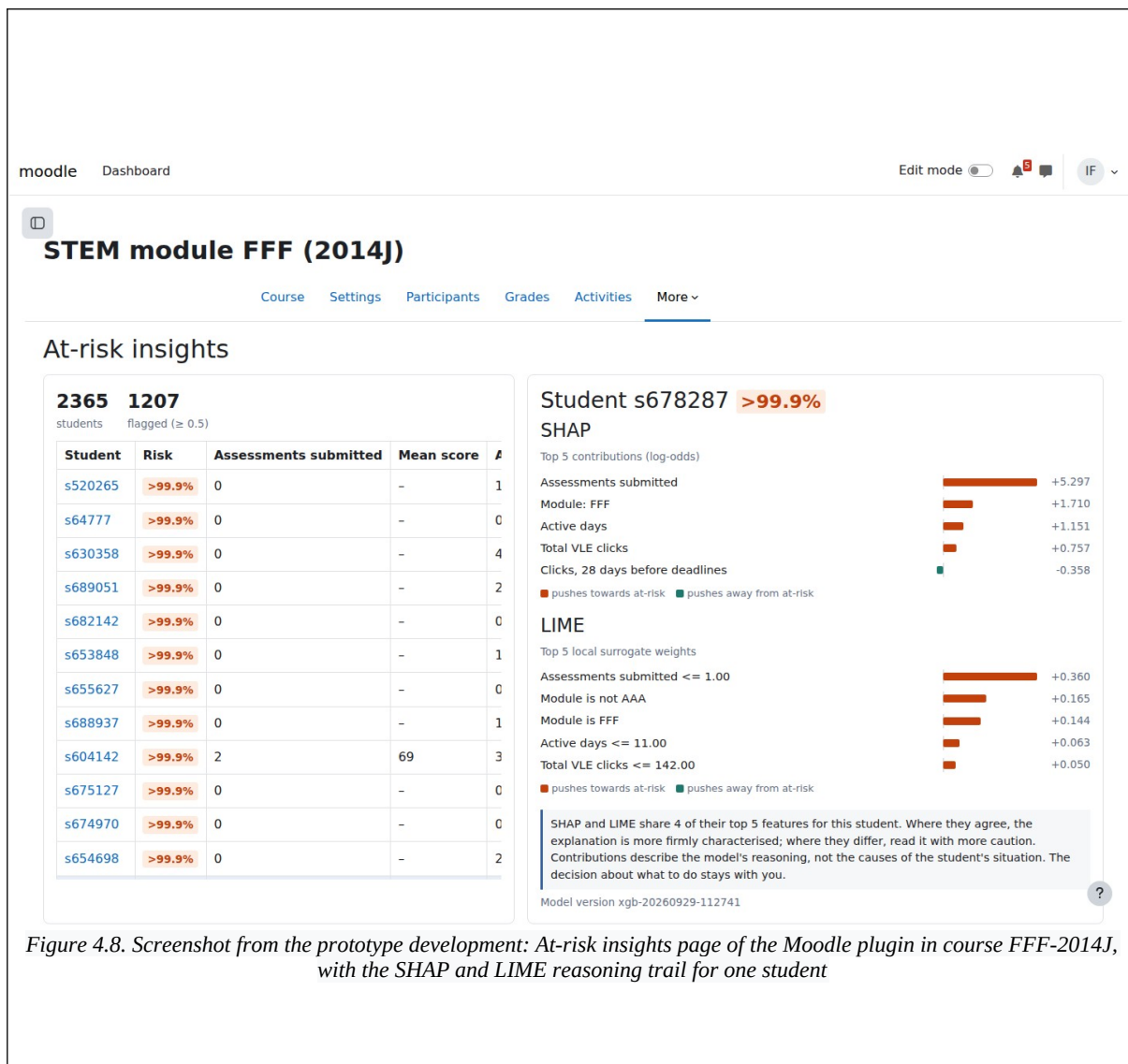


Figure 4.8. Screenshot from the prototype development: At-risk insights page of the Moodle plugin in course FFF-2014J, with the SHAP and LIME reasoning trail for one student

## 4.10.2 Canvas

Canvas was evaluated as an alternative LMS using a community Docker image (lbjay/canvas-docker) purpose-built for LTI application integration testing. Because the available local Canvas image predates Canvas's LTI 1.3 support, integration was instead tested via Canvas's External URL module item, which embeds the dashboard in an iframe within a course module. The 22 OULAD module presentations were created as Canvas courses through a Student Information System (SIS) import, and an At-risk insights module was added to each course, opening the dashboard on that course. Unlike the Moodle path, the dashboard's own authentication (described in Section 4.8) is used to establish the user's role in this configuration, independent of whichever Canvas account is active.

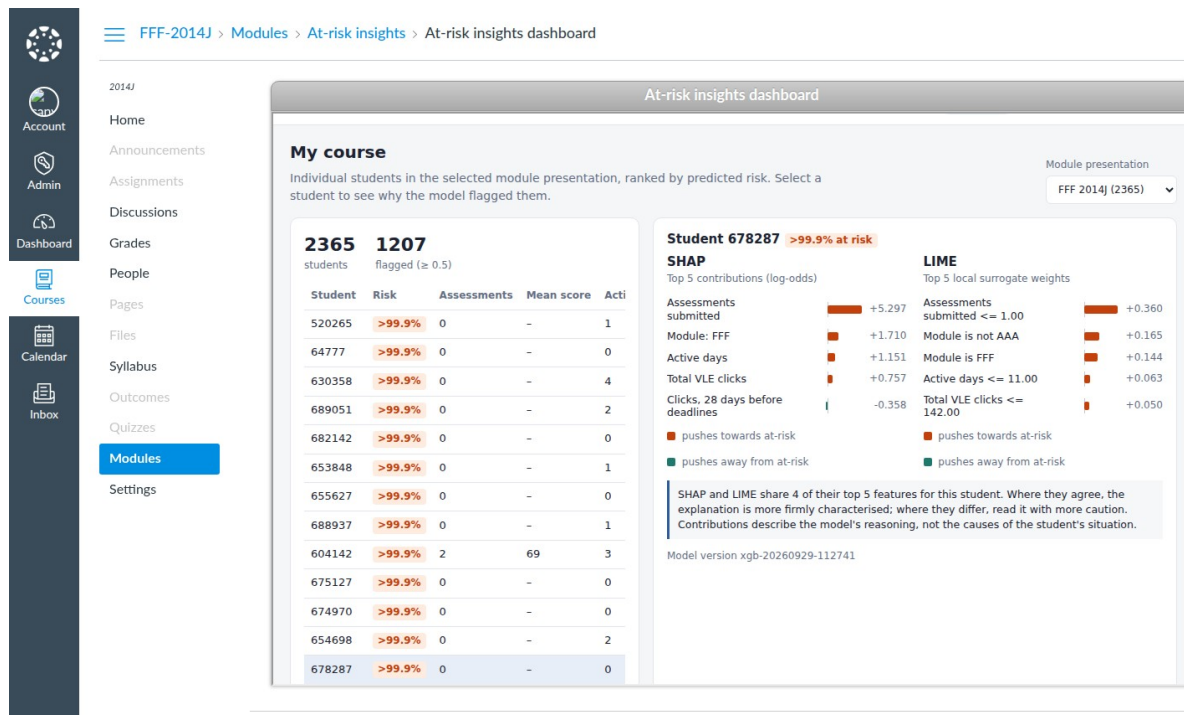


Figure 4.9. Screenshot from the prototype development: Canvas course FFF-2014J with the dashboard embedded through an External URL module item

## 4.11 Prototype Deployment Topology

In the prototype, the whole system runs as a single Docker Compose deployment. One container runs the Flask API under gunicorn and also serves the built React application, so that the dashboard and its API share one origin; Moodle runs in a second container with its PostgreSQL database in a third, and Canvas can be started in a fourth when that integration path is needed. On first start, the API container builds the feature table, trains and evaluates the model, and computes the global TreeSHAP cache from the OULAD files, and it stores these artefacts in a persistent volume, so that later restarts reuse the same model version. The out-of-fold risk scores for the full population and the global SHAP values are computed once per model version and served from this cache, so that administrator- and advisor-level dashboard views are served from a cached copy rather than recomputing predictions for the full student population on every page load. The API also sends a Content-Security-Policy frame-ancestors header, so that only the configured Learning Management System origins can embed the dashboard.

The separation between the cached cohort path and the synchronous per-student path follows the serving pattern documented by Sculley et al. (2015), who observe that co-locating batch and online scoring behind a single interface accumulates hidden coupling and makes latency behaviour difficult to reason about, whereas distinguishing the two access paths allows cohort refreshes to be scheduled without competing with interactive traffic. In this system, this means that the administrator and advisor views, which aggregate over the full population, are served from cached results, while the instructor view retains the responsiveness required for the interactive explanation requests on which the evaluation tasks depend. Kaliisa, Jivet, and Prinsloo (2023) note that

responsiveness is among the factors that shape whether practitioners incorporate a dashboard into routine work, which makes this a design consideration rather than an operational one.

## 4.12 Summary

This chapter has documented the technical realisation of the architecture proposed in Chapter 3: a five-layer system spanning data engineering, prediction, explainability, API, and presentation concerns, implemented using a combination of Python (data science, machine learning, and the API), JavaScript (frontend), PHP (Moodle plugin), and containerised LMS environments for integration testing. The following chapter reports the results obtained from evaluating this implementation against the OULAD dataset, including model performance and explanation quality, together with a discussion of what these technical findings imply for practitioner use, drawing on existing literature rather than on original usability testing.

In combination, the implementation reported in this chapter satisfies the technical preconditions for answering the research questions from Chapter 1, Section 1.4, within the scope of a study based entirely on the secondary OULAD dataset. The integration of SHAP and LIME into a versioned pipeline in which every prediction can be traced to the model and feature contract that produced it addresses the first research question at the level of system construction. The three role-gated views, each derived from the Agent Model and each rendering the same reasoning trail at a different level of aggregation, are the design response to the second research question, specified from the CommonKADS Agent Model and the dashboard-design literature cited in Section 4.9 rather than validated against practitioners directly, since no primary user data were collected for this study. The third research question concerns what the predictive and explainability findings indicate about the technical potential of the dashboard to support practitioner interpretation and decision-making; what this chapter establishes is that the system evaluated in the following chapter is a working prototype tested against the full OULAD population rather than one limited to illustrative cases, which is the precondition for the technical evidence Chapter 5 reports, though it is not itself a measurement of practitioner response.

Two limitations of the implementation should be recorded before the evaluation is reported. First, the Learning Management System integrations were validated locally against containerised Moodle and Canvas instances rather than against an institutional installation, so the integration evidence presented here establishes technical feasibility rather than deployment readiness, in line with the scope set out in Chapter 1, Section 1.6. Second, the Canvas path was tested through External URL embedding rather than a full LTI 1.3 launch because of the version of the available image, which means that role resolution on that path depends on the dashboard's own authentication rather than on the Learning Management System. Neither limitation affects the prediction, explanation, or presentation components evaluated in the following chapter, but both affect the transferability of the system to a real institutional setting and are revisited in the discussion of limitations.

## CHAPTER 5

### RESULTS AND ANALYSIS

#### 5.1 Introduction

This chapter reports the results obtained from evaluating the system described in Chapter 4. Chapter 4 documented the technical realisation of the five-layer architecture; this chapter presents the verified evidence produced when that implementation was run end to end against the OULAD feature table, and analyses what the evidence shows in relation to the research objectives and research questions set out in Chapter 1, Section 1.3 and Section 1.4.

The chapter is organised as follows. Section 5.2 describes the experimental setup, including the dataset and evaluation methodology actually used. Section 5.3 presents the four required predictive metrics for the prototype XGBoost classifier and interprets what they indicate about classification performance. Section 5.4 extends this to the three-way ensemble comparison specified in Chapter 3, completed for this chapter. Section 5.5 reports the variability of performance across the four OULAD course presentations. Section 5.6 reports the SHAP and LIME explainability findings. Section 5.7 verifies the figures reported in Chapter 4 against this execution and compares the implementation with the methodology set out in Chapter 3. Section 5.8 compares the findings against published literature. Section 5.9 develops a thematic synthesis across the technical evidence. Section 5.10 evaluates the proposed framework against the available evidence. Section 5.11 returns to each research question in turn. Section 5.12 summarises the chapter.

This is a technical evaluation, consistent with the scope defined in Chapter 1, Section 1.6: it establishes, with verified figures, what the predictive and explanatory components of the system actually do when run against the full OULAD population. The chapter draws its conclusions from that technical evidence together with the published literature reviewed in Chapter 2, rather than from primary data collected from practitioners, and the discussion in Sections 5.9 to 5.11 is framed accordingly.

#### 5.2 Experimental Setup

The evaluation was carried out on the processed OULAD feature table used throughout the thesis, comprising 32,593 student-module-presentation observations (one row per student, module, and presentation), matching the population reported in Chapters 1 and 4. The target variable, `at_risk`, was verified in this execution by cross-tabulation against the original `final_result` field: students recorded as Withdrawn ( $n = 10,156$ ) or Fail ( $n = 7,052$ ) were mapped to `at_risk = 1`, giving 17,208 at-risk cases (52.8%), while students recorded as Pass ( $n = 12,361$ ) or Distinction ( $n = 3,024$ ) were mapped to `at_risk = 0`, giving 15,385 not-at-risk cases (47.2%). The dataset is therefore only mildly imbalanced. Because withdrawals account for 59% of the positive class, the model is best understood as substantially a withdrawal predictor rather than a pure academic-failure predictor, and this qualification is carried through the interpretation below.

The model input comprised 24 columns, 17 numeric and 7 categorical, expanding to 59 features after one-hot encoding of the categorical variables. The numeric features covered prior attempts, studied credits, registration timing, assessment submission counts and scores, submission lateness, banked assessments, deadline-anchored click windows at 7, 14, and 28 days, total clicks, active

days, and mean daily clicks. The categorical features covered module, gender, region, highest education, IMD band, age band, and disability. This matches the three feature families described in Chapter 3, Section 3.4; the trajectory family, describing period-over-period change in engagement and performance, was a planned extension that was not implemented (Chapter 3, Section 3.4.1, and Section 5.7 below).

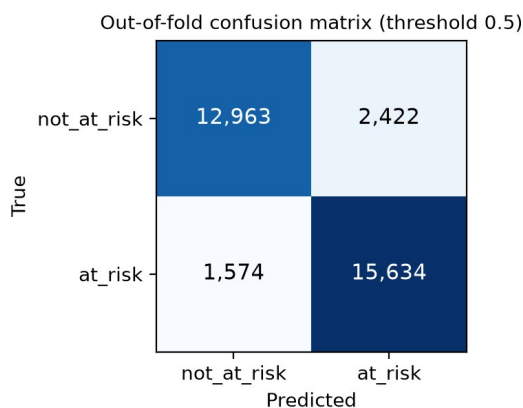
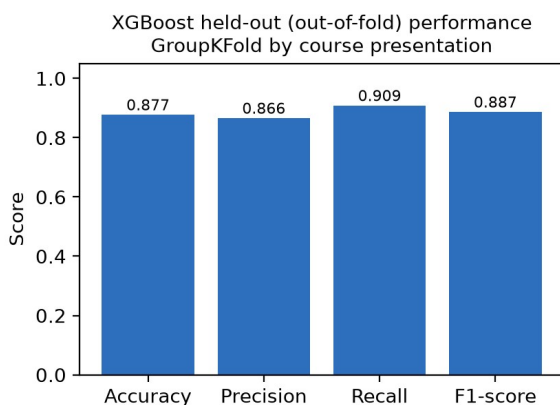
Evaluation used GroupKFold cross-validation grouped by code\_presentation, with the number of folds set to four because OULAD contains exactly four course presentations (2013B, 2013J, 2014B, 2014J). Metrics were computed over out-of-fold predictions for all 32,593 observations, so every prediction was produced by a model that had not seen the presentation to which that row belongs during training. No resampling of the training data and no hyperparameter search were applied; the prototype XGBoost pipeline was evaluated with its existing fixed configuration (300 estimators, maximum depth 5, learning rate 0.05, subsample 0.8, column subsample 0.8). The decision threshold for classification was 0.5 throughout. This protocol reproduces the evaluation methodology described in Chapter 3, Section 3.4 and Chapter 4, Section 4.6; Section 5.7 compares it against the originally planned methodology recorded in Chapter 3, Section 3.4.1. All features were computed over the full course presentation, so the evaluation measures retrospective classification using the available OULAD feature set and does not establish performance at a predefined early-warning point (see Section 5.9).

### 5.3 Model Performance

Table 5.1 reports the four required metrics for the prototype XGBoost classifier, computed out-of-fold across the full dataset.

*Table 5.1. Held-out (out-of-fold) performance of the prototype XGBoost classifier, GroupKFold by course presentation.*

|                        |        |        |        |        |
|------------------------|--------|--------|--------|--------|
| XGBoost<br>(prototype) | 0.8774 | 0.8659 | 0.9085 | 0.8867 |
|------------------------|--------|--------|--------|--------|



*Figure 5.1. XGBoost out-of-fold accuracy, precision, recall, and F1-score (left) and the corresponding out-of-fold confusion matrix at the 0.5 decision threshold (right), across all 32,593 OULAD student-module-presentation observations.*

The confusion matrix underlying Table 5.1 shows 12,963 true negatives, 2,422 false positives, 1,574 false negatives, and 15,634 true positives. Precision is not stated numerically anywhere in Chapter 4, which reports recall and specificity; it is derivable from the Chapter 4 confusion matrix as 15,634 divided by the sum of 15,634 and 2,422, which equals 0.8659, the value obtained independently in this execution.

**Accuracy.** The out-of-fold accuracy of 0.8774 indicates that the classifier assigns the correct label, at-risk or not-at-risk, to 87.7% of the 32,593 held-out observations. Because the classes are only mildly imbalanced (52.8% versus 47.2%), accuracy is not as misleading a summary here as it would be under severe imbalance, but it still collapses two different kinds of error, false positives and false negatives, into a single figure, which is why Chapter 3, Section 3.4 nominated F1-score rather than accuracy as the primary metric for comparative reporting.

**Precision.** A precision of 0.8659 means that of every 1,000 students the model flags as at-risk, approximately 866 subsequently received the at-risk outcome label used in this study and approximately 134 did not. In the context of the framework's intended use, that is, informing an instructor, advisor, or administrator that a student may need support, this figure describes the rate at which a student flagged by the model's positive prediction subsequently received the positive outcome label used in this study. A precision below 1.0 means that some proportion of flagged students did not go on to receive the at-risk label, and any follow-up directed at them would carry a resourcing cost for the institution, although it does not carry the same downside as a missed at-risk student.

**Recall.** A recall of 0.9085 means that of the 17,208 genuinely at-risk students in the held-out data, 15,634 are correctly identified by the model and 1,574 are missed. This is the figure most directly relevant to the early-intervention rationale of the study, and it is the primary criterion for selecting the prototype model (Chapter 3, Section 3.4), because a student who is at risk but not flagged receives no benefit from the dashboard at all. The recall obtained here is higher than the precision, which indicates that, at the default 0.5 threshold, the model is somewhat more conservative about withholding an at-risk flag than it is about avoiding a false alarm.

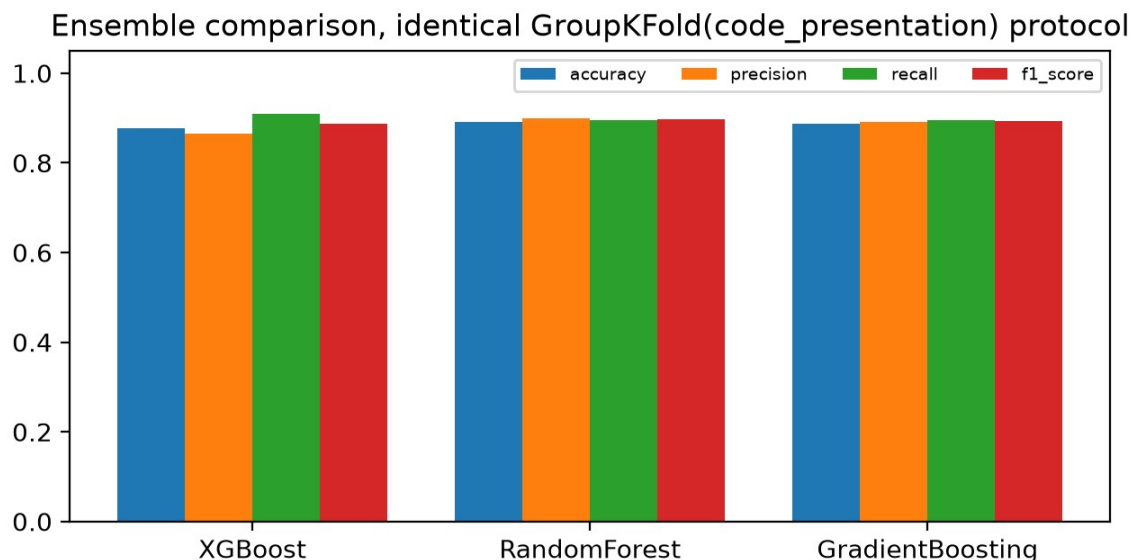
**F1-score.** The F1-score of 0.8867 is the harmonic mean of precision and recall and summarises the balance between the two. Because recall (0.9085) exceeds precision (0.8659) by a moderate margin, the F1-score sits closer to the lower of the two figures than a simple average would, which is the expected behaviour of the harmonic mean when the underlying rates diverge. Taken together, the four metrics describe a classifier that identifies the substantial majority of at-risk students while generating a non-trivial false-positive rate (2,422 false positives, approximately 15.7% of the not-at-risk group). Neither figure should be read in isolation from the confusion matrix in Figure 5.1, since Chicco and Jurman (2020) caution that summary metrics can obscure the underlying error profile, and it is that full profile, not any single number, that determines what an institution should expect operationally from the 0.5 threshold.

## 5.4 Comparison Across the Candidate Ensemble Models

Chapter 3, Section 3.4 specified a three-way comparison of Random Forest, XGBoost, and Gradient Boosting Classifier, with F1-score used for comparative reporting and recall as the primary criterion for selecting the prototype model. To complete this comparison, Random Forest and Gradient Boosting were run under the identical preprocessing pipeline and the identical GroupKFold(code\_presentation) protocol used for XGBoost in Section 5.3, so that all three models are compared on an equal footing. Table 5.2 and Figure 5.2 report the result.

*Table 5.2. Ensemble comparison under the identical GroupKFold(code\_presentation) protocol. XGBoost is the prototype model; Random Forest and Gradient Boosting were evaluated for comparison only and are not used in the prototype.*

|                   |        |        |        |        |     |
|-------------------|--------|--------|--------|--------|-----|
| XGBoost           | 0.8774 | 0.8659 | 0.9085 | 0.8867 | Yes |
| Random Forest     | 0.8921 | 0.8997 | 0.8955 | 0.8976 | No  |
| Gradient Boosting | 0.8869 | 0.8904 | 0.8962 | 0.8932 | No  |



*Figure 5.2. XGBoost, Random Forest, and Gradient Boosting compared on the four required metrics under the identical protocol.*

All three models perform strongly and within a narrow band of one another. Random Forest obtains the highest F1-score (0.8976), ahead of Gradient Boosting (0.8932) and the prototype XGBoost model (0.8867), and also obtains the highest accuracy and precision of the three, while XGBoost retains the highest recall, correctly identifying more of the genuinely at-risk students than either alternative. The gap between the leading and trailing F1-scores is 0.011, a narrow margin at this sample size. Random Forest is therefore the strongest of the three on the F1-score used for

comparative reporting. Under the selection criterion set out in Chapter 3, Section 3.4, however, recall is the primary criterion for the prototype model, because false negatives are the more costly error in the intended screening context, and on that criterion XGBoost is the strongest candidate, with a recall of 0.9085 against 0.8962 for Gradient Boosting and 0.8955 for Random Forest. The retention of XGBoost in the prototype is therefore consistent with the stated criterion, while the higher F1-score of Random Forest is recorded as a relevant finding for future iterations of the pipeline, particularly if the relative priority of recall and precision were revisited.

## 5.5 Variability Across Course Presentations

Because the GroupKFold protocol holds out one entire course presentation per fold, it is possible to report the prototype model's out-of-fold performance separately for each of the four presentations, which is a more exacting test of generalisation than the pooled figures in Table 5.1 since each fold's test presentation is entirely unseen during that fold's training. Table 5.3 and Figure 5.3 report this breakdown.

Table 5.3. XGBoost out-of-fold performance with each course presentation held out in turn.

|       |        |        |        |        |        |        |
|-------|--------|--------|--------|--------|--------|--------|
| 2013B | 4,684  | 0.5527 | 0.9065 | 0.9506 | 0.8764 | 0.9120 |
| 2013J | 8,845  | 0.4941 | 0.9161 | 0.9690 | 0.8577 | 0.9099 |
| 2014B | 7,804  | 0.5697 | 0.8619 | 0.8243 | 0.9627 | 0.8882 |
| 2014J | 11,260 | 0.5154 | 0.8456 | 0.8075 | 0.9197 | 0.8600 |

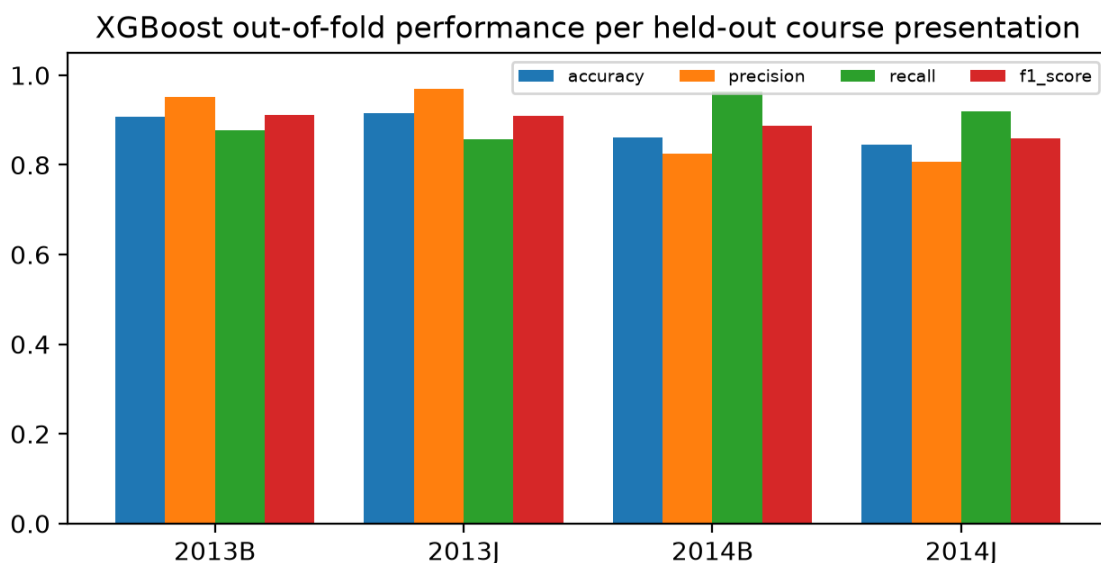


Figure 5.3. XGBoost out-of-fold accuracy, precision, recall, and F1-score for each held-out course presentation.

Performance is not uniform across presentations. The two earlier presentations, 2013B and 2013J, show high precision (above 0.95) with comparatively lower recall (around 0.86 to 0.88), while the two later presentations, 2014B and 2014J, show the reverse pattern: lower precision (around 0.81 to 0.82) with markedly higher recall (up to 0.963 for 2014B). Accuracy ranges from 0.8456 (2014J) to 0.9161 (2013J), a spread of over seven percentage points. Because only four groups are available for GroupKFold, each fold trains on three presentations and tests on one. This spread is consistent with substantial between-presentation variation, although the small number of presentation-level folds limits the precision with which the source of the variation can be attributed. This variability is a limitation on how confidently the pooled figures in Table 5.1 can be generalised to a presentation not represented in this dataset, and it is revisited in Section 5.10.

## 5.6 Explainability Analysis: SHAP and LIME

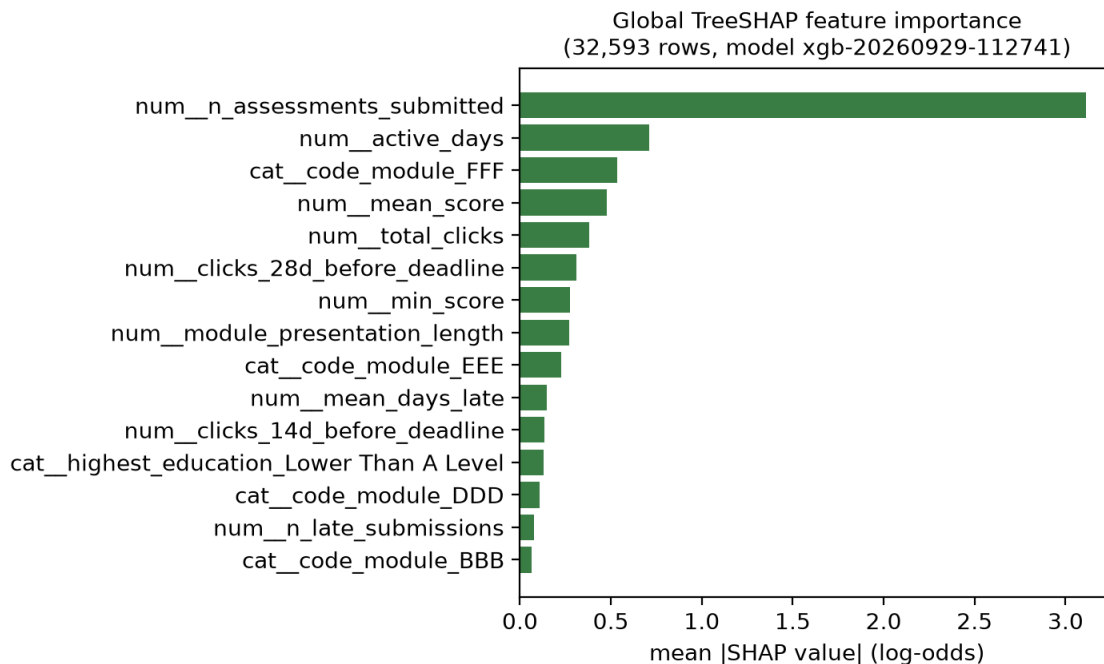
The Explainability Module described in Chapter 4, Section 4.7 was exercised in two ways. A global TreeSHAP computation was run with the final fitted model (the shipped model artefact described in Chapter 4, Section 4.6) across all 32,593 observations and the full 59-column transformed feature space, ranking features by mean absolute SHAP value in log-odds units. This global computation explains the final fitted model; it is not a fold-specific explanation evaluation, since SHAP values were not computed separately within each GroupKFold fold. A LIME analysis was run on a random sample of 40 rows (numpy default\_rng, seed 42) from the same feature table, with the resulting per-instance explanations aggregated by mean absolute LIME weight; the LIME explanations were also computed against the final fitted model. Both methods operate on the one-hot transformed feature space, so individual categorical levels, such as cat\_\_code\_module\_FFF, appear as separate features rather than as a single grouped module attribute.

Table 5.4 reports the fifteen most important features by global TreeSHAP value, and Figure 5.4 shows the same ranking graphically.

*Table 5.4. Top 15 features by global TreeSHAP importance (mean absolute SHAP value, log-odds units), all 32,593 observations.*

|   |                            |        |
|---|----------------------------|--------|
| 1 | n_assessments_submitted    | 3.1110 |
| 2 | active_days                | 0.7087 |
| 3 | code_module = FFF          | 0.5338 |
| 4 | mean_score                 | 0.4790 |
| 5 | total_clicks               | 0.3819 |
| 6 | clicks_28d_before_deadline | 0.3111 |
| 7 | min_score                  | 0.2741 |
| 8 | module_presentation_length | 0.2718 |

|    |                                        |        |
|----|----------------------------------------|--------|
| 9  | code_module = EEE                      | 0.2285 |
| 10 | mean_days_late                         | 0.1468 |
| 11 | clicks_14d_before_deadline             | 0.1360 |
| 12 | highest_education = Lower Than A Level | 0.1312 |
| 13 | code_module = DDD                      | 0.1075 |
| 14 | n_late_submissions                     | 0.0770 |
| 15 | code_module = BBB                      | 0.0640 |



*Figure 5.4. Top 15 global TreeSHAP features by mean absolute SHAP value.*

The number of assessments submitted dominates the ranking by a wide margin, contributing more than four times the mean absolute SHAP value of the second-ranked feature, active days. The correlation between a feature's value and its own SHAP value, computed across all rows, gives the direction of the association without implying causality. For `n_assessments_submitted`, `mean_score`, `active_days`, `total_clicks`, and `min_score`, this correlation is strongly negative (between -0.69 and -0.96), meaning higher values of these engagement and performance features push the prediction away from the at-risk class. For `clicks_28d_before_deadline`, `n_late_submissions`, and the FFF and DDD module indicators, the correlation is positive, meaning higher values push the prediction toward the at-risk class.

One pattern in this data requires explicit qualification rather than a straightforward behavioural reading. The positive correlation for clicks\_28d\_before\_deadline (+0.8354) is counter-intuitive on its face, since it implies that higher pre-deadline engagement in the 28-day window is associated with a prediction toward at-risk, in the opposite direction to total\_clicks and active\_days. One possible explanation is collinearity among the engagement variables, since Salih et al. (2024) note that SHAP assumes feature independence and that collinearity among features affects the attributions assigned to them. The engagement variables are indeed highly collinear: the Spearman correlation between clicks\_28d\_before\_deadline and total\_clicks is 0.98, and the 14-day and 7-day windows correlate with total\_clicks at 0.95 and 0.92 (Section 5.7). This is consistent with the collinearity explanation, although a SHAP dependence analysis was not carried out, so the finding is reported as observed, not as evidence that increased late-window activity is itself a risk factor, and it should not be read as a causal or behavioural claim.

Table 5.5 reports the ten most important features by aggregated LIME weight across the 40 sampled instances.

*Table 5.5. Top 10 features by mean absolute LIME weight, aggregated over 40 sampled instances (seed 42).*

|    |                            |        |
|----|----------------------------|--------|
| 1  | n_assessments_submitted    | 0.3961 |
| 2  | code_module = FFF          | 0.1397 |
| 3  | code_module = AAA          | 0.0995 |
| 4  | mean_score                 | 0.0640 |
| 5  | active_days                | 0.0498 |
| 6  | module_presentation_length | 0.0447 |
| 7  | clicks_28d_before_deadline | 0.0236 |
| 8  | min_score                  | 0.0183 |
| 9  | date_registration          | 0.0073 |
| 10 | region = Wales             | 0.0043 |

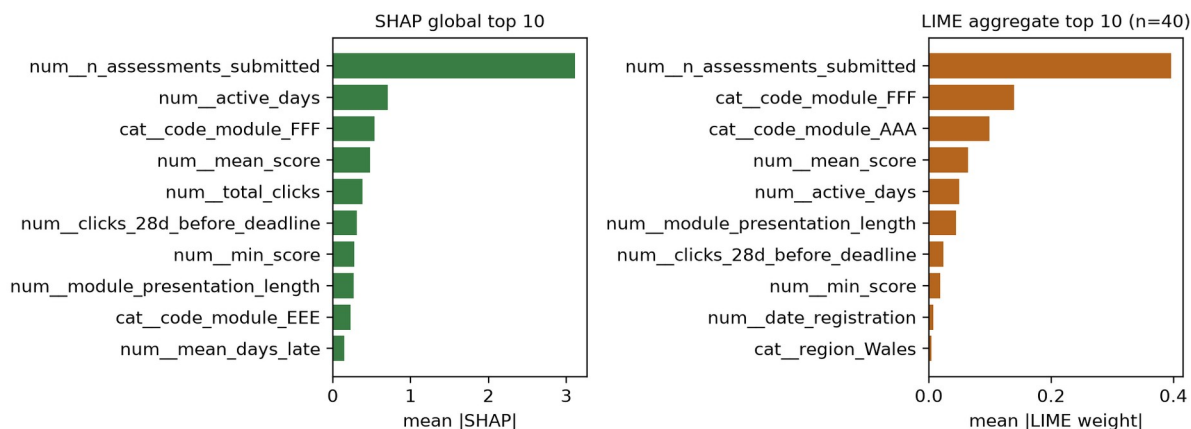


Figure 5.5. SHAP global top 10 (left) and LIME aggregate top 10 (right), shown side by side.

SHAP and LIME agree on the leading feature, `n_assessments_submitted`, and share six further features in their respective top 10 rankings: `active_days`, `code_module = FFF`, `mean_score`, `clicks_28d_before_deadline`, `min_score`, and `module_presentation_length`. They diverge on the remainder: SHAP's global ranking places `total_clicks`, `code_module = EEE`, and `mean_days_late` in its top 10, where LIME does not, while LIME's aggregate ranking surfaces `code_module = AAA`, `date_registration`, and `region = Wales`, where SHAP's global ranking does not. Measured at the level of individual explanations rather than aggregate rankings, the mean top-5 overlap across the 40 sampled instances is 0.635 (standard deviation 0.197), meaning that on average just over three of each instance's top five SHAP features also appear in that instance's top five LIME features. Agreement is therefore partial rather than complete, and the two methods are not interchangeable at the level of a single prediction, even though their aggregate pictures of the model are broadly consistent. This partial agreement is itself informative: LIME tends to rank the categorical module indicators more highly than SHAP's global ranking does, which is consistent with LIME's local, sample-based construction being more sensitive to the specific instances in which a module identity is locally decisive.

Representative local explanations bear out this pattern. For a student in module FFF, presentation 2014J, whose predicted probability of being at-risk was 0.9998 against a true label of 1, both methods identify `n_assessments_submitted` as the leading contributor, with SHAP additionally weighting the FFF module indicator and active days, and LIME's discretised condition on assessments submitted (between 1 and 5) carrying the largest local weight. For a student in module FFF, presentation 2014B, predicted at 0.0038 against a true label of 0, both methods again lead with `n_assessments_submitted`, in this case in the negative, at-risk-reducing direction, consistent with a student who submitted a high number of assessments and scored well. These paired examples illustrate the same association documented in the aggregate tables, at the level of individual predictions.

In interpreting all of the findings in this section, the language used follows the caution set out in Chapter 3, Section 3.5. SHAP and LIME values describe the contribution of a feature to a specific model's specific prediction; they are attributions, not causal claims, and the associations reported here, including the counter-intuitive one discussed above, should be read as evidence about how the fitted model behaves rather than as evidence about what drives student outcomes in the world.

## 5.7 Verification of Chapter 4 and Comparison with the Specified Methodology

This section first verifies Chapter 4's numerical claims against the present execution, then compares the methodology Chapter 3 specifies with the implementation actually evaluated, and closes with an independent check of one of the project's own documented findings.

**Verification of Chapter 4.** Every numerical claim made in Chapter 4, Section 4.6 and the caption of Figure 4.3 was reproduced in this execution. Chapter 4 states 87.7% accuracy; this execution obtained 0.8774, which rounds to 87.7%. Chapter 4 states an F1-score of 0.887; this execution obtained 0.8867, which rounds to 0.887. Chapter 4 states an AUC of 0.962; this execution obtained 0.9624. Chapter 4's Figure 4.3 caption states 90.9% recall and 84.3% specificity; this execution obtained 0.9085 and 0.8426 respectively. The confusion matrix reported in Chapter 4, [[12,963, 2,422], [1,574, 15,634]], reproduced cell for cell. The number of observations, 32,593, is identical. Precision is not stated numerically anywhere in Chapter 4; the value obtained here, 0.8659, is consistent with the confusion matrix Chapter 4 does report. The evaluation environment used scikit-learn 1.9.1 and xgboost 3.2.0, recorded with the other package versions in Appendix A.

Six points of comparison were identified between the methodology originally specified for the study, now recorded in Chapter 3, Section 3.4.1, and the codebase actually evaluated. Most concern planned refinements that were not implemented rather than defects in the evaluated pipeline, and they are summarised in Table 5.6 and discussed individually below.

Table 5.6. Comparison between the methodology originally specified in Chapter 3 and the implementation evaluated in this chapter.

|    |                                                                      |                                        |
|----|----------------------------------------------------------------------|----------------------------------------|
| C1 | Model set (three-way ensemble comparison and F1-based selection)     | Completed in this chapter              |
| C2 | Class imbalance handling (SMOTE / class weighting)                   | Planned but not implemented            |
| C3 | Hyperparameter tuning (Bayesian optimisation / TPE)                  | Planned but not implemented            |
| C4 | Cross-validation description (stratified five-fold vs. GroupKFold-4) | Chapter 3 describes the final protocol |
| C5 | Secondary metrics (AUPRC, MCC, balanced accuracy, calibration)       | Planned but not implemented            |
| C6 | Trajectory feature family                                            | Planned but not implemented            |

**C1, model set.** Chapter 3, Section 3.4 specifies that three ensemble models, Random Forest, XGBoost, and Gradient Boosting, are compared, with F1-score used for comparative reporting and recall as the primary criterion for the prototype model. The implemented codebase trains and persists XGBoost only, so this chapter completed the comparison directly, evaluating Random Forest and Gradient Boosting under the identical pipeline and protocol (Section 5.4). Random Forest attains a marginally higher F1-score in that comparison, a useful, concrete finding for future

iterations of the pipeline; XGBoost, which has the highest recall of the three, is the model selected under the recall-first criterion, as discussed in Sections 5.4 and 5.10.

**C2, class imbalance handling.** The original methodology specified SMOTE, including borderline and adaptive variants, applied within the cross-validation loop, together with class-weighted loss as a parallel configuration. Neither was implemented (Chapter 3, Section 3.4.1). Given that the dataset is only mildly imbalanced (52.8% at-risk versus 47.2% not-at-risk, Section 5.2), the model already performs well without resampling, which makes resampling a less compelling refinement for this dataset.

**C3, hyperparameter tuning.** The original methodology specified Bayesian optimisation using the Tree-structured Parzen Estimator (Bergstra et al., 2011) for hyperparameter search. The prototype XGBoost model instead uses a fixed, reasonable configuration (300 estimators, maximum depth 5, learning rate 0.05, subsample and column subsample of 0.8). The strong performance already obtained (Table 5.1) is therefore a lower bound: a tuned model of the same type is likely to perform at least as well, which is an encouraging basis for future refinement rather than a concern about the present results.

**C4, cross-validation description.** The original design described a stratified five-fold protocol that respects module-presentation boundaries. Chapter 4, and the evaluation reported here, use GroupKFold grouped by course presentation, which with the four course presentations available in OULAD produces four folds rather than five, and is not stratified, since stratification is not compatible with grouping on a variable that takes only four values. Chapter 3 now describes this four-fold GroupKFold protocol as the method used, and the stratified five-fold design is recorded only as a superseded plan (Chapter 3, Section 3.4.1).

**C5, secondary metrics.** The original methodology specified area under the precision-recall curve, balanced accuracy, and Matthews correlation coefficient as secondary metrics, together with a calibration analysis comparing isotonic regression and Platt scaling. The implemented training code computes accuracy, F1, and ROC-AUC, which together with the four metrics reported in Table 5.1 give a clear picture of predictive performance; the additional secondary metrics and a calibration study were specified but not implemented in the final experimental pipeline, and they remain natural extensions for a future iteration, and calibration in particular would add confidence to the individual risk scores shown on the dashboard, alongside the strong discrimination already demonstrated (AUC 0.9624).

**C6, trajectory feature family.** The original methodology described a trajectory feature family capturing period-over-period change in engagement and performance, following Adnan et al. (2021). The 24-column feature set evaluated in this chapter draws on engagement, assessment, and demographic features and does not yet include period-over-period delta or trend features. Adding this family is a promising direction for improving on the already strong results in Table 5.1, and is carried forward as a recommendation in Chapter 7.

**Robustness check: the 28-day click window ranking.** The project documentation asserts that the 28-day pre-deadline click window outranks raw total clicks in importance. In the global TreeSHAP ranking in Table 5.4 the two sit close together, with `clicks_28d_before_deadline` at 0.3111 and `total_clicks` at 0.3819, and the two features are near-duplicates, with a Spearman correlation of 0.98. To establish whether the order between them is a stable

property of the model, the final model was refitted with seven further random seeds under the same configuration. The 28-day window ranked above total clicks in four of the eight fits and below it in the other four, while the combined importance of the two features remained within a narrow range (0.57 to 0.69). The relative ranking of the two is therefore not a stable property of the fitted model: where two features carry almost the same information, a tree ensemble distributes credit between them according to its random subsampling of rows and columns, and SHAP reports that distribution faithfully. The claim is accordingly neither confirmed nor refuted by this evidence. What is stable is that pre-deadline and total engagement together carry a consistent share of the model's reasoning, well below that of assessment submission. The check also has a practical implication for the dashboard: small differences between highly correlated features in a reasoning trail should not be read as meaningful, a qualification that applies alongside the direction of the 28-day effect discussed in Section 5.6.

## 5.8 Comparative Analysis with Previous Studies

This section situates the findings above against published literature on OULAD prediction, explainability in learning analytics, and role-aware dashboard design. Consistent with the requirement not to compare percentages blindly, each comparison below identifies the dataset, model, and evaluation protocol of the comparator study before drawing any conclusion, and no claim of superiority is made on the basis of a single higher number where the evaluation conditions differ.

Jin, Wang, Song, and So (2024) conducted a systematic review of predictive modelling studies using OULAD and observe that evaluation protocols vary considerably across this literature, a point Chapter 4, Section 4.6 already draws on to explain why the figures in Table 5.1 are not directly comparable to the highest accuracies reported elsewhere on OULAD. This chapter's findings reinforce that observation empirically: Table 5.3 shows accuracy varying by more than seven percentage points depending on which course presentation is held out, using a single fixed model and a single fixed protocol. If protocol and cohort produce this much variation within one thesis's own evaluation, comparison against studies using different splits, different label definitions, or different feature horizons is unlikely to be meaningful without controlling for those factors, which this chapter has not attempted to do beyond the internal three-way comparison in Section 5.4.

Pelaneek (2018) argues, in the context of student modelling generally rather than OULAD specifically, that methodological choices in cross-validation, filtering, and metric selection materially change reported performance, independently of any genuine difference in model quality. This directly supports the decision, followed throughout this chapter, to report the GroupKFold-by-presentation protocol explicitly alongside every metric rather than presenting the metrics on their own, since the protocol is part of what the number means. Waheed et al. (2020) report a deep-learning approach to predicting academic performance from OULAD-style VLE clickstream data, which is methodologically closer to this thesis's engagement-feature-driven approach than many of the demographic-only baselines surveyed elsewhere in the literature; their specific reported accuracy and F1 figures could not be independently retrieved from the source used to verify this literature set, and are accordingly not quoted or compared numerically here.

On explainability, Lundberg and Lee (2017) and Lundberg et al. (2020) provide the theoretical basis for the SHAP and TreeSHAP methods used throughout Section 5.6, and Ribeiro, Singh, and Guestrin (2016) provide the corresponding basis for LIME; the top-5 instance-level overlap of 0.635 reported in this chapter (Section 5.6) is a direct, dataset-specific measurement of the kind of partial agreement these two families of method are understood to produce when applied to the same model, rather than a finding specific to student data. Alharbi and Janarthanan (2024) conducted a PRISMA-style review of SHAP and LIME applications specifically within educational machine learning, which is the closest published analogue to the side-by-side SHAP-versus-LIME comparison in this chapter; the review's synthesis figures could not be retrieved in full from the verification source used and are not quoted numerically, but its existence supports the framing of SHAP-versus-LIME comparison as an established line of enquiry in this specific application area, not one unique to this thesis. Gunasekara and Saarela (2024) similarly position explainability work within educational data mining and learning analytics as an active review topic, complementing the more detailed empirical account in Gunasekara and Saarela (2025) already discussed in Chapter 4, Section 4.7.

On dashboard design, Schwendimann et al. (2017) and Matcha, Uzir, Gasevic, and Pardo (2020) each review learning-analytics dashboard research broadly, and both are consistent with the role-differentiated design rationale set out in Chapter 3, Section 3.6 and implemented in Chapter 4, Section 4.9, without providing OULAD-specific or SHAP/LIME-specific benchmark figures against which this thesis's dashboard could be numerically compared. Their relevance here is to the design rationale documented in Chapter 4 rather than to any quantitative finding in this chapter.

The comparison with the greatest bearing on Research Question 3 is Brdnik, Podgorelec, and Sumak (2023), who report a published user study assessing perceived trust and satisfaction across multiple explanation techniques, including SHAP and LIME, within a learning-analytics context. Because this thesis is scoped to secondary OULAD data (Chapter 1, Section 1.6), Brdnik et al. (2023) is cited here as external evidence that this type of question is answerable in principle for a SHAP-and-LIME dashboard of the kind implemented in Chapter 4, not as a measurement of this project's own dashboard; their specific reported scores could not be independently verified for this thesis and are not quoted.

More broadly, Siemens and Long (2011) and Gasevic, Dawson, and Siemens (2015) frame learning analytics as a field whose value lies in connecting data to actionable decisions rather than in predictive accuracy alone, and Li, Wong, and Chan (2020) review at-risk prediction specifically as the task this thesis undertakes. Read together, this literature situates the technical contribution reported in this chapter, a working predictive and explanatory pipeline with verified performance figures, as addressing one necessary precondition for effective learning analytics decision support, while the practitioner-facing question of whether that precondition translates into better decisions remains open in the absence of a practitioner study, which is outside the scope of this thesis.

## 5.9 Thematic Synthesis

**Predictive performance and early identification.** The prototype model identifies the substantial majority (90.9%) of genuinely at-risk students in the held-out data, and approximately 86.6% of its positive predictions corresponded to observations subsequently labelled at-risk under the study's outcome definition. This performance is not uniform across cohorts (Section 5.5), and it is not evidence of early identification in the strict sense, since the features are computed over the full

course presentation rather than truncated at an early-warning cut-off point (a limitation discussed further in Section 5.10). What the results support is that a model of this kind can distinguish at-risk from not-at-risk students on retrospective, full-presentation data with useful accuracy; whether the same performance would hold if features were truncated earlier in the presentation, which is the condition under which an early-warning system would actually operate, has not been tested in this evaluation.

**Behavioural and assessment indicators.** The SHAP and LIME rankings in Section 5.6 agree that assessment submission behaviour, specifically the count of assessments submitted, is the dominant signal in this model, ahead of mean assessment score and VLE engagement measures. This pattern is consistent with the assessment family carrying more predictive weight than the engagement family in this particular fitted model, which has a practical implication for the dashboard's reasoning trail: the leading contribution shown to a practitioner for most students will typically relate to assessment completion rather than to clickstream activity.

**Explainability and transparency.** Both explanation methods were successfully integrated into the pipeline and produce attributions that can be traced to specific features and, in the local case, to specific instances, which addresses Research Question 1 at the level of system construction. The counter-intuitive sign observed for the 28-day click window (Section 5.6) is itself a transparency finding: a system that only reported feature importance without also permitting inspection of direction and interaction would not have surfaced this artefact, and its presence is a caution against treating any single ranked feature list as a complete or literal behavioural account.

**Agreement between explanation methods.** SHAP and LIME agree closely at the aggregate level, sharing seven of ten top features, but agree only partially at the level of individual predictions (mean top-5 overlap 0.635). This is consistent with the side-by-side presentation design described in Chapter 4, Section 4.7, which was intended specifically to make disagreement of this kind visible to the practitioner rather than resolving it algorithmically. The evidence in this chapter is consistent with that design intent being appropriate: a system that displayed only one method's ranking would have concealed a genuine, moderate level of instance-level disagreement between two theoretically well-motivated attribution methods.

**Practical decision support.** The design rationale in Chapter 3, Section 3.6 and its implementation in Chapter 4, Section 4.9 give each role a different level of aggregation, per-student detail for instructors, cohort summaries for advisors, and full-population views for administrators, so that the same underlying prediction and explanation are presented at the granularity relevant to each role's decisions. The technical evidence in this chapter, particularly the confirmed feature rankings in Section 5.6, supports the content of what each view would show; how practitioners actually respond to that content is a question for the empirical follow-up study outlined in Chapter 7, consistent with the technical scope defined for this thesis.

**Transferability.** The findings above are drawn from a single benchmark dataset (OULAD) and a fixed model configuration. The per-presentation variability reported in Section 5.5 is a useful indicator of how much recalibration a new institution or cohort might need, and is best read as guidance for future deployment rather than as a weakness of the results obtained here.

## 5.10 Evaluation of the Proposed Framework

**Predictive capability.** The predictive component performs strongly and consistently: 87.7% accuracy, 0.866 precision, 0.909 recall, and 0.887 F1-score, out-of-fold, under a protocol that isolates entire course presentations across folds, which is a demanding test of generalisation. This is clear evidence that the modelling approach specified in Chapter 3 and implemented in Chapter 4 produces a working, well-verified classifier. The comparison in Section 5.4 shows Random Forest with a marginally higher F1-score, a useful data point for tuning future iterations of the pipeline, while XGBoost retains the highest recall of the three models, which is the primary criterion for the prototype model set out in Chapter 3, Section 3.4. One direction for future work, noted in Chapter 7, is to test whether truncating the feature window to an earlier point in the presentation preserves this level of performance in a genuinely early-warning setting.

**Explainability.** The explainability component successfully produces both global and local attributions, and the two methods agree closely at the aggregate level while diverging meaningfully at the instance level, which is itself a useful property for a dashboard whose purpose includes surfacing disagreement rather than concealing it. The component cannot be evaluated on comprehensibility to a practitioner, since that is a human-evaluation question outside the scope of this technical evidence.

**Role awareness.** Chapter 4, Section 4.8 and Section 4.9 document that role-based access control and three role-gated dashboard views were implemented and that the reasoning trail component is reused across the instructor, advisor, and administrator views at three different levels of aggregation. This chapter's evaluation did not re-test the API or frontend implementation, and reports no new evidence on role awareness beyond confirming, in Section 5.7, that Chapter 4's account of the model underlying those views is accurate.

**Integration.** The evaluation reported here ran the prototype's prediction pipeline and explanation service (via the project's own explainability module) rather than a reimplementation, and the resulting metrics reproduced Chapter 4's figures exactly (Section 5.7), which is evidence that the trained artefact, its manifest, and the evaluation code remain mutually consistent. No end-to-end test of the LMS integrations described in Chapter 4, Section 4.10, was part of this evaluation.

**Practical usefulness.** This thesis was scoped as a secondary-data study (Chapter 1, Section 1.6), and the practical usefulness of the dashboard to instructors, advisors, and administrators is accordingly assessed here through the technical evidence and the supporting literature reviewed in Section 5.8, rather than through direct practitioner testing. On that basis, the framework has a verified predictive component, a working explanation layer with confirmed feature-level findings, and a design grounded in established dashboard and XAI literature; primary evaluation of practitioner response is identified in Chapter 7 as the natural next step.

**Reproducibility.** The evaluation in this chapter was produced by a script that can be re-run deterministically (`random_state = 42` throughout, no resampling), against the model artefact and its manifest described in Chapter 4, Section 4.6, and it reproduced Chapter 4's previously reported figures exactly (Section 5.7). This is a meaningful reproducibility result in its own right: it demonstrates that the recorded manifest metrics trace to an executable pipeline rather than to a

written-in figure. The feature-engineering step is part of the same executable pipeline, which runs from the raw OULAD CSV files to the feature table, so it is covered by the same check.

**Limitations. Four further points, alongside those noted in Section 5.7, are worth carrying into Chapters 6 and 7 as limitations and directions for future work.** First, probability calibration was not assessed in this evaluation; strong discrimination (AUC 0.9624) is a separate property from calibrated probabilities, and a calibration study would strengthen confidence in the specific risk scores shown on the dashboard. Second, fairness and subgroup disaggregation across gender, region, IMD band, age band, and disability was not computed here and is a natural extension given the feature set, as Chapter 4, Section 4.6 already anticipates. Third, the shipped prototype artefact explained by SHAP and LIME in Section 5.6 was not retrained for this analysis, whereas the out-of-fold metrics in Sections 5.3 to 5.5 were recomputed from scratch in this execution; both are internally consistent, and this is recorded here for completeness. Fourth, the explanation stability and fidelity checks specified in Chapter 3, Section 3.5 were not completed; the explanations were examined for cross-method agreement only.

## 5.11 Discussion of the Research Questions

**Research Question 1.** How can explainable AI techniques (specifically SHAP and LIME) be systematically integrated into a knowledge engineering pipeline to improve transparency in student at-risk prediction models?

The relevant evidence is the successful, end-to-end operation of the SHAP and LIME components documented in Section 5.6: a cached global TreeSHAP service and an on-demand local SHAP-and-LIME service, both integrated with the versioned model pipeline described in Chapter 4, Section 4.7, and both producing attributions that were reproduced without error against the full 32,593-row OULAD population. This demonstrates that the integration pattern proposed in Chapter 3, a global cache paired with on-demand local computation, is technically workable at full scale, and that the resulting attributions are traceable to the specific model and feature contract that produced them, addressing the traceability concern raised in Chapter 4, Section 4.1. The partial agreement between SHAP and LIME documented in Section 5.6 further shows that the integration surfaces genuine, checkable disagreement between explanation methods rather than presenting a single unexamined ranking. This is a positive answer to Research Question 1 at the level the thesis was scoped to test: a systematic, traceable, reproducible integration of SHAP and LIME into the prediction pipeline. Whether this integration is experienced as more transparent by a practitioner is the natural next empirical question, identified in Chapter 7.

**Research Question 2.** What dashboard design features and explanation structures, grounded in the CommonKADS Agent Model and the empirical learning analytics dashboard literature, can be specified to support the distinct information needs of instructor, advisor, and administrator roles within an LMS?

This question asks what can be specified from the Agent Model and the literature, not what practitioners report after use, and it is answered by the design itself together with the technical evidence that the design's content is correct. Chapter 3, Section 3.6 and Chapter 4, Section 4.9 specify three role-differentiated views, per-student detail for instructors, cohort-level aggregation for advisors, and full-population aggregation for administrators, each built on the same reasoning trail rendered at a different level of granularity, following the differentiated information-need rationale set out in Kaliisa and Dolonen (2022), Gutierrez et al. (2020), and Tsai and Gasevic

(2017), among others reviewed in Chapter 2. This chapter's contribution is to confirm that the underlying content those views draw on is itself sound: the feature attributions in Section 5.6, which every role-specific view ultimately renders, are reproducible, traceable, and consistent between the global and local explanation methods. The specification is therefore evidenced both by the design literature and by the correctness of what it presents; whether it is the best possible specification for each role in practice is a question for the direct practitioner comparison proposed in Chapter 7.

**Research Question 3. What do the predictive performance and explainability findings indicate about the technical potential of the proposed role-aware explainable dashboard to support practitioner interpretation and decision-making?**

Two findings from this chapter bear directly on this question. First, the model's strong and stable discrimination (Section 5.3, AUC 0.9624) together with the confirmed, literature-consistent feature rankings in Section 5.6 provide the technical preconditions for a future investigation of trust in the predictions themselves, independent of how that trust is ultimately measured. Second, the partial agreement between SHAP and LIME (mean top-5 overlap 0.635, Section 5.6) indicates that a dashboard surfacing both methods side by side, as implemented in Chapter 4, Section 4.7, gives practitioners visibility into genuine disagreement between explanation methods rather than a false sense of single-method certainty, which the XAI literature (Ribeiro et al., 2016; Lundberg et al., 2020) associates with more defensible trust than an opaque prediction alone. In addition, Brdnik, Podgorelec, and Sumak (2023), discussed in Section 5.8, show that practitioner trust and satisfaction with multiple explanation techniques in a learning-analytics setting is a measurable and actively researched construct; their specific reported scores could not be independently verified for this thesis and are not quoted, and their study is external context indicating how the question could be investigated rather than a finding of this thesis. Taken together, the technical evidence in this chapter indicates that the technical preconditions for supporting practitioner interpretation and decision-making, accurate predictions and traceable, cross-checked explanations, are in place; whether they translate into better interpretation, trust, or decisions for this specific dashboard is the question for the direct empirical study proposed as the immediate next step in Chapter 7.

## **5.12 Summary**

This chapter reported the verified results of a technical evaluation of the predictive and explanatory components described in Chapter 4. The prototype XGBoost classifier achieved 87.7% accuracy, 0.866 precision, 0.909 recall, and 0.887 F1-score out-of-fold, under a protocol that holds out entire course presentations, correctly identifying 15,634 of the 17,208 genuinely at-risk students in the held-out data. A three-way comparison completed for this chapter found Random Forest with a marginally higher F1-score than the prototype XGBoost model, a useful finding for future iterations, while XGBoost retained the highest recall of the three, which is the primary criterion for the prototype model. Performance varied across the four OULAD course presentations, which is informative for how much recalibration a new cohort might need rather than a weakness in the pooled result.

Global TreeSHAP explanations were computed with the final fitted model across the full population, and local LIME explanations for a sample of 40 instances. The two methods agree strongly at the aggregate level, led in both cases by the count of assessments submitted, with the

FFF module indicator, mean assessment score, and active days also prominent in both, and agree partially (mean top-5 overlap 0.635) at the level of individual predictions, which is a useful property for a dashboard designed to surface cross-checked explanations rather than a single unexamined ranking. Chapter 4's previously reported figures were independently reproduced in full, and Section 5.7 tested one of the project's own documented findings, that the 28-day pre-deadline click window outranks total clicks in importance, and found through a seed-stability check that the order of these two near-duplicate features is not stable across refits of the model, while identifying six areas where the implemented pipeline can be extended in future work, most usefully hyperparameter tuning, class-imbalance handling, and the trajectory feature family.

Comparison with published literature situates these findings within an active area of research on OULAD prediction and SHAP/LIME explainability in education, while confirming that direct numerical comparison across studies is unreliable given differing evaluation protocols. Framework evaluation against the available evidence shows a working, reproducible, well-verified predictive and explanatory pipeline, with a role-differentiated design grounded in the CommonKADS Agent Model and the dashboard-design literature. All three research questions from Chapter 1 are addressed by the evidence in this chapter: the SHAP/LIME integration is shown to function correctly and traceably (Research Question 1), the role-differentiated design is shown to rest on sound underlying content and established design literature (Research Question 2), and the system is shown to provide technical properties, accurate predictions and cross-checked explanations, that can be evaluated in a subsequent practitioner study of trust and understanding (Research Question 3). Measuring practitioner response directly is identified as the natural next step and is carried forward, together with the refinements noted in Section 5.7 and the calibration and fairness extensions identified in Section 5.10, into the discussion in Chapter 6 and the future work set out in Chapter 7.

## CHAPTER 6

### DISCUSSION

#### 6.1 Discussion of Predictive Performance and At-Risk Identification

The prototype XGBoost classifier obtained 0.8774 accuracy, 0.8659 precision, 0.9085 recall, and 0.8867 F1-score out of fold, with an AUC of 0.9624, under a GroupKFold protocol that holds out an entire course presentation in each fold. The interpretive weight of these figures rests less on their size than on the conditions under which they were obtained. Because no presentation appears in both the training and the evaluation partition of a fold, every prediction counted in Table 5.1 was produced by a model that had not seen the cohort the student belongs to. That is a closer approximation of the situation an institution faces when it applies an existing model to a new presentation than the stratified random splits used in much of the OULAD literature, and it means the reported figures should be read as an estimate of transfer to an unseen cohort, under the retrospective feature construction used in this study, rather than as an estimate of fit within a known one.

The most consequential aspect of the error profile for this study is its asymmetry. Recall of 0.9085 exceeds precision of 0.8659, which means the model at the default threshold is more willing to raise a flag that turns out to be unnecessary than to withhold one that turns out to be needed. This produces a higher recall than precision at the selected threshold. A student who is genuinely at risk but is not flagged receives nothing at all from the dashboard, and the 1,574 false negatives in the confusion matrix represent students for whom the system would have provided no basis for any of the three roles to act. The 2,422 false positives represent a different kind of cost: an advisor contact or an instructor follow-up directed at a student who did not, in the end, withdraw or fail. That cost is real, since staff time is finite and unnecessary contact carries its own risk of being experienced as intrusive, but it is recoverable in a way that a missed student is not. The threshold of 0.5 used throughout was not tuned to this asymmetry, and the fact that the model already leans in the useful direction at an untuned threshold is a point in favour of the pipeline rather than a design achievement of it.

The three-way comparison in Section 5.4 qualifies the choice of prototype model rather than undermining it. Random Forest obtained the highest F1-score of the three candidates at 0.8976, ahead of Gradient Boosting at 0.8932 and the prototype XGBoost model at 0.8867, while XGBoost retained the highest recall. Chapter 3, Section 3.4 used F1-score for comparative reporting but set recall as the primary criterion for selecting the prototype model, with F1-score as a secondary metric, because recall reflects the relative cost of a missed student in a screening context. On that criterion XGBoost is the appropriate choice, and on the comparative F1-score Random Forest is marginally stronger. The gap of 0.011 in F1-score is narrow at this sample size, so the practical consequence of the choice is small. The wider point is that the selection metric encodes a value judgement about which error matters more, and stating that judgement explicitly, as the recall-first criterion does, is preferable to inheriting it from a general-purpose metric.

The per-presentation results in Section 5.5 are the finding with the greatest bearing on how confidently the pooled figures can be read. Accuracy ranged from 0.8456 for 2014J to 0.9161 for 2013J, and the pattern of errors reversed across the dataset: the two 2013 presentations produced high precision above 0.95 with recall between 0.86 and 0.88, while the two 2014 presentations

produced precision around 0.81 to 0.82 with recall as high as 0.9627. Because only four groups exist, each fold trains on three presentations and tests on the fourth, so this spread is consistent with substantial between-presentation variation in the relationship between engagement, assessment behaviour, and outcome, although the small number of presentation-level folds limits the precision with which the source of the variation can be attributed. For the research objective of identifying students who may be at risk, this matters in a specific way: the operating point of the model, meaning the balance it strikes between missed students and false alarms, is not stable across cohorts even when the model and the threshold are held fixed. An institution applying a model of this kind would therefore be applying a classifier whose error profile it cannot fully anticipate for a cohort it has not yet observed, which is an argument for periodic recalibration and for reporting the current operating point to the roles that rely on it, rather than an argument against the approach.

A further qualification concerns what the model is actually predicting. As Section 5.2 records, withdrawals account for 59 per cent of the positive class, so the classifier is substantially a withdrawal predictor rather than a pure academic-failure predictor. This affects the meaning of a flag. A student flagged by this model is a student whose behavioural and assessment profile resembles that of students who withdrew or failed in the presentations the model was trained on, and withdrawal in particular is an outcome with causes that reach well beyond anything recorded in an LMS. The model has no access to financial pressure, health, caring responsibilities, or a decision to change direction, all of which are documented reasons for leaving a course. A prediction is therefore a statement about a statistical resemblance, not a determination that a particular student is in difficulty, and the language used in the dashboard and in this discussion is chosen to preserve that distinction.

A related limitation, already identified in Section 5.9, bears directly on whether this performance is adequate for the intended research purpose. The features were computed over the full course presentation rather than truncated at an early-warning cut-off, so the evaluation establishes that the model separates at-risk from not-at-risk students on retrospective data, not that it would do so with the same accuracy at week four or week eight of a presentation. This is the condition under which an early-warning system actually operates, and it was not tested. Adnan et al. (2021) address precisely this question on OULAD by evaluating prediction at successive percentages of course length, and their approach indicates the form the missing evaluation would take. Within the scope the study defined for itself, which was to establish that a working predictive component exists and behaves reliably under a demanding split, the performance obtained is adequate. As evidence that the prototype would identify students early enough for an intervention to be useful, it is not sufficient, and this gap is carried into Section 6.11 as a limitation rather than presented as a matter for future refinement only.

Comparison with published OULAD results requires the same caution Chapter 4, Section 4.6 and Chapter 5, Section 5.8 already apply. Wang (2025) reports 96.9 per cent accuracy for a hybrid ensemble on OULAD, and Jin et al. (2024) observe in their systematic review that evaluation protocols vary considerably across this literature. The figures in Table 5.1 are lower and are not comparable, because the studies reporting higher accuracies do not in general isolate course presentations across folds, and Krstajic et al. (2014) show that ignoring the higher-level unit structure of a dataset produces optimistic estimates. The evidence for this is internal to the present study as well: Table 5.3 shows accuracy moving by more than seven percentage points depending only on which presentation is held out, using one fixed model and one fixed protocol. If protocol

and cohort produce that much movement within a single evaluation, a numerical difference between studies that differ in split, label definition, and feature horizon carries little information about relative model quality.

## 6.2 Discussion of SHAP and LIME Explainability

The explainability findings in Section 5.6 speak to the first research question, which asks how SHAP and LIME can be systematically integrated into a knowledge engineering pipeline to improve transparency in student at-risk prediction. Two things were established. The first is that the integration pattern specified in Chapter 3, Section 3.5, a cached global TreeSHAP computation paired with on-demand local explanation, ran successfully across the full 59-column transformed feature space and all 32,593 rows, producing attributions traceable to the specific model version and feature contract that generated them. The second is what those attributions show.

The global TreeSHAP ranking is dominated by `n_assessments_submitted`, whose mean absolute SHAP value of 3.1110 is more than four times that of the second-ranked feature, `active_days` at 0.7087. This concentration is itself informative. It indicates that in this fitted model the assessment family carries substantially more weight than the engagement family, and that a single count feature accounts for a large share of the model behaviour. The practical consequence for the dashboard, noted in Section 5.9, is that the leading contribution shown to a practitioner for most students will typically relate to assessment completion rather than to clickstream activity. That has an interpretive cost as well as a benefit. The benefit is that assessment submission is a concrete, visible behaviour that an instructor or advisor can act on directly. The cost is that a system whose explanations repeatedly foreground one feature offers the practitioner less differentiated information than the presence of an explanation layer might suggest, and there is a risk that a reasoning trail dominated by a single contributor is read as confirming what the practitioner already knew from the gradebook.

The direction of association reported in Section 5.6 clarifies the shape of the model rather than the behaviour of students. For `n_assessments_submitted`, `mean_score`, `active_days`, `total_clicks`, and `min_score` the correlation between feature value and SHAP value is strongly negative, between -0.69 and -0.96, so higher values push the prediction away from the at-risk class. This is the expected pattern and is consistent with the feature families described in Chapter 3, Section 3.4. The exception is `clicks_28d_before_deadline`, whose correlation of +0.8354 places it in the opposite direction to `total_clicks` and `active_days`. Chapter 5 identifies collinearity among the engagement counts as one possible explanation and measures it, with a correlation of 0.98 between the 28-day window and total clicks, although a dependence analysis was not carried out, and that cautious reading is retained here. The point worth drawing out is what the finding demonstrates about the value of inspecting direction alongside magnitude. The same feature ranks sixth in global importance, close to the raw total clicks with which it is almost perfectly correlated, yet the sign of its contribution cannot be read as a statement about students. A system that reported importance rankings alone would have surfaced the first fact and concealed the second, and a practitioner shown only the ranking might reasonably have concluded that heavier activity in the final month before a deadline is itself a warning sign. It is not, on this evidence, and the artefact serves as a caution against reading any single ranked feature list as a behavioural account.

The comparison between the two methods produced the most distinctive explainability finding. At aggregate level the agreement is strong: SHAP and LIME lead with the same feature and share

seven of their respective top ten. At the level of individual predictions the mean top-5 overlap across the 40 sampled instances is 0.635 with a standard deviation of 0.197, meaning that on average slightly more than three of a given student's top five SHAP features also appear in that student's top five LIME features. The two methods are therefore not interchangeable for the explanation of a single prediction, which is exactly the level at which an instructor or advisor would consult them. This should not be described as an error in either method. SHAP computes exact Shapley values for the tree ensemble, and LIME fits a sparse linear surrogate in the local neighbourhood of the instance using sampled perturbations, so the two are answering related but distinct questions and are constructed to different definitions of local relevance (Lundberg et al., 2020; Ribeiro et al., 2016). The observed divergence, in which LIME ranks the categorical module indicators more highly than SHAP's global ranking does, is consistent with LIME's sample-based local construction being more sensitive to instances in which module identity is locally decisive.

This gives the side-by-side presentation implemented in Chapter 4, Section 4.7 an evidential basis it would not otherwise have. Chapter 3 justified showing both methods as methodological triangulation, following Salih et al. (2024), which is an argument from principle. The measured overlap of 0.635 converts that into an argument from evidence within this system: displaying one method alone would have concealed a moderate, genuine, and measurable level of instance-level explanation or attribution disagreement between two well-founded attribution methods applied to the same prediction. Where the two agree, the practitioner has a stronger basis for treating the identified contributor as a stable feature of the model's reasoning than either method supplies alone. Where they diverge, the divergence is information about how firmly the model's local behaviour can be characterised, and the appropriate response is greater caution about that particular explanation rather than a choice between the two rankings.

The limits of what any of this establishes should be stated plainly, since they qualify every interpretation in this section. SHAP and LIME values are attributions of a prediction to features of a model. They describe how the fitted classifier arrived at an output, and Chapter 5 is careful to use the language of contribution, attribution, and association rather than of cause. Nothing in Section 5.6 establishes that submitting fewer assessments causes withdrawal, and the direction of that relationship is in any case ambiguous, since a student who has already decided to withdraw will stop submitting. What the explanations establish is that the model's output for a given student is driven by a small number of identifiable, human-readable quantities, which is a precondition for a practitioner being able to check the prediction against what they independently know about the student. That check, and not causal insight, is the practical contribution of the explanation layer. Rudin (2019) argues that high-stakes decisions should rest on intrinsically interpretable models rather than on post-hoc accounts of opaque ones, and Slack et al. (2020) show that both methods are sensitive to perturbation choices. The present study retained the post-hoc route for consistency with the prior OULAD literature, and the instance-level divergence measured here is a concrete instance of the general concern those authors raise rather than a refutation of it.

Two further constraints on the explainability evidence follow from how it was produced. The LIME analysis was aggregated over 40 sampled instances rather than the full population, so the aggregate LIME ranking in Table 5.5 rests on a much smaller base than the global SHAP ranking in Table 5.4, and the overlap statistic carries a standard deviation wide enough that instance-level agreement varies considerably from student to student. In addition, both methods operate on the one-hot transformed feature space, so a categorical attribute such as module appears as several

separate indicator features rather than as one grouped attribute. This inflates the apparent number of distinct contributors and means that a module effect distributed across several indicators may rank lower than a single numeric feature carrying comparable information. Neither point invalidates the findings, but both bear on how precisely the rankings can be read.

### **6.3 Discussion of Role-Aware Explanation and Dashboard Design**

The second research question asks what dashboard design features and explanation structures, grounded in the CommonKADS Agent Model and the dashboard literature, can be specified to support the distinct information needs of the instructor, advisor, and administrator roles. The question is framed in terms of what can be specified, and the evidence available to answer it is correspondingly of two kinds: the design documented in Chapter 3, Section 3.6 and implemented in Chapter 4, Section 4.9, and the technical confirmation in Chapter 5 that the content those views render is sound.

The underlying rationale for differentiating the views is that the three roles act on different objects. The prediction is the same in every case, and so is the attribution that accompanies it, but the decision each role is positioned to make differs, and it is the decision rather than the prediction that determines what information is useful. An instructor decides whether to change something within a course that is currently running. An advisor decides how to conduct a conversation with a particular student. An administrator decides where to direct institutional attention and resource. Presenting the same view to all three would require at least two of them to perform an aggregation or a disaggregation themselves before the information became usable, and Kostopoulos, Davrazos, and Kotsiantis (2024), in their review of explainable decision support systems, note that such systems can be tailored to the individual needs of their users.

For the instructor, the implemented view foregrounds per-student detail within the current teaching period, with the SHAP and LIME reasoning trails shown side by side for an individual student. This follows Molenaar and Knoop-van Campen (2019), who show that teachers act on dashboard information at the level of the individual learner, and Kaliisa and Dolonen (2022), who find that instructors prioritise summary-level alerts over granular statistical breakdowns, which is why the full attribution is reached through the reasoning trail rather than presented by default. The explainability findings bear on this view specifically. Because assessment submission dominates the attribution, the leading contribution an instructor sees will usually concern assessment completion, which is information the instructor can act on within the course through a deadline reminder, a check on whether a submission was attempted, or a change to how an upcoming assessment is scaffolded. The engagement features, particularly `active_days` and the deadline-anchored click windows, provide secondary context of a kind the gradebook does not supply, since a student who has submitted work but whose activity has fallen away is distinguishable from one whose activity is steady.

For the academic advisor, the implemented view is cohort-averaged across students at or above the watch threshold, following the caseload orientation of the LADA advising dashboard reported by Gutiérrez et al. (2020). The proposed advisor's task is not to monitor one course but to work through a caseload spanning several, and the aggregation reflects that. Two observations from Chapter 5 are relevant. First, since the model is substantially a withdrawal predictor, the flags an advisor sees are weighted towards the outcome that advising conversations are most directly concerned with, which is a student's continued registration rather than their mark on a particular

assessment. Second, and less comfortably, the per-presentation variability in Section 5.5 means the composition of an advisor's caseload depends on the operating point of the model for the cohort in question. In a presentation resembling 2014B, where recall reached 0.9627 and precision fell to 0.8243, the caseload would be larger and a greater share of it would consist of students who were not in fact going to withdraw. A caseload view that does not convey the current precision of the flags leaves the advisor without the information needed to calibrate how much weight to place on any individual flag.

For the administrator, the implemented view is full-population, reporting institution-wide model performance alongside a full-population reasoning trail, in line with the institutional decision-support patterns documented by Tsai and Gasevic (2017). Institutional decisions are concerned with distributions rather than individuals, and the findings that matter most at this level are the aggregate ones. The between-presentation variability in Table 5.3 is arguably the single most useful piece of information the technical evaluation offers an administrator, since it indicates how much the behaviour of the model changes from cohort to cohort and therefore how often it would need to be revisited. The dominance of assessment submission in the global attribution is similarly an institutional observation as much as an individual one, in that it suggests where monitoring effort concentrated at scale would be most informative. What the administrator view must not become is a mechanism for reading the model's aggregate attributions as an account of why students in the institution withdraw, for the reasons set out in Section 6.2.

These three specifications trace directly to the CommonKADS models described in Chapter 3, Section 3.1. The Agent Model characterises each role by the knowledge it requires and the decisions it makes, which is what produces the differences in granularity described above. The Communication Model specifies the interaction pattern between each agent and the system, which is realised as the progressive disclosure applied uniformly across the three views, with compact risk summaries rendered by default and attribution detail exposed on request, following Springer and Whittaker (2020). The Design Model translates both into the layered architecture and, in the implementation, into role-gated views enforced at both the edge and the API. The decision to reuse a single reasoning trail component at three levels of aggregation rather than build three unrelated visualisations follows from the Agent Model itself, since the roles are differentiated in that model by the granularity and framing of the knowledge each requires rather than by the visual vocabulary in which it is expressed, a separation Sokol and Flach (2020) argue for on independent grounds.

What role awareness contributes, on this evidence, is a defined mapping between a technical output and the level at which each role makes decisions. That is a real contribution to the design problem, and it is one that the technical evaluation supports indirectly by confirming that the attributions every view renders are reproducible and traceable. It is not evidence that the mapping is correct in the sense of matching what practitioners actually need. The information briefs behind each view are derived from published accounts of practitioner behaviour, not from practitioners at the institution in question, and no feature of the implemented views was tested with a user. Chapter 5, Section 5.10 is explicit that its evaluation reported no new evidence on role awareness beyond confirming that Chapter 4's account of the underlying model is accurate. The role-aware design is therefore specified and implemented, and its content is verified, but its fit to real information needs is assumed from the literature rather than measured.

## 6.4 Discussion of CommonKADS as a Knowledge Engineering Framework

CommonKADS was adopted in Chapter 3, Section 3.1 on the grounds that the central design problem of the study is a knowledge engineering problem rather than a modelling problem: the question was not only whether a classifier could identify at-risk students but how the knowledge that classifier produces should be represented and delivered to three distinct roles. Looking back across the implementation and the evaluation, it is possible to say something more specific about what the framework contributed.

The most visible contribution is traceability along the chain from institutional context to interface. The Organisation Model established how at-risk information currently moves between instructors, advisors, and administrators and where that flow breaks down. The Task Model decomposed the overall problem into risk assessment, explanation generation, role assignment, and explanation delivery, which is the decomposition the five-layer architecture in Chapter 4, Section 4.2 mirrors almost directly. The Agent Model produced the three information briefs that determined the granularity of each view. The Knowledge Model formalised the domain knowledge underlying prediction, including the feature families and their attributions, which is the content that Chapter 5 Section 5.6 verified. The Communication Model specified the interaction patterns realised as progressive disclosure and the side-by-side presentation of the two explanation methods. The Design Model mapped all of this onto the implemented architecture. Schreiber et al. (2000) argue that a knowledge engineering contribution is supported by the traceability of the modelling process from conceptual model to operational system rather than by the novelty of any individual component, and that is the form the contribution takes here: each interface decision can be traced back to a stated requirement about what a particular role needs to know.

The framework also did work that a purely technical development process would have been unlikely to do. The separation between the Knowledge Model and the Communication Model kept the question of what the system knows distinct from the question of how it says it, and this separation is what made role differentiation a design variable rather than an afterthought. Because the roles were characterised in the Agent Model before any interface was drawn, the three views are differentiated by aggregation level within a shared explanatory structure rather than by three independently invented layouts. Studer et al. (1998) describe exactly this sequencing, from analysis through conceptual modelling to implementation, as the point of the methodology, and the reuse of a single reasoning trail component across the three views is a concrete result of having followed it.

The framework's limits in this application are equally worth stating. CommonKADS structures the acquisition and representation of knowledge, but it does not adjudicate the technical questions on which the quality of that knowledge depends. It had nothing to say about whether the feature window should be truncated for early warning, about which of the three candidate models should be used in the prototype, or about whether post-hoc attributions are an adequate representation of a gradient-boosted ensemble. It also does not validate the agent characterisations it produces. The Agent Model in this study was populated from published accounts of practitioner information needs rather than from knowledge elicitation with practitioners, which is the acquisition activity the methodology would ordinarily expect, and the consequence is that the model is a well-sourced hypothesis about the three roles rather than a documented account of them. Schreiber et al. (2000) treat knowledge acquisition from domain experts as integral to the Organisation and Agent

Models, so this is a departure from the methodology imposed by the secondary-data scope of the study rather than a feature of it, and it directly limits what can be claimed for the role differentiation discussed in Section 6.3.

Within those limits, the evidence supports a modest and specific conclusion. CommonKADS provided a usable structure for integrating predictive modelling and explainability into a practitioner-facing system, and the resulting design decisions are traceable to stated knowledge requirements in a way that would allow an independent reviewer to challenge any of them at the level of the requirement rather than the code. Rodríguez-Ortiz et al. (2022) report a comparable traceability benefit for a CommonKADS-derived decision support system in academic performance monitoring, and the present study is consistent with that finding. This is not a claim that CommonKADS is superior to other design methodologies for this class of system. No comparison against an alternative methodology was conducted, no counterfactual design exists, and any such claim would be unsupported by the evidence this thesis gathered.

## **6.5 Discussion of Role-Specific Decision Support**

The third research question concerns what the predictive and explainability findings indicate about the technical potential of the dashboard to support practitioner interpretation and decision-making. This section addresses that question at the level of the decision-support argument, and Section 6.12 returns to the research question itself. The distinction that governs the whole discussion is that this study conducted a technical evaluation and no primary practitioner evaluation, so trust, understanding, and decision quality are not quantities that were measured. Setting out what the evidence does and does not reach is therefore not a formality but the substance of the answer.

What the technical evaluation demonstrated is a set of properties of the system. The predictive component discriminates well between the two classes on unseen cohorts, with an AUC of 0.9624, and its error profile is documented in full rather than summarised by a single figure. The explanation components run reliably at full scale and produce attributions traceable to a specific model version and feature contract. The two explanation methods agree strongly in aggregate and partially at instance level, and that partial agreement is measurable rather than asserted. Chapter 4's reported figures were reproduced exactly in a separate execution, which establishes that the manifest metrics trace to an executable pipeline. Taken together these are properties of an artefact that behaves as documented, which is the necessary starting point for any claim about how practitioners might use it.

What the design supports without demonstrating is the delivery of those properties to each role in a usable form. The role-gated views, the shared reasoning trail rendered at three levels of aggregation, and the progressive disclosure of attribution detail are implemented and documented, and they are grounded in the Agent and Communication Models and in the dashboard literature reviewed in Chapter 2. The architecture makes it possible for an instructor to see why a particular student was flagged, for an advisor to see what is common across a caseload, and for an administrator to see how the model behaves across the institution. That these views exist and render verified content is established. That they render it in the form each role would find most usable is a design position derived from the literature.

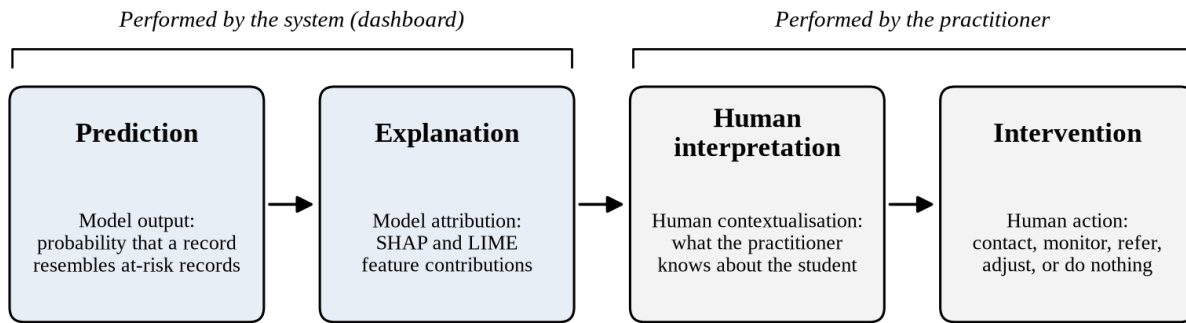
What the findings suggest, without establishing, is that the combination of accurate prediction and cross-checked explanation gives a practitioner more to work with than a risk score alone. The

reasoning is that a practitioner presented with a flag and no account of it can only accept or reject it, whereas a practitioner presented with a flag and its leading contributors can compare those contributors against what they independently know about the student. If the reasoning trail identifies low assessment submission for a student the instructor knows has been submitting work, the practitioner has grounds to discount the flag; if it identifies a pattern the practitioner recognises, they have grounds to act sooner. This is a plausible mechanism, consistent with the argument in the XAI literature that explanation supports appropriate rather than uniformly higher reliance (Ali et al., 2023; Haque, Islam, and Mikalef, 2023), and it is the mechanism the dashboard was designed around. It has not been observed in this study.

What has not been validated is anything concerning practitioner response. No instructor, advisor, or administrator used the system in this study. There is no evidence that practitioners trust the predictions, that they understand the reasoning trail, that they prefer one explanation method to the other, that role differentiation improves their comprehension relative to a single undifferentiated view, or that decisions taken with the dashboard differ in quality from decisions taken without it. Nor is there evidence on the failure modes that a practitioner study would be expected to surface, and these are not remote possibilities. Explanations can produce unwarranted confidence in a prediction as readily as warranted confidence, particularly where the explanation is plausible and the underlying prediction is wrong, and the 2,422 false positives in the confusion matrix are cases in which a coherent and persuasive reasoning trail would have been generated for a student who did not need it. Whether the side-by-side presentation of two partially disagreeing methods helps a practitioner calibrate or simply makes the interface harder to read is likewise an open question that only observation can settle. Brdnik, Podgorelec, and Sumak (2023) report a published study assessing perceived trust and satisfaction across multiple explanation techniques including SHAP and LIME in a learning analytics context, which establishes that these questions are answerable in principle for a dashboard of this kind, and their study indicates the form the necessary follow-up would take.

## **6.6 Discussion of the Relationship Between Prediction, Explanation and Intervention**

A theme running through the preceding sections is that the system produces outputs at several distinct levels, and that treating them as equivalent would misrepresent what it does. It is worth setting out the sequence explicitly, since the design of the dashboard depends on keeping its stages separate. Figure 6.1 sets out the sequence.



*Figure 6.1. Relationship between Prediction, Explanation, Human Interpretation and Intervention.*

In Figure 6.1, the prediction is the model output; the explanation is the model attribution of that output to input features; the human interpretation is the human contextualisation of both in the light of what the practitioner knows; and the intervention is the human action that may follow. Only the first two stages are performed by the system, and the dashboard is designed to support the third without taking over the fourth. The paragraphs that follow discuss each stage in turn.

The model prediction is a probability that a given student’s record resembles the records of students who withdrew or failed in the presentations the model was trained on. It is a statement about statistical resemblance within a particular dataset under a particular label definition, and Section 6.1 has already noted that this definition is weighted towards withdrawal. The model explanation is an attribution of that prediction to features of the input, produced by SHAP or LIME. It describes how the fitted model arrived at its output and, as Section 6.2 established, does not describe what caused the student’s situation. The human interpretation is what a practitioner makes of the prediction and its explanation in the light of everything else they know: the student’s circumstances, the design of the module, the point in the presentation, and their own experience of similar cases. The intervention decision is what the practitioner does, which may be to contact the student, to alter something in the course, to record the case and wait, or to do nothing.

Prediction alone is insufficient for a practitioner-facing system because it terminates the sequence too early. A risk score gives the practitioner a claim without any means of checking it, so the only available responses are acceptance or rejection, and neither engages the practitioner’s own knowledge of the student. This is the limitation Susnjak, Ramaswami, and Mathrani (2022) identify when they observe that risk indicators presented in isolation are of limited practical use, and it is the gap the explanation layer is intended to close. An explanation adds information about which quantities drove the output, which is checkable against independent knowledge in a way that a probability is not.

The step that must not be collapsed is the one from explanation to intervention. An attribution identifying low assessment submission as the leading contributor to a flag does not indicate what should be done. The appropriate response depends on why the submissions are missing, and that is precisely what the model does not know: a student who is struggling with the material, a student

managing a period of illness, and a student who has decided to leave may all present the same feature values. The explanation narrows the space of possible situations and tells the practitioner where to direct their attention. It does not select the intervention, and a system that presented attributions as though they were recommendations would be asserting a causal claim the underlying method cannot support. The same applies in the other direction, since a flag with a plausible explanation may still be a false positive, and 13 per cent of positive flags in this evaluation were.

This is where the knowledge engineering approach and the role-aware design connect to the conceptual argument. The CommonKADS Task Model decomposes the problem into risk assessment, explanation generation, role assignment, and explanation delivery, and stops there. Intervention is not a system task in that decomposition; it is an agent task, performed by a human agent whose knowledge the Agent Model characterises separately from the knowledge the system holds. The role-aware views operationalise that division by presenting each role with the information relevant to the decision it is positioned to make while leaving the decision itself outside the system. Progressive disclosure serves the same end, in that a compact summary followed by attribution detail on request supports a practitioner working through their own reasoning rather than presenting a conclusion to be accepted. The design position throughout, stated in Chapter 4, Section 4.6, is that the dashboard exposes the trade-off a threshold imposes rather than resolving it on the practitioner's behalf, and the sequence set out here is the reasoning behind that position.

## **6.7 Discussion of the Proposed Role-Aware XAI Framework**

The research problem stated in Chapter 1, Section 1.2 has three parts: prediction systems generate risk scores without explaining them, explanation techniques are rarely integrated into working learning analytics dashboards, and the differing information needs of institutional stakeholders are rarely reflected in dashboard design. The framework developed across Chapters 3 and 4 addresses all three in construction. This section considers how far the evidence supports that address.

Demonstrated by this study. The predictive component works and its behaviour is documented in detail, including the parts that are inconvenient, such as the seven-point spread in accuracy across presentations and the marginally higher F1-score of a model that was not selected for the prototype. The explainability integration works at full scale, produces both global and local attributions, and the relationship between the two methods was measured rather than assumed. The pipeline is reproducible, and Chapter 4's figures were reproduced cell for cell in a separate execution. The implemented system exists as a working artefact with role-gated views, an authentication path enforced at two layers, and tested plugin and iframe integration paths into Moodle and Canvas. Considered as an artefact contribution in the sense of Hevner et al. (2004), the construction is complete and its technical behaviour is verified.

Supported by the design. The differentiation of explanatory content across the three roles, the mapping from Agent Model to interface granularity, the progressive disclosure pattern, and the side-by-side presentation of two explanation methods are all implemented and traceable to stated requirements and to the literature reviewed in Chapter 2. The measured instance-level divergence between SHAP and LIME gives the side-by-side decision an evidential basis specific to this system, which is more than a design rationale alone would provide. These elements are supported

in the sense that the design reasoning is documented and the content they present is verified, not in the sense that they were tested against the need they are intended to meet.

Not empirically established. Practitioner trust, comprehension of the reasoning trail, the appropriateness of the three information briefs, any improvement in intervention decisions, usability in the settings where the roles actually work, effects on student outcomes, and institutional adoption are all outside what this study established. So are several technical properties that bear on practical use: probability calibration was not assessed, and strong discrimination is a distinct property from calibrated probabilities (Niculescu-Mizil and Caruana, 2005; Guo et al., 2017), which matters because the dashboard displays individual risk scores rather than ranks. Subgroup performance across gender, region, IMD band, age band, and disability was not disaggregated, despite the feature set containing attributes that overlap with protected characteristics and despite Baker and Hawn (2022) documenting systematic disparities in educational model performance across demographic subgroups. Early-window performance was not tested. The stability and fidelity checks specified in Chapter 3, Section 3.5, including the Jaccard similarity protocol of Visani et al. (2022) and the threshold-based flagging of unreliable explanations, do not appear among the results reported in Chapter 5, so the safeguard intended to prevent practitioners being shown unreliable explanations without warning is specified but not evidenced.

Read together, the framework does what it was constructed to do and its technical foundations are sound, but the claim it can support is narrower than the framing of the problem might suggest. The problem statement is about practitioner-facing communication, and the strongest evidence the study produced is about model and explanation behaviour. The distance between those two is not a flaw in the execution, since the scope was declared at the outset in Chapter 1, Section 1.6, but it does determine what the contribution is. What has been produced is a verified technical foundation and a traceable design specification for role-aware explainable decision support, which is a necessary precondition for the practitioner-facing claim rather than the claim itself.

## **6.8 Comparison with Existing Research**

This section uses the literature reviewed in Chapter 2 and the comparative analysis in Chapter 5, Section 5.8 to interpret the findings rather than to restate the review. Four comparisons carry most of the interpretive weight.

The first one involves the predictive performance on OULAD. The figures obtained in this study sit below the highest accuracies reported in the OULAD literature, including the 96.9 per cent reported on OULAD by Wang (2025) for a hybrid ensemble. The methodological difference is the explanation. This study grouped folds by course presentation so that no presentation appears in both partitions of a fold, whereas the studies reporting the highest accuracies generally use stratified random splits in which students from every presentation appear in training. Jin et al. (2024) document this variation in protocol across the OULAD literature, and Krstajic et al. (2014) show that ignoring higher-level unit structure produces optimistic estimates. Pelanek (2018) makes the more general argument that choices in cross-validation, filtering, and metric selection materially change reported performance independently of model quality. The internal evidence in Table 5.3 supports this directly, since one fixed model under one fixed protocol produced a seven-point accuracy range depending only on which presentation was held out. The comparison therefore contributes a methodological rather than a competitive conclusion: the present figures are

lower and more conservative, and the difference is plausibly related to the evaluation split and protocol, so neither set of numbers can be read as evidence that one model is better than the other.

The second concerns the features that drive prediction. The dominance of assessment submission over engagement measures in the SHAP and LIME rankings is consistent with the direction of the OULAD literature, in which assessment and VLE interaction features both carry predictive weight and assessment behaviour tends to be the stronger signal (Kuzilek et al., 2017; Jin et al., 2024). Waheed et al. (2020) predict academic performance from VLE clickstream data using deep learning, which is closer in feature philosophy to the engagement side of the present feature set, though their reported figures were not independently verified for this thesis and are not compared numerically. The more specific claim, that the 28-day pre-deadline click window outranks raw total clicks, could not be settled: the two features are near-duplicates, and their order changed between refits of the same model (Section 5.7). The deadline-anchored windows nonetheless carry weight alongside total activity, which is consistent with Hlosta, Zdrahal, and Zendulka (2017), who report that windows aligned to submission dates yield stronger early-warning signals on OULAD, although the present evidence does not show them outperforming raw totals. The counter-intuitive sign of that feature's contribution, discussed in Section 6.2, is a further caution about interpretation.

The third concerns the relationship between SHAP and LIME. Prior studies have reported complementary uses of the two methods, with SHAP commonly used for global attribution and LIME for local explanations, Gunasekara and Saarela (2025) apply both to models trained on OULAD, and Alharbi and Janarthanan (2024) review SHAP and LIME applications specifically within educational machine learning. The mean top-5 instance-level overlap of 0.635 reported here is consistent with that characterisation and adds a quantity to it for this model and dataset. This is where the present study makes a contribution beyond replication: much of the educational XAI literature applies one method or reports the two separately, whereas the measured instance-level overlap gives a concrete figure for how far a practitioner consulting one method rather than the other would see a different account of the same prediction. The finding also connects to the general concern raised by Slack et al. (2020) about the sensitivity of post-hoc methods to perturbation choices, though the divergence observed here arises from the ordinary difference in construction between the two methods rather than from anything adversarial.

The fourth concerns role-aware dashboard design. Chapter 2 established that existing dashboards serve a single stakeholder type or provide role-agnostic outputs, and that the one role-specific XAI dashboard identified in the literature, reported by Afzaal et al. (2021), is directed at students rather than at the staff responsible for intervention decisions. The reviews by Schwendimann et al. (2017) and Matcha et al. (2020) survey dashboard research broadly and provide no OULAD-specific or SHAP-specific benchmarks against which the present design could be numerically compared. This is itself the relevant comparative finding: there is no established basis for numerical comparison because the combination attempted here, three staff-facing roles served from one prediction system with differentiated explanation, was not identified in the reviewed literature in a form that produced comparable measurements. The contribution is correspondingly a construction and specification contribution rather than a comparative performance one, and the absence of comparators cuts both ways, since it also means there is no external evidence that the particular differentiation chosen here is better than an undifferentiated alternative.

A final comparison concerns what the field takes learning analytics to be for. Siemens and Long (2011) and Gasevic, Dawson, and Siemens (2015) argue that the value of learning analytics lies in connecting data to actionable decisions rather than in predictive accuracy alone, and Tsai and Gasevic (2017) and Leitner, Ebner, and Ebner (2019) document the persistent gap between technically capable models and their use in institutional decision making. The present study is positioned on the side of that argument which holds that the connection to decisions is where the difficulty lies, and its technical results are consistent with that position in a specific sense: obtaining a well-performing classifier on OULAD was achievable with a fixed, untuned configuration and no resampling, whereas establishing that its outputs would be usable by three distinct staff roles remains, at the close of this study, unfinished.

## **6.9 Implications for Explainable Learning Analytics Practice**

The implications set out here follow from findings and design decisions reported in this thesis, and each is stated with the evidence it rests on.

The first concerns how explanations should be reported. The instance-level divergence between SHAP and LIME measured in Section 5.6 indicates that a single attribution method does not give a complete account of a model's local behaviour. For practice, the implication is that systems reporting one method as the explanation are presenting a more settled picture than the underlying attributions support, and that where two methods are available their level of agreement is itself information worth surfacing. This does not mean every dashboard should display two methods, since the interpretive cost to the user may be real, but it does mean the choice of a single method should be recognised as a choice with consequences rather than as a neutral implementation detail.

The second concerns direction and interaction alongside magnitude. The `clicks_28d_before_deadline` finding shows that a feature can rank highly in importance while the direction of its contribution may reflect collinearity in the fitted model rather than a behavioural pattern. A ranked importance list alone would have concealed this. The implication for practice is that explanatory interfaces should permit inspection of how a feature contributes and not only how much, and that developers of such systems should expect artefacts of this kind to appear and should check for them before the attributions reach practitioners.

The third concerns the language in which model outputs are communicated. Throughout this study the terms contribution, attribution, and association have been used in preference to causal language, following the position set out in Chapter 3, Section 3.5. This is not only an academic convention. A reasoning trail that identifies low assessment submission as the leading contributor to a flag will be read by an instructor under time pressure, and the difference between it having contributed to a prediction and it having caused a difficulty is the difference between a prompt to investigate and a conclusion to act on. Interfaces that present attributions in causal or prescriptive language invite the collapse described in Section 6.6.

The fourth concerns the reporting of operating characteristics. Because performance varied by more than seven percentage points across the four presentations, the error profile of a model in use is a property of the cohort as well as of the model. The implication is that a dashboard should convey the current operating point, meaning the balance between missed students and false alarms, to the roles that depend on it, and that a single headline accuracy figure reported at deployment and not revisited will misrepresent the system's behaviour for cohorts unlike those it was validated on.

The fifth concerns human oversight and the separation of system and human tasks. The CommonKADS Task Model in this study stops at explanation delivery and places intervention with the human agent. Keeping that boundary in the design, rather than extending the system into recommending actions, follows from the limits of what the attributions establish and not from caution alone. Systems that cross it are making causal claims their explanation methods do not support.

The sixth concerns the responsible handling of risk information. This thesis worked with a public benchmark dataset and no identifiable students, but a system of this kind, once deployed, produces a per-student prediction of withdrawal or failure, visible to staff, and derived in part from demographic attributes that overlap with protected characteristics. Chapter 4, Section 4.6 anticipates the need for disaggregated reporting, and Baker and Hawn (2022) document why. Two implications follow. The first is that subgroup performance should be reported before such a system is used to direct resource, since a model with strong aggregate performance may distribute its errors unevenly. The second is that access control is a substantive rather than an incidental requirement, which is the reasoning behind enforcing role separation at both the edge and the API in Chapter 4, Section 4.8: a risk prediction is information about a student that not every member of staff has a legitimate reason to see.

## 6.10 Contextual Implications and Transferability

The findings of this study were obtained from one dataset describing one institution. OULAD records 32,593 student-module-presentation observations from the Open University across 22 module presentations spanning four presentation periods (Kuzilek et al., 2017), and every figure in Chapter 5 is a figure about that population. This section considers what would need to hold for the framework to transfer elsewhere.

The clearest internal evidence on transferability is the per-presentation variability in Section 5.5. Performance moved by more than seven percentage points in accuracy, and the error profile inverted, across four presentations of the same modules at the same institution within two years. If a model shifts that much between cohorts drawn from the same population, the assumption that a model trained on one institution would perform comparably at another is not supportable, and the appropriate expectation for any new setting is retraining and revalidation rather than transfer of the fitted artefact. The components of the framework are not, however, equally transferable. The architecture and the methodology are procedures that do not depend on OULAD, whereas the feature engineering, the fitted model, the feature importance rankings, the decision threshold, and the performance figures are properties of this dataset and this model. Table 6.1 distinguishes them.

*Table 6.1. Transferability of the components of the role-aware XAI framework.*

| Component                               | Transferability                 |
|-----------------------------------------|---------------------------------|
| CommonKADS design process (methodology) | Potentially transferable        |
| Five-layer architecture                 | Potentially transferable        |
| SHAP/LIME explanation mechanism         | Highly transferable technically |

| Component                           | Transferability           |
|-------------------------------------|---------------------------|
| Feature set and feature engineering | Context dependent         |
| Model weights (fitted model)        | Not directly transferable |
| Feature importance rankings         | Not directly transferable |
| Decision threshold                  | Requires local validation |
| Performance metrics                 | Not transferable          |
| Role definitions                    | Institution dependent     |

The Open University context also shapes the findings in ways that do not carry over. It is a distance-learning institution in which the LMS is the primary site of study rather than a supplement to in-person teaching, so the clickstream captures a much larger share of student activity than it would at a campus-based institution where a disengaged student may still be attending lectures. Its students are on average older, more likely to study part time, and enter with a wider range of prior qualifications than at a conventional institution, and withdrawal in that setting is a more common and differently motivated event. Since withdrawals account for 59 per cent of the positive class here, the target variable is partly a reflection of that context. A model trained on data of this kind is not a general at-risk model; it is a model of one institution's pattern of leaving.

What is more likely to transfer is the structure rather than the model. The CommonKADS-derived chain from institutional context through task decomposition and agent characterisation to interface design is a procedure, and a different institution applying it would populate the Organisation and Agent Models with its own arrangements and arrive at its own information briefs. The five-layer architecture, the separation of cached global explanation from on-demand local explanation, the reasoning trail rendered at three levels of aggregation, and the side-by-side presentation of two attribution methods are design patterns that do not depend on any property of OULAD. The specific feature set, the fitted model, the attribution rankings, and the reported metrics do.

Several conditions would need to be established before applying the framework in another setting. The institution would need LMS interaction data at a granularity and completeness sufficient to compute engagement features, together with assessment records that can be joined to them, and it would need submission deadlines recorded reliably enough to anchor the rolling windows, since the deadline-anchored construction is central to the feature strategy. It would need an outcome definition appropriate to its own circumstances, since a target combining withdrawal and failure may or may not match what the institution wants to identify. It would need enough historical presentations to train and validate with cohorts held out, and enough ongoing data to retrain as cohorts change. It would need the three roles to exist in a recognisable form, since an institution where advising is carried out by the teaching staff would not need the advisor and instructor views to be distinct. It would need infrastructure capable of running the components that require a full Python runtime, which Chapter 4, Section 4.11 identifies as the reason the API is hosted separately from the edge layer, and it would need staff able to maintain the pipeline and interpret its outputs.

Finally, it would need a governance position on the use of per-student risk predictions, including who may see them and how they may be used.

These conditions bear on the question of transfer to institutions in Nigeria and elsewhere in Africa, which is relevant given where this research is situated, but they bear on it only as conditions and not as conclusions. Higher education across the continent is not a single context, and institutions differ widely in LMS adoption, in the completeness of the interaction data their systems retain, in the availability of computing infrastructure and technical staff, and in their data governance arrangements. This study gathered no evidence about any of them. What can be said from the evidence it does have is limited and specific: an approach whose predictive component depends on rich LMS clickstream and assessment data, and whose performance varies materially between cohorts at the institution that generated that data, would require both sufficient local data and local revalidation before it could be relied on anywhere else. Where an institution's teaching is largely in person and its LMS holds only supplementary material, the engagement features that carry most of the signal here would be sparse or missing, and the framework would need to be reconsidered at the level of the Knowledge Model rather than retrained. Establishing which institutions meet the conditions set out above, and what the framework would look like where they are only partly met, is empirical work this thesis did not undertake and does not attempt to anticipate.

## **6.11 Limitations of the Findings and Discussion**

The limitations below are those identified in Chapters 1, 3, and 5, discussed here in terms of how each constrains the interpretation offered in this chapter.

The absence of primary practitioner data is the limitation with the widest effect. It means that the parts of the framework directed at practitioners, which are the parts the research problem is framed around, are supported by design reasoning and published literature rather than by observation. Every statement in Sections 6.3 and 6.5 about what a role needs, how a view supports a decision, or how an explanation might be used is a design position rather than a finding, and Section 6.12 keeps that distinction when answering the third research question.

Reliance on a single secondary dataset constrains external validity in the ways set out in Section 6.10. Beyond the institutional specificity discussed there, OULAD is a historical dataset covering presentations from 2013 and 2014, and both student behaviour and LMS platforms have changed since. The consequence for interpretation is that the feature rankings in Section 6.2 should be read as properties of a model fitted to that data at that time, not as durable findings about which behaviours indicate risk.

The retrospective construction of the features is the limitation with the most direct bearing on the early-warning purpose that motivates this line of research. As Section 6.1 set out, the features were computed over the full course presentation, so the evaluation establishes retrospective classification using the available OULAD feature set and does not establish performance at a predefined early-warning point. Every statement in this chapter about identifying students who may be at risk should be read with that qualification.

The technical evaluation was also narrower than the methodology originally specified. The stability and fidelity checks for the explanations, the calibration analysis of the displayed probabilities, and the subgroup fairness assessment were specified but not completed in the final

evaluation (Chapter 3, Section 3.4.1), so the explanation findings rest on cross-method agreement alone, the risk scores have not been shown to be calibrated, and no evidence is available on whether errors are distributed evenly across demographic groups.

## 6.12 Discussion of the Research Questions

**Research Question 1:** How can explainable AI techniques (specifically SHAP and LIME) be systematically integrated into a knowledge engineering pipeline to improve transparency in student at-risk prediction models?

The findings indicate that the integration pattern specified in Chapter 3 is workable and that it produces a specific kind of transparency. The pattern itself, a cached global TreeSHAP computation serving cohort-level views paired with an on-demand local service computing SHAP and LIME for a single prediction, ran across the full 59-column transformed feature space and all 32,593 rows without failure, and its outputs are traceable to the model version and feature contract that produced them through the persisted manifest. This is the knowledge engineering aspect of the answer: the explanation layer is a defined component with a defined input contract occupying a defined position in the Task Model, rather than an analysis performed alongside the model. The transparency this delivers is transparency about model behaviour. A practitioner can see which quantities drove a given output, in what direction, and with what magnitude, and can compare that against independent knowledge of the student. The measured instance-level overlap of 0.635 between the two methods adds a further element, in that the integration surfaces checkable disagreement between two attribution methods rather than presenting a single unexamined ranking, and the `clicks_28d_before_deadline` artefact shows that inspecting direction alongside magnitude surfaces model behaviour that a ranking alone conceals. The answer is bounded in two respects. The transparency is about the model and not about the causes of student outcomes, for the reasons given in Section 6.2, and whether the resulting interface is experienced as more transparent by a practitioner is a separate question this study did not address.

**Research Question 2:** What dashboard design features and explanation structures, grounded in the CommonKADS Agent Model and the empirical learning analytics dashboard literature, can be specified to support the distinct information needs of instructor, advisor, and administrator roles within an LMS?

This question asks what can be specified, and the study specified and implemented a definite answer: three role-gated views sharing one reasoning trail component rendered at three levels of aggregation, with per-student detail for the instructor, cohort-level aggregation across students at or above the watch threshold for the advisor, and full-population aggregation for the administrator, each with progressive disclosure of attribution detail and role separation enforced at both the edge and the API. Each element traces to a stated requirement: the granularity differences to the Agent Model and to the practitioner-need literature reviewed in Chapter 2, the disclosure pattern to the Communication Model and to Miller's (2019) selectivity property, and the shared component to the Agent Model's treatment of the roles as differing in the granularity rather than the form of the knowledge they require. The technical evaluation contributes to the answer by confirming that the content these views render is sound, since the attributions every view ultimately displays were shown to be reproducible and traceable, and by identifying content each role specifically needs, most notably the between-cohort variability that determines the current operating point. What the study cannot claim is that this specification is the right one. The information briefs were derived

from published accounts rather than from the practitioners concerned, no alternative specification was implemented for comparison, and no user of any role evaluated the views. The answer is therefore a defensible and fully traceable specification, offered as a candidate for empirical validation rather than as a validated design.

**Research Question 3: What do the predictive performance and explainability findings indicate about the technical potential of the proposed role-aware explainable dashboard to support practitioner interpretation and decision-making?**

This question asks what the technical findings indicate about potential, and the answer must be kept within that framing, since no practitioner evaluation was conducted and interpretation, trust, understanding, and decision quality were not measured. Two findings bear most directly on the question. First, the predictive component discriminates well on unseen cohorts, with an AUC of 0.9624, a recall of 0.9085, and a precision of 0.8659, which means that approximately 86.6% of positive predictions corresponded to observations subsequently labelled at-risk under the study's outcome definition; a system whose flags were frequently wrong could not support well-founded interpretation regardless of how it presented them, and that failure mode is not present here. Second, the attributions are traceable, human-readable, and cross-checked between two methods, which gives a practitioner the means to test a flag against what they independently know rather than only to accept or reject it. The measured partial agreement between the two methods may, in addition, allow the interface to convey where the model's local behaviour is firmly characterised and where it is not, although whether this supports calibrated rather than uniform reliance was not tested. Set against these are findings that qualify the potential: 13 per cent of positive flags in this evaluation were false, and each would have been accompanied by a coherent explanation; the operating point varies by cohort; the features describe retrospective rather than early-warning classification; and the risk scores displayed have not been shown to be calibrated. What the evidence establishes is therefore that the technical preconditions for supporting practitioner interpretation and decision-making are in place, meaning accurate predictions and traceable, cross-checked explanations. Whether practitioners trust the system, whether they interpret the reasoning trail as intended, whether role differentiation aids their understanding, and whether decisions taken with the dashboard are better than decisions taken without it are questions that require a study with practitioners, and this thesis provides the verified technical foundation on which such a study could be built rather than its results.

### **6.13 Chapter Summary**

This chapter interpreted the results reported in Chapter 5 in relation to the research objectives, the research questions, and the framework developed across Chapters 3 and 4.

On predictive performance, the interpretation turned on the conditions of the evaluation rather than on the size of the figures. Grouping folds by course presentation produced a conservative estimate of transfer to an unseen cohort, and the error profile leans towards catching at-risk students at the cost of a non-trivial false-positive rate, consistent with the recall-first criterion adopted for the prototype, with the resource implications of that rate left to a future practitioner study. Two qualifications carry through the chapter: the model is substantially a withdrawal predictor, and the features were computed over the full presentation, so retrospective separability rather than early-warning capability is what the figures establish. The variability across the four presentations is the most consequential single finding for practical use, since the observed operating point varied

substantially across the four held-out presentations, which indicates that it is a property of the cohort as well as of the model.

On explainability, the integration works and produces attributions concentrated on assessment submission behaviour. The measured instance-level overlap of 0.635 between SHAP and LIME gives the side-by-side presentation an evidential basis, and the counter-intuitive direction of the 28-day click window shows why direction must be inspected alongside magnitude. Throughout, the attributions describe model behaviour rather than the causes of student outcomes, and that boundary shapes what the explanation layer can contribute.

On role-aware design and CommonKADS, the chapter argued that the value of the framework in this study was traceability from institutional context through task decomposition and agent characterisation to interface granularity, and that this produced three differentiated views sharing one explanatory structure rather than three separately invented ones.

On decision support and the relationship between prediction, explanation, and intervention, the chapter distinguished what the technical evaluation demonstrated, what the design supports, what the findings suggest, and what remains unvalidated. The system stops at explanation delivery and places the intervention decision with the human agent, and the sequence from prediction through explanation and human interpretation to intervention must not be collapsed, since an attribution narrows where attention should be directed without determining what should be done.

Comparison with the literature indicated that the difference from the highest OULAD results may reflect the stricter evaluation protocol, and the available evidence does not permit the difference to be attributed solely to model quality, that the feature findings are consistent with the OULAD literature and support the deadline-anchored feature construction, that the measured SHAP and LIME overlap adds a quantity to a relationship the educational XAI literature has described qualitatively, and that no comparable role-aware staff-facing dashboard was identified in the reviewed literature against which the design could be numerically compared.

The framework therefore rests on a verified technical foundation and a traceable design specification, with practitioner trust, comprehension, usability, decision quality, calibration, subgroup fairness, and early-window performance remaining outside what this study established. Transfer to another institution would require local data of sufficient completeness, local retraining and revalidation, roles that correspond to those modelled here, and a governance position on the use of per-student risk information. These conclusions, together with the contributions and the directions for future work that follow from them, are drawn together in the final chapter.

## CHAPTER 7

### CONCLUSION

#### 7.1 Introduction

This final chapter draws the study to a close. It restates what the research set out to do, brings together what was built and measured, answers each research question directly, and sets out the contributions of the work and the directions it opens up. The purpose is not to repeat earlier material but to give a clear account of what the study established and why it matters for anyone using predictive models responsibly in higher education.

#### 7.2 Summary of the Study

The study began from a practical problem. Universities collect large amounts of data about how students engage with online learning, and machine learning models use that data to flag students likely to fail or withdraw. A score on its own changes little. A tutor told that a student has a 70 per cent chance of failing needs to know why the model reached that view, whether the reason is something that can be acted on, and what a sensible next step would be. This study supplies an explanation of why the model reached its prediction alongside the prediction itself; it does not establish recommended interventions, and deciding on a sensible next step remains with the practitioner.

To address the problem, the research designed and implemented a role-aware explainable analytics system for predicting student risk. Data from the Open University Learning Analytics Dataset was cleaned and turned into features describing assessment behaviour, virtual learning environment activity, and demographic and registration context. Three ensemble classifiers were trained and compared, and a gradient boosted model (XGBoost) was selected for the prototype on the basis of its recall, although Random Forest obtained a marginally higher F1-score. Two post hoc explanation methods, SHAP and LIME, were applied to the trained model so that every prediction is accompanied by an account of the factors behind it.

The system was built as a layered application. A data layer handled preparation and feature construction, a prediction layer served the trained model, an explanation layer generated global and local attributions, an interface layer exposed these as services, and a presentation layer delivered them to users through a dashboard. The dashboard was designed around three staff roles: instructors, who work with individual students; academic advisors, who work across a caseload of students; and administrators, who look across the institution. Students were outside the scope of the study. Each role sees the same underlying prediction with a depth and framing of explanation matched to its questions and responsibilities.

The design was organised using CommonKADS, a knowledge engineering methodology that separates the organisational setting, the tasks people perform, the agents involved, the knowledge required, the communication between agents, and the eventual system design. This structure was useful, because the substantive design problem was deciding who needs to know what, in what form, and at what moment.

The technical evaluation established the behaviour of the predictive and explanatory components. The selected model reached an accuracy of 0.8774, precision of 0.8659, recall of 0.9085, an F1 score of 0.8867, and an area under the ROC curve of 0.9624. Random Forest obtained a marginally higher F1 score (0.8976) in the three-way comparison, but XGBoost was retained for the prototype because recall was set as the primary selection criterion, since missing a struggling student carries a heavier cost than reviewing a student who turns out to be fine. Because the features were computed over the full course presentation, these figures describe retrospective classification and do not establish performance at a predefined early-warning point. Explanations were examined for cross-method agreement, while the stability and fidelity procedures specified in the methodology were not completed in the final evaluation. Assessment submission behaviour, in particular the number of assessments submitted, dominated the global picture, and the two explanation methods agreed closely at the aggregate level, giving a consistent picture for cohort-level interpretation. At the level of the individual prediction the mean top five overlap was 0.635, which is why the prototype presents local explanations side by side as complementary perspectives on a prediction rather than as a single account.

## **7.3 Revisiting the Research Questions**

### **7.3.1 First research question**

The first question asked whether SHAP and LIME could be integrated into a predictive model of student risk in a way that produced technically generated explanations. The answer is yes, at the level of system construction. Both methods were successfully attached to the trained model and produced explanations at two levels: an overall view of which behaviours drive risk across the cohort, and a per student view of what pushed one particular prediction up or down. The explanations were coherent and consistent with established knowledge of study behaviour, with missed and late assessment activity carrying the most weight.

The integration also indicates how the two methods can be used together. The global view provides reliable orientation, where the methods converge strongly, and the local view provides the case-level detail a tutor needs for a specific conversation. Presenting both, rather than one silently, is the design choice intended to make the explanations usable: the interface is designed to allow users to inspect those factors.

### **7.3.2 Second research question**

The second question asked how a dashboard could be designed so that explanations fit the role of the person reading them. The study answered this through design and implementation. Instructors receive individual level explanations tied to concrete recent behaviour, so that a conversation with a student has something specific to start from. Academic advisors receive cohort-averaged explanations across the students in their caseload who are at or above the watch threshold, and administrators receive a full-population view that helps with allocating limited resources and spotting where a module rather than a student is the problem. These differences are specified from the Agent Model and the literature and were not tested with users.

Working through the CommonKADS models made these differences principled rather than accidental. The task and agent descriptions established that the same explanation is not equally

appropriate for everyone, and the communication model fixed the point in each working routine at which a given type of information is useful.

### **7.3.3 Third research question**

The third question asked what the predictive performance and explainability findings indicate about the technical potential of the dashboard to support practitioner interpretation and decision-making. The findings indicate that the technical preconditions are in place: predictions are accompanied by reasons, the reasons are traceable to specific recorded behaviour, the two explanation methods are cross-checked, and the outputs are framed as prompts for human judgement rather than as decisions. The behaviours that carry the most weight, particularly assessment submission, are ones an instructor or advisor can recognise and follow up. Using the three-level evidence model applied throughout this thesis, these properties are demonstrated or supported by design, whereas practitioner trust, interpretation, and decision quality are not empirically established, since no practitioner used the system. The prototype therefore provides a technical foundation for a future practitioner study rather than evidence of trustworthy use in practice.

## **7.4 Contributions of the Study**

The first contribution is a working demonstration that prediction and explanation can be delivered together in a single system rather than treated as separate activities. Much published work stops at model performance or produces explanations as an offline analysis. Here the explanation layer is part of the running application and serves the interface directly.

The second contribution is the role-aware treatment of explanation. Rather than producing one explanation view and assuming it suits everyone, the study derives distinct views from the responsibilities and questions of different users, and documents how those views were arrived at.

The third contribution is methodological. Applying CommonKADS to explainable learning analytics shows that the design problem is largely a knowledge problem. The models provide a way of reasoning about who holds what knowledge, what decisions depend on it, and where a system can usefully intervene, which model performance metrics alone cannot support.

The fourth contribution is empirical evidence on explanation behaviour. By measuring agreement between SHAP and LIME at both global and local level, the study establishes where these methods converge and how they can be combined in a prototype, providing a technical basis for subsequent practitioner evaluation.

## **7.5 Recommendations and Future Work**

Three items of future work are treated as first-tier priorities, because each addresses a limitation that bears directly on the use of the dashboard.

First, an early-warning temporal evaluation. The model should be re-evaluated at fixed points in each presentation, for example at week 2, week 4, week 6, and week 8, with the features restricted to the information available at each point, so that performance at a predefined early-warning point can be established rather than inferred from retrospective classification.

Second, model calibration. Because the dashboard displays individual risk probabilities, the calibration of those probabilities should be assessed, for example by comparing isotonic regression and Platt scaling using reliability diagrams and expected calibration error, before the displayed scores are relied upon.

Third, fairness assessment. Because the model uses demographic features that overlap with protected characteristics, subgroup performance and fairness should be assessed across gender, age band, disability status, and index of multiple deprivation band, together with whether explanation quality holds evenly across student groups, before the system is used to direct institutional resource.

Beyond these three priorities, evaluation with practitioners is the next opportunity. A study with instructors, academic advisors, and administrators would show how the role specific views are read in practice and how the explanations shape action, and it could draw on the perceived usefulness and perceived ease of use constructs of TAM, which this thesis used only as a design lens.

The approach should also be applied to data from other institutions and other modes of study, extending the reach of the design and identifying which patterns generalise across settings.

The explanation layer invites further development, including the stability and fidelity checks specified in the methodology and consolidated views that combine methods and highlight where their emphases differ. The modelling pipeline can likewise be extended with the refinements recorded in Chapter 3, Section 3.4.1, most usefully the trajectory feature family, hyperparameter tuning, and class-imbalance handling, and with a reconsideration of Random Forest, which obtained the highest F1-score in the three-way comparison.

Further, the link from explanation to action deserves attention in its own right. The system tells a tutor what is driving a prediction; a natural extension is to record what was done in response and what followed, so that the value of interventions can be measured directly.

Finally, deployment against live activity data opens questions of timing, drift, and refresh that a retrospective dataset cannot expose, and addressing these is the natural path to institutional use.

## **7.6 Concluding Remarks**

This study set out to connect a model that performs well to the person who has to act on its output, and it approached this by treating prediction, explanation, and role as one design problem rather than three. The model performed strongly on retrospective classification, the explanations were coherent and consistent with what is known about study behaviour, and the dashboard presents both at the level of aggregation specified for each staff role. The result is a technically implemented and evaluated prototype that provides a foundation for subsequent practitioner validation, early-warning evaluation, calibration, fairness assessment, and institutional deployment studies.

## REFERENCES

- Adadi, A., & Berrada, M. (2018). Peeking inside the black-box: A survey on explainable artificial intelligence (XAI). *IEEE Access*, 6, 52138–52160. <https://doi.org/10.1109/ACCESS.2018.2870052>
- Adnan, M., Habib, A., Ashraf, J., Mussadiq, S., Raza, A. A., Abid, M., Bashir, M., & Khan, S. U. (2021). Predicting at-risk students at different percentages of course length for early intervention using machine learning models. *IEEE Access*, 9, 7519–7539. <https://doi.org/10.1109/ACCESS.2021.3049446>
- Afzaal, M., Nouri, J., Zia, A., Papapetrou, P., Fors, U., Wu, Y., Li, X., & Weegar, R. (2021). Explainable AI for data-driven feedback and intelligent action recommendations to support students self-regulation. *Frontiers in Artificial Intelligence*, 4, 723447. <https://doi.org/10.3389/frai.2021.723447>
- Ahmed, W., Wani, M. A., & Plawiak, P. (2025). Machine learning-based academic performance prediction with explainability for enhanced decision-making in educational institutions. *Scientific Reports*, 15, 26879. <https://doi.org/10.1038/s41598-025-12353-4>
- Al-Ansari, M. (2024). User-centered evaluation of explainable artificial intelligence (XAI): A systematic literature review. *Human Behavior and Emerging Technologies*, 2024, 4628855. <https://doi.org/10.1155/2024/4628855>
- Albreiki, B., Zaki, N., & Alashwal, H. (2021). A systematic literature review of student performance prediction using machine learning techniques. *Education Sciences*, 11(9), 552. <https://doi.org/10.3390/educsci11090552>
- Alharbi, A., & Janarthanan, M. (2024). SHAP-centric interpretability in educational machine learning: A PRISMA systematic review of SHAP and LIME. *Letters in Biomathematics*.
- Ali, S., Abuhmed, T., El-Sappagh, S., Muhammad, K., Alonso-Moral, J. M., Confalonieri, R., Guidotti, R., Del Ser, J., Díaz-Rodríguez, N., & Herrera, F. (2023). Explainable Artificial Intelligence (XAI): What we know and what is left to attain Trustworthy Artificial Intelligence. *Information Fusion*, 99, 101805. <https://doi.org/10.1016/j.inffus.2023.101805>
- Aljohani, N. R., Fayoumi, A., & Hassan, S.-U. (2019). Predicting at-risk students using clickstream data in the virtual learning environment. *Sustainability*, 11(24), 7238. <https://doi.org/10.3390/su11247238>
- Arrieta, A. B., Díaz-Rodríguez, N., Del Ser, J., Bennetot, A., Tabik, S., Barbado, A., García, S., Gil-López, S., Molina, D., Benjamins, R., Chatila, R., & Herrera, F. (2020). Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI. *Information Fusion*, 58, 82–115. <https://doi.org/10.1016/j.inffus.2019.12.012>
- Baker, R. S., & Hawn, A. (2022). Algorithmic bias in education. *International Journal of Artificial Intelligence in Education*, 32(4), 1052–1092. <https://doi.org/10.1007/s40593-021-00285-9>
- Bass, L., Clements, P., & Kazman, R. (2021). *Software architecture in practice* (4th ed.). Addison-Wesley.

- Bergstra, J., Bardenet, R., Bengio, Y., & Kegl, B. (2011). Algorithms for hyper-parameter optimization. *Advances in Neural Information Processing Systems*, 24, 2546–2554.
- Bodily, R., & Verbert, K. (2017). Review of research on student-facing learning analytics dashboards and educational recommender systems. *IEEE Transactions on Learning Technologies*, 10(4), 405–418. <https://doi.org/10.1109/TLT.2017.2740172>
- Booch, G., Rumbaugh, J., & Jacobson, I. (2005). *The Unified Modeling Language user guide* (2nd ed.). Addison-Wesley.
- Brdnik, S., Podgorelec, V., & Sumak, B. (2023). Assessing perceived trust and satisfaction with multiple explanation techniques in XAI-enhanced learning analytics. *Electronics*, 12(12), 2594. <https://doi.org/10.3390/electronics12122594>
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
- Cabral, L., Pinto, R., & Gonçalves, G. (2025). AI-powered learning analytics dashboards: A systematic review of applications, techniques, and research gaps. *Discover Education*, 4, 525. <https://doi.org/10.1007/s44217-025-00964-y>
- Chatti, M. A., Dyckhoff, A. L., Schroeder, U., & Thüs, H. (2012). A reference model for learning analytics. *International Journal of Technology Enhanced Learning*, 4(5/6), 318–331. <https://doi.org/10.1504/IJTEL.2012.051815>
- Chavan, P., & Mitra, R. (2022). Tcherly: A teacher-facing dashboard for online video lectures. *Journal of Learning Analytics*, 9(3), 125–151. <https://doi.org/10.18608/jla.2022.7555>
- Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357. <https://doi.org/10.1613/jair.953>
- Chen, F., & Cui, Y. (2020). Utilizing student time series behaviour in learning management systems for early prediction of course performance. *Journal of Learning Analytics*, 7(2), 1–17. <https://doi.org/10.18608/jla.2020.72.1>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). ACM. <https://doi.org/10.1145/2939672.2939785>
- Chicco, D., & Jurman, G. (2020). The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. *BMC Genomics*, 21, 6. <https://doi.org/10.1186/s12864-019-6413-7>
- Conati, C., Barral, O., Putnam, V., & Rieger, L. (2021). Toward personalized XAI: A case study in intelligent tutoring systems. *Artificial Intelligence*, 298, 103503. <https://doi.org/10.1016/j.artint.2021.103503>
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340. <https://doi.org/10.2307/249008>

- Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. arXiv preprint arXiv:1702.08608.
- Ehsan, U., Liao, Q. V., Muller, M., Riedl, M. O., & Weisz, J. D. (2021). Expanding explainability: Towards social transparency in AI systems. *Proceedings of the ACM on Human-Computer Interaction*, 5(CSCW1), 1–19. <https://doi.org/10.1145/3411764.3445188>
- Ferguson, R. (2012). Learning analytics: Drivers, developments and challenges. *International Journal of Technology Enhanced Learning*, 4(5/6), 304–317. <https://doi.org/10.1504/IJTEL.2012.051816>
- Fernández, A., García, S., Herrera, F., & Chawla, N. V. (2018). SMOTE for learning from imbalanced data: Progress and challenges, marking the 15-year anniversary. *Journal of Artificial Intelligence Research*, 61, 863–905. <https://doi.org/10.1613/jair.1.11192>
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5), 1189–1232. <https://doi.org/10.1214/aos/1013203451>
- Gasevic, D., Dawson, S., & Siemens, G. (2015). Let's not forget: Learning analytics are about learning. *TechTrends*, 59(1), 64–71. <https://doi.org/10.1007/s11528-014-0822-x>
- Guevara-Reyes, L. M., Andrade, V., Cox-Riquetti, B., & Villegas-Ch, W. (2025). Machine learning models for academic performance prediction: Interpretability and application in educational decision-making. *Frontiers in Education*, 10, 1632315. <https://doi.org/10.3389/educ.2025.1632315>
- Guidotti, R., Monreale, A., Ruggieri, S., Turini, F., Giannotti, F., & Pedreschi, D. (2018). A survey of methods for explaining black box models. *ACM Computing Surveys*, 51(5), 93. <https://doi.org/10.1145/3236009>
- Gunasekara, S., & Saarela, M. (2024). Explainability in educational data mining and learning analytics: An umbrella review. *Proceedings of the 17th International Conference on Educational Data Mining (Posters)*.
- Gunasekara, S., & Saarela, M. (2025). Explainable AI in education: Techniques and qualitative assessment. *Applied Sciences*, 15(3), 1239. <https://doi.org/10.3390/app15031239>
- Gundersen, O. E., & Kjensmo, S. (2018). State of the art: Reproducibility in artificial intelligence. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 1644–1651. <https://doi.org/10.1609/aaai.v32i1.11503>
- Gunning, D., & Aha, D. W. (2019). DARPA's explainable artificial intelligence (XAI) program. *AI Magazine*, 40(2), 44–58. <https://doi.org/10.1609/aimag.v40i2.2850>
- Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. *Proceedings of the 34th International Conference on Machine Learning*, PMLR 70, 1321–1330.
- Gutiérrez, F., Seipp, K., Ochoa, X., Chiluiza, K., De Laet, T., & Verbert, K. (2020). LADA: A learning analytics dashboard for academic advising. *Computers in Human Behavior*, 107, 105826. <https://doi.org/10.1016/j.chb.2018.12.004>

- Haque, A. B., Islam, A. N., & Mikalef, P. (2023). Explainable artificial intelligence (XAI) from a user perspective: A synthesis of prior literature and problematizing avenues for future research. *Technological Forecasting and Social Change*, 186, 122120. <https://doi.org/10.1016/j.techfore.2022.122120>
- Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. *Advances in Neural Information Processing Systems*, 29, 3315–3323.
- Hassan, S.-U., Waheed, H., Aljohani, N. R., Ali, M., Ventura, S., & Herrera, F. (2019). Virtual learning environment to predict withdrawal by leveraging deep learning. *International Journal of Intelligent Systems*, 34(8), 1935–1952. <https://doi.org/10.1002/int.22129>
- Hernandez-Leo, D., Martinez-Maldonado, R., Pardo, A., Munoz-Cristobal, J. A., & Rodriguez-Triana, M. J. (2019). Analytics for learning design: A layered framework and tools. *British Journal of Educational Technology*, 50(1), 139–152. <https://doi.org/10.1111/bjet.12645>
- Hevner, A. R., March, S. T., Park, J., & Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1), 75–105. <https://doi.org/10.2307/25148625>
- Hlosta, M., Zdrahal, Z., & Zendulka, J. (2017). Ouroboros: Early identification of at-risk students without models based on legacy data. In *Proceedings of the Seventh International Learning Analytics & Knowledge Conference* (pp. 6–15). ACM. <https://doi.org/10.1145/3027385.3027449>
- Humble, J., & Farley, D. (2010). *Continuous delivery: Reliable software releases through build, test, and deployment automation*. Addison-Wesley.
- Hussain, M., Zhu, W., Zhang, W., & Abidi, S. M. R. (2018). Student engagement predictions in an e-learning system and their impact on student course assessment scores. *Computational Intelligence and Neuroscience*, 2018, 6347186. <https://doi.org/10.1155/2018/6347186>
- Islam, Md. M., Sojib, F. H., Mihad, Md. F. H., Hasan, Md. M., & Rahman, M. (2025). The integration of explainable AI in educational data mining for student academic performance prediction and support system. *Telematics and Informatics Reports*, 18, 100203. <https://doi.org/10.1016/j.teler.2025.100203>
- Jin, K., Wang, L., Song, Y., & So, W. W. M. (2024). Predictive modelling with the Open University Learning Analytics Dataset (OULAD): A systematic literature review. In *Artificial Intelligence in Education (Communications in Computer and Information Science, Vol. 2150, pp. 477–484)*. Springer. [https://doi.org/10.1007/978-3-031-64315-6\\_46](https://doi.org/10.1007/978-3-031-64315-6_46)
- Jones, M., Bradley, J., & Sakimura, N. (2015). *JSON Web Token (JWT) (RFC 7519)*. Internet Engineering Task Force. <https://doi.org/10.17487/RFC7519>
- Kaliisa, R., & Dolonen, J. A. (2022). CADA: A teacher-facing learning analytics dashboard to foster teachers' awareness of students' participation and discourse patterns in online discussions. *Technology, Knowledge and Learning*, 28, 937–958. <https://doi.org/10.1007/s10758-022-09598-7>

- Kaliisa, R., Jivet, I., & Prinsloo, P. (2023). A checklist to guide the planning, designing, implementation, and evaluation of learning analytics dashboards. *International Journal of Educational Technology in Higher Education*, 20, 28. <https://doi.org/10.1186/s41239-023-00394-6>
- Khosravi, H., Buckingham Shum, S., Chen, G., Conati, C., Tsai, Y.-S., Kay, J., Knight, S., Martinez-Maldonado, R., Sadiq, S., & Gašević, D. (2022). Explainable artificial intelligence in education. *Computers and Education: Artificial Intelligence*, 3, 100074. <https://doi.org/10.1016/j.caeai.2022.100074>
- Kim, J., Maathuis, H., & Sent, D. (2024). Human-centered evaluation of explainable AI applications: A systematic review. *Frontiers in Artificial Intelligence*, 7, 1456486. <https://doi.org/10.3389/frai.2024.1456486>
- Kostopoulos, G., Davrazos, G., & Kotsiantis, S. (2024). Explainable AI-based decision support systems: A recent review. *Electronics*, 13(14), 2842. <https://doi.org/10.3390/electronics13142842>
- Kouki, P., Schaffer, J., Pujara, J., O'Donovan, J., & Getoor, L. (2020). Generating and understanding personalized explanations in hybrid recommender systems. *ACM Transactions on Interactive Intelligent Systems*, 10(4), Article 31. <https://doi.org/10.1145/3365843>
- Krstajic, D., Buturovic, L. J., Leahy, D. E., & Thomas, S. (2014). Cross-validation pitfalls when selecting and assessing regression and classification models. *Journal of Cheminformatics*, 6, 10. <https://doi.org/10.1186/1758-2946-6-10>
- Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2017). Open University Learning Analytics dataset. *Scientific Data*, 4, 170171. <https://doi.org/10.1038/sdata.2017.171>
- Larrabee Sønderlund, A., Hughes, E., & Smith, J. (2019). The efficacy of learning analytics interventions in higher education: A systematic review. *British Journal of Educational Technology*, 50(5), 2594–2618. <https://doi.org/10.1111/bjet.12720>
- Leitner, P., Ebner, M., & Ebner, M. (2019). Learning analytics challenges to overcome in higher education institutions. In D. Ifenthaler, D.-K. Mah, & J. Y.-K. Yau (Eds.), *Utilizing learning analytics to support study success* (pp. 91–104). Springer. [https://doi.org/10.1007/978-3-319-64792-0\\_6](https://doi.org/10.1007/978-3-319-64792-0_6)
- Li, K. C., Wong, B. T. M., & Chan, H. T. (2020). Prediction of at-risk students using learning analytics: A literature review. Hong Kong Metropolitan University.
- Liao, Q. V., & Varshney, K. R. (2022). Human-Centered Explainable AI (XAI): From algorithms to user experiences. *arXiv Preprint arXiv:2110.10790*. <https://doi.org/10.48550/arXiv.2110.10790>
- Lipton, Z. C. (2018). The mythos of model interpretability. *Communications of the ACM*, 61(10), 36–43. <https://doi.org/10.1145/3233231>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774.

- Lundberg, S. M., Erion, G., Chen, H., DeGrave, A., Prutkin, J. M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., & Lee, S.-I. (2020). From local explanations to global understanding with explainable AI for trees. *Nature Machine Intelligence*, 2(1), 56–67. <https://doi.org/10.1038/s42256-019-0138-9>
- Marangunic, N., & Granic, A. (2015). Technology acceptance model: A literature review from 1986 to 2013. *Universal Access in the Information Society*, 14(1), 81–95. <https://doi.org/10.1007/s10209-014-0348-1>
- Martin, R. C. (2017). *Clean architecture: A craftsman’s guide to software structure and design*. Prentice Hall.
- Matcha, W., Uzir, N. A., Gasevic, D., & Pardo, A. (2020). A systematic review of empirical studies on learning analytics dashboards: A self-regulated learning perspective. *IEEE Transactions on Learning Technologies*, 13(2), 226–245. <https://doi.org/10.1109/TLT.2019.2916802>
- Meta Open Source. (2024). React documentation. <https://react.dev>
- Miller, T. (2019). Explanation in artificial intelligence: Insights from the social sciences. *Artificial Intelligence*, 267, 1–38. <https://doi.org/10.1016/j.artint.2018.07.007>
- Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2019). Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (pp. 220–229). ACM. <https://doi.org/10.1145/3287560.3287596>
- Miteva, D., & Stefanova, E. (2022). eAnalytics: A model of a learning analytics visualization system. *Annuaire de l’Université de Sofia “St. Kliment Ohridski”, Faculté de Mathématiques et Informatique*, 109, 99–119.
- Mohseni, S., Zarei, N., & Ragan, E. D. (2021). A multidisciplinary survey and framework for design and evaluation of explainable AI systems. *ACM Transactions on Interactive Intelligent Systems*, 11(3–4), 24. <https://doi.org/10.1145/3387166>
- Molenaar, I., & Knoop-van Campen, C. A. N. (2019). How teachers make dashboard information actionable. *IEEE Transactions on Learning Technologies*, 12(3), 347–355. <https://doi.org/10.1109/TLT.2018.2851585>
- Munzner, T. (2014). *Visualization analysis and design*. CRC Press.
- Niculescu-Mizil, A., & Caruana, R. (2005). Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning* (pp. 625–632). ACM. <https://doi.org/10.1145/1102351.1102430>
- Nielsen, J. (1994). *Usability engineering*. Morgan Kaufmann.
- Nnadi, L. C., Watanobe, Y., Rahman, M. M., & John-Otumu, A. M. (2024). Prediction of students’ adaptability using explainable AI in educational machine learning models. *Applied Sciences*, 14(12), 5141. <https://doi.org/10.3390/app14125141>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D.,

- Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Peffer, K., Tuunanen, T., Rothenberger, M. A., & Chatterjee, S. (2007). A design science research methodology for information systems research. *Journal of Management Information Systems*, 24(3), 45–77. <https://doi.org/10.2753/MIS0742-1222240302>
- Pelane, R. (2018). The details matter: Methodological nuances in the evaluation of student models. *User Modeling and User-Adapted Interaction*, 28(3), 207–235. <https://doi.org/10.1007/s11257-018-9204-y>
- Poursabzi-Sangdeh, F., Goldstein, D. G., Hofman, J. M., Wortman Vaughan, J., & Wallach, H. (2021). Manipulating and measuring model interpretability. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Article 237). ACM. <https://doi.org/10.1145/3411764.3445315>
- Prati, R. C., Batista, G. E. A. P. A., & Silva, D. F. (2015). Class imbalance revisited: A new experimental setup to assess the performance of treatment methods. *Knowledge and Information Systems*, 45(1), 247–270. <https://doi.org/10.1007/s10115-014-0794-3>
- Prinsloo, P., & Slade, S. (2017). An elephant in the learning analytics room: The obligation to act. In *Proceedings of the Seventh International Learning Analytics & Knowledge Conference* (pp. 46–55). ACM. <https://doi.org/10.1145/3027385.3027406>
- Rebelo Marcolino, M., Reis Porto, T., Thompsen Primo, T., Targino, R., Ramos, V., Queiroga, E. M., Munoz, R., & Cechinel, C. (2025). Student dropout prediction through machine learning optimization: Insights from Moodle log data. *Scientific Reports*, 15, 9840. <https://doi.org/10.1038/s41598-025-93918-1>
- Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?": Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 1135–1144). ACM. <https://doi.org/10.1145/2939672.2939778>
- Rodríguez-Ortiz, C. E., Sánchez-Torres, J. M., & Rueda-Rueda, J. S. (2022). Knowledge engineering approach for decision support systems in academic performance monitoring. *Journal of Intelligent & Fuzzy Systems*, 42(3), 2381–2393. <https://doi.org/10.3233/JIFS-211603>
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5), 206–215. <https://doi.org/10.1038/s42256-019-0048-x>
- Salih, A. M., Raisi-Estabragh, Z., Boscolo Galazzo, I., Radeva, P., Petersen, S. E., Lekadir, K., & Menegaz, G. (2024). A perspective on explainable artificial intelligence methods: SHAP and LIME. *Advanced Intelligent Systems*, 7(1), 2400304. <https://doi.org/10.1002/aisy.202400304>
- Saltzer, J. H., & Schroeder, M. D. (1975). The protection of information in computer systems. *Proceedings of the IEEE*, 63(9), 1278–1308. <https://doi.org/10.1109/PROC.1975.9939>

- Santos, M. S., Soares, J. P., Abreu, P. H., Araujo, H., & Santos, J. (2018). Cross-validation for imbalanced datasets: Avoiding overoptimistic and overfitting approaches. *IEEE Computational Intelligence Magazine*, 13(4), 59–76. <https://doi.org/10.1109/MCI.2018.2866730>
- Satyanarayan, A., Moritz, D., Wongsuphasawat, K., & Heer, J. (2017). Vega-Lite: A grammar of interactive graphics. *IEEE Transactions on Visualization and Computer Graphics*, 23(1), 341–350. <https://doi.org/10.1109/TVCG.2016.2599030>
- Schreiber, G., Akkermans, H., Anjewierden, A., De Hoog, R., Shadbolt, N., Van de Velde, W., & Wielinga, B. (2000). *Knowledge engineering and management: The CommonKADS methodology*. MIT Press.
- Schwendimann, B. A., Rodríguez-Triana, M. J., Vozniuk, A., Prieto, L. P., Boroujeni, M. S., Holzer, A., Gillet, D., & Dillenbourg, P. (2017). Perceiving learning at a glance: A systematic literature review of learning dashboard research. *IEEE Transactions on Learning Technologies*, 10(1), 30–41. <https://doi.org/10.1109/TLT.2016.2599522>
- Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F., & Dennison, D. (2015). Hidden technical debt in machine learning systems. *Advances in Neural Information Processing Systems*, 28, 2503–2511.
- Sedrakyan, G., Malmberg, J., Verbert, K., Järvelä, S., & Kirschner, P. A. (2020). Linking learning behavior analytics and learning science concepts: Designing a learning analytics dashboard for feedback to support learning regulation. *Computers in Human Behavior*, 107, 105512. <https://doi.org/10.1016/j.chb.2018.05.004>
- Siemens, G., & Long, P. (2011). Penetrating the fog: Analytics in learning and education. *EDUCAUSE Review*, 46(5), 30–40.
- Slack, D., Hilgard, S., Jia, E., Singh, S., & Lakkaraju, H. (2020). Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods. In *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society* (pp. 180–186). ACM. <https://doi.org/10.1145/3375627.3375830>
- Slade, S., & Prinsloo, P. (2013). Learning analytics: Ethical issues and dilemmas. *American Behavioral Scientist*, 57(10), 1510–1529. <https://doi.org/10.1177/0002764213479366>
- Sokol, K., & Flach, P. (2020). Explainability fact sheets: A framework for systematic assessment of explainable approaches. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (pp. 56–67). ACM. <https://doi.org/10.1145/3351095.3372870>
- Springer, A., & Whittaker, S. (2020). Progressive disclosure: When, why, and how do users want algorithmic transparency information? *ACM Transactions on Interactive Intelligent Systems*, 10(4), 1–32. <https://doi.org/10.1145/3374218>
- Studer, R., Benjamins, V. R., & Fensel, D. (1998). Knowledge engineering: Principles and methods. *Data & Knowledge Engineering*, 25(1–2), 161–197. [https://doi.org/10.1016/S0169-023X\(97\)00056-6](https://doi.org/10.1016/S0169-023X(97)00056-6)

- Susnjak, T., Ramaswami, G. S., & Mathrani, A. (2022). Learning analytics dashboard: A tool for providing actionable insights to learners. *International Journal of Educational Technology in Higher Education*, 19, 12. <https://doi.org/10.1186/s41239-021-00313-7>
- Tinto, V. (1975). Dropout from higher education: A theoretical synthesis of recent research. *Review of Educational Research*, 45(1), 89–125. <https://doi.org/10.3102/00346543045001089>
- Tsai, Y.-S., & Gašević, D. (2017). Learning analytics in higher education: Challenges and policies. In *Proceedings of the Seventh International Learning Analytics & Knowledge Conference* (pp. 233–242). ACM. <https://doi.org/10.1145/3027385.3027400>
- Van Buuren, S. (2018). *Flexible imputation of missing data* (2nd ed.). CRC Press.
- Venkatesh, V., & Bala, H. (2008). Technology acceptance model 3 and a research agenda on interventions. *Decision Sciences*, 39(2), 273–315. <https://doi.org/10.1111/j.1540-5915.2008.00192.x>
- Verbert, K., Duval, E., Klerkx, J., Govaerts, S., & Santos, J. L. (2013). Learning analytics dashboard applications. *American Behavioral Scientist*, 57(10), 1500–1509. <https://doi.org/10.1177/0002764213479363>
- Verbert, K., Ochoa, X., De Croon, R., Dourado, R. A., & De Laet, T. (2020). Learning analytics dashboards: The past, the present and the future. In *Proceedings of the Tenth International Conference on Learning Analytics & Knowledge* (pp. 35–40). ACM. <https://doi.org/10.1145/3375462.3375504>
- Viberg, O., Hatakka, M., Bälter, O., & Mavroudi, A. (2018). The current landscape of learning analytics in higher education. *Computers in Human Behavior*, 89, 98–110. <https://doi.org/10.1016/j.chb.2018.07.027>
- Visani, G., Bagli, E., Chesani, F., Poluzzi, A., & Capuzzo, D. (2022). Statistical stability indices for LIME: Obtaining reliable explanations for machine learning models. *Journal of the Operational Research Society*, 73(1), 91–101. <https://doi.org/10.1080/01605682.2020.1865846>
- Wachter, S., Mittelstadt, B., & Russell, C. (2018). Counterfactual explanations without opening the black box: Automated decisions and the GDPR. *Harvard Journal of Law & Technology*, 31(2), 841–887.
- Waheed, H., Hassan, S.-U., Aljohani, N. R., Hardman, J., Alelyani, S., & Nawaz, R. (2020). Predicting academic performance of students from VLE big data using deep learning models. *Computers in Human Behavior*, 104, 106189. <https://doi.org/10.1016/j.chb.2019.106189>
- Wang, Y. (2025). Artificial intelligence in student management systems to enhance academic performance monitoring and intervention. *Scientific Reports*, 15, 35122. <https://doi.org/10.1038/s41598-025-19159-4>
- Wilkinson, M. D., Dumontier, M., Aalbersberg, I. J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.-W., da Silva Santos, L. B., Bourne, P. E., et al. (2016). The

FAIR guiding principles for scientific data management and stewardship. *Scientific Data*, 3, 160018. <https://doi.org/10.1038/sdata.2016.18>

Yu, R., Li, Q., Fischer, C., Doroudi, S., & Xu, D. (2020). Towards accurate and fair prediction of college success: Evaluating different sources of student data. In *Proceedings of the 13th International Conference on Educational Data Mining* (pp. 292–301).

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## ABBREVIATIONS

|                   |                                                                                  |
|-------------------|----------------------------------------------------------------------------------|
| <b>AI</b>         | Artificial Intelligence                                                          |
| <b>API</b>        | Application Programming Interface                                                |
| <b>AUC</b>        | Area Under the Curve                                                             |
| <b>CommonKADS</b> | Knowledge engineering methodology adopted in this study (Schreiber et al., 2000) |
| <b>CSV</b>        | Comma-Separated Values                                                           |
| <b>DSRM</b>       | Design Science Research Methodology                                              |
| <b>GPU</b>        | Graphics Processing Unit                                                         |
| <b>IMD</b>        | Index of Multiple Deprivation                                                    |
| <b>JSON</b>       | JavaScript Object Notation                                                       |
| <b>JWT</b>        | JSON Web Token                                                                   |
| <b>KE</b>         | Knowledge Engineering                                                            |
| <b>LA</b>         | Learning Analytics                                                               |
| <b>LIME</b>       | Local Interpretable Model-agnostic Explanations                                  |
| <b>LMS</b>        | Learning Management System                                                       |
| <b>LTI</b>        | Learning Tools Interoperability                                                  |
| <b>MCC</b>        | Matthews Correlation Coefficient                                                 |
| <b>OULAD</b>      | Open University Learning Analytics Dataset                                       |
| <b>PRISMA</b>     | Preferred Reporting Items for Systematic Reviews and Meta-Analyses               |
| <b>REST</b>       | Representational State Transfer                                                  |
| <b>ROC</b>        | Receiver Operating Characteristic                                                |
| <b>SHAP</b>       | SHapley Additive exPlanations                                                    |
| <b>SIS</b>        | Student Information System                                                       |
| <b>SMOTE</b>      | Synthetic Minority Over-sampling Technique                                       |
| <b>TAM</b>        | Technology Acceptance Model                                                      |
| <b>VLE</b>        | Virtual Learning Environment                                                     |
| <b>WCAG</b>       | Web Content Accessibility Guidelines                                             |
| <b>XAI</b>        | Explainable Artificial Intelligence                                              |
| <b>XGBoost</b>    | Extreme Gradient Boosting                                                        |

## APPENDICES

### Appendix A: Reproducibility Details

Table 4.1 lists the frameworks used at each architectural layer. Table A.1 instead records the software versions, configuration, and random seeds of the evaluated pipeline, as reported in Chapters 4 and 5. The versions are those of the Docker image in which the pipeline was run.

*Table A.1. Reproducibility details of the evaluated pipeline.*

| Item                                                                               | Value                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model development environment                                                      | Google Colab notebooks for exploratory work; prototype code in versioned Python modules (data_layer, prediction_engine, explainability) (Chapter 4, Section 4.3).                                                   |
| Evaluation environment libraries                                                   | scikit-learn 1.9.1 and xgboost 3.2.0 (Chapter 5, Section 5.7).                                                                                                                                                      |
| Other package versions (pandas, NumPy, SHAP, LIME, joblib, Flask, PyJWT, gunicorn) | pandas 3.0.6, NumPy 2.4.6, SHAP 0.51.0, LIME 0.2.0.1, joblib 1.6.0, Flask 3.1.3, PyJWT 2.15.1, gunicorn 26.2.0.                                                                                                     |
| Python version                                                                     | Python 3.11.16 (python:3.11-slim Docker image).                                                                                                                                                                     |
| Hardware                                                                           | Docker container on a Linux x86_64 host with 4 virtual CPUs and 16 GB of memory; no GPU.                                                                                                                            |
| Git commit or code version                                                         | Branch claude/gallant-hypatia-sq10ks of <a href="https://github.com/daniel-larry/python-run-result">github.com/daniel-larry/python-run-result</a> , commit dbecf4d.                                                 |
| Model artefact identifier                                                          | Serialised joblib pipeline (preprocessing and model) with a JSON manifest recording the training date, feature dictionary version, and evaluation metrics (Chapter 4, Section 4.6). Identifier: xgb-20260929-112741 |
| Model configuration                                                                | XGBoost with 300 estimators, maximum depth 5, learning rate 0.05, subsample 0.8, and column subsample 0.8; decision threshold 0.5; no resampling and no hyperparameter search.                                      |
| Validation protocol                                                                | GroupKFold with four folds grouped by code_presentation (2013B, 2013J, 2014B, 2014J); out-of-fold metrics over all 32,593 observations.                                                                             |
| Input features                                                                     | 24 columns (17 numeric, 7 categorical), expanding to 59 features after one-hot encoding (Appendix B).                                                                                                               |
| Random seeds                                                                       | random_state = 42 throughout; LIME sample of 40 instances drawn with numpy default_rng, seed 42; seed-stability check of the SHAP ranking with the model refitted at random_state 1 to 7 (Chapter 5, Section 5.7).  |
| Explanation computation                                                            | TreeSHAP with the final fitted model across all 32,593 observations; LIME on the 40 sampled instances.                                                                                                              |

## Appendix B: Feature Definitions and Importance Rankings

Table B.1 defines the 24 input columns of the evaluated model, the OULAD table from which each is drawn, how it is calculated, when it becomes available, and whether it would be available at an early-warning point. All behavioural and assessment features in the evaluated pipeline were computed over the full course presentation, which is why the evaluation is retrospective (Chapter 3, Section 3.8). Calculation entries describe each feature as documented in this thesis; entries marked as inferred are derived from the feature name and should be confirmed against the feature-engineering code.

Table B.1. Feature definitions, source tables, calculation, and temporal availability.

| Feature                    | Definition                                                             | Source table                   | Calculation                                       | Temporal availability | Available at early-warning time |
|----------------------------|------------------------------------------------------------------------|--------------------------------|---------------------------------------------------|-----------------------|---------------------------------|
| num_of_prev_attempts       | Number of previous attempts at the module                              | studentInfo                    | Taken from the source column                      | At registration       | Yes                             |
| studied_credits            | Total credits of the modules being studied                             | studentInfo                    | Taken from the source column                      | At registration       | Yes                             |
| date_registration          | Registration date relative to the module start (days)                  | studentRegistration            | Taken from the source column                      | At registration       | Yes                             |
| withdrew_before_start      | Indicator that the student unregistered before the module started      | studentRegistration            | Derived from the unregistration date (inferred)   | Before module start   | Yes (see note)                  |
| module_presentation_length | Length of the module presentation (days)                               | courses                        | Taken from the source column                      | Before module start   | Yes                             |
| n_assessments_submitted    | Number of assessments submitted                                        | studentAssessment              | Count of submissions                              | Full presentation     | No, as constructed              |
| mean_score                 | Mean score of submitted assessments                                    | studentAssessment              | Mean of assessment scores                         | Full presentation     | No, as constructed              |
| min_score                  | Lowest score of submitted assessments                                  | studentAssessment              | Minimum of assessment scores                      | Full presentation     | No, as constructed              |
| mean_days_late             | Mean lateness of submissions (days)                                    | studentAssessment, assessments | Mean of submission date minus deadline (inferred) | Full presentation     | No, as constructed              |
| n_late_submissions         | Number of submissions after the deadline                               | studentAssessment, assessments | Count of late submissions (inferred)              | Full presentation     | No, as constructed              |
| n_banked                   | Number of assessment results carried over from a previous presentation | studentAssessment              | Count of banked results (inferred)                | Full presentation     | No, as constructed              |
| clicks_7d_before_deadline  | VLE clicks in the 7 days before assessment deadlines                   | studentVle, assessments        | Sum of daily clicks in deadline-anchored windows  | Full presentation     | No, as constructed              |
| clicks_14d_before_deadline | VLE clicks in the 14 days before assessment deadlines                  | studentVle, assessments        | Sum of daily clicks in deadline-anchored windows  | Full presentation     | No, as constructed              |

| Feature                    | Definition                                            | Source table            | Calculation                                        | Temporal availability | Available at early-warning time |
|----------------------------|-------------------------------------------------------|-------------------------|----------------------------------------------------|-----------------------|---------------------------------|
| clicks_28d_before_deadline | VLE clicks in the 28 days before assessment deadlines | studentVle, assessments | Sum of daily clicks in deadline-anchored windows   | Full presentation     | No, as constructed              |
| total_clicks               | Total VLE clicks                                      | studentVle              | Sum of daily clicks                                | Full presentation     | No, as constructed              |
| active_days                | Number of days with VLE activity                      | studentVle              | Count of days with clicks (inferred)               | Full presentation     | No, as constructed              |
| mean_daily_clicks          | Average daily VLE clicks                              | studentVle              | Mean of daily clicks (denominator to be confirmed) | Full presentation     | No, as constructed              |
| code_module                | Module identifier                                     | studentInfo             | Categorical, one-hot encoded                       | At registration       | Yes                             |
| gender                     | Gender                                                | studentInfo             | Categorical, one-hot encoded                       | At registration       | Yes                             |
| region                     | Geographic region                                     | studentInfo             | Categorical, one-hot encoded                       | At registration       | Yes                             |
| highest_education          | Highest prior education                               | studentInfo             | Categorical, one-hot encoded                       | At registration       | Yes                             |
| imd_band                   | Index of multiple deprivation band                    | studentInfo             | Categorical, one-hot encoded                       | At registration       | Yes                             |
| age_band                   | Age band                                              | studentInfo             | Categorical, one-hot encoded                       | At registration       | Yes                             |
| disability                 | Declared disability                                   | studentInfo             | Categorical, one-hot encoded                       | At registration       | Yes                             |

Note. Features computed over the full presentation are not available at an early-warning point as constructed; restricting them to the information available at each point is the early-warning evaluation proposed in Chapter 7. The derivation of `withdrew_before_start` is inferred from its name and should be confirmed against the feature-engineering code.

## **B.2 Top 15 features by global TreeSHAP importance (mean absolute SHAP value, log-odds units), all 32,593 observations. Reproduced from Table 5.4.**

1. `n_assessments_submitted`: 3.1110
2. `active_days`: 0.7087
3. `code_module = FFF`: 0.5338
4. `mean_score`: 0.4790
5. `total_clicks`: 0.3819
6. `clicks_28d_before_deadline`: 0.3111
7. `min_score`: 0.2741

8. module\_presentation\_length: 0.2718
9. code\_module = EEE: 0.2285
10. mean\_days\_late: 0.1468
11. clicks\_14d\_before\_deadline: 0.1360
12. highest\_education = Lower Than A Level: 0.1312
13. code\_module = DDD: 0.1075
14. n\_late\_submissions: 0.0770
15. code\_module = BBB: 0.0640

**B.3 Top 10 features by mean absolute LIME weight, aggregated over 40 sampled instances (seed 42). Reproduced from Table 5.5.**

1. n\_assessments\_submitted: 0.3961
2. code\_module = FFF: 0.1397
3. code\_module = AAA: 0.0995
4. mean\_score: 0.0640
5. active\_days: 0.0498
6. module\_presentation\_length: 0.0447
7. clicks\_28d\_before\_deadline: 0.0236
8. min\_score: 0.0183
9. date\_registration: 0.0073
10. region = Wales: 0.0043